

BUGBOTS • ICE CREAM BOTS • TERROR BOTS

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MAGAZINE

February 2009

MEGABOT "ASPIRES" TO MOVE

*Using A Laptop For
Your Robot's Brain*

Combat Zone

*Building A Better
Battery System*

Creative Armor

*Meet The Grand
"Tourinho"*



♦ A Practical
Quadrature Encoder

♦ Implementing A
Digital PID Controller

♦ BLDC Motor Project

A photograph of Kristin O'Friel and Che-Wei Wang standing in the main hall of Grand Central Terminal. Kristin, on the left, is wearing a dark coat and a yellow scarf, with her hand on her hip. Che-Wei, on the right, is wearing a grey sweater and jeans, holding a red knitted Momo robot. The background shows the iconic architecture of the terminal, including large arched windows and a clock. A small, metallic, dome-shaped Momo robot is on the floor near Che-Wei's feet.

Let your geek shine.

Meet Kristin O'Friel and Che-Wei Wang, inventors of Momo. Momo is a haptic GPS navigational device born out of a physical computing class project at ITP. Kristin and Che-Wei used SparkFun products to develop their Momo prototype.

The tools are out there - from GPS modules to microcontrollers, tutorials to forums. Find the resources you need and let your geek shine too.



Sharing Ingenuity

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For more information on Momo visit www.momobots.com. For more information on ITP and the Tisch School of the Arts at NYU visit itp.nyu.edu/itp.



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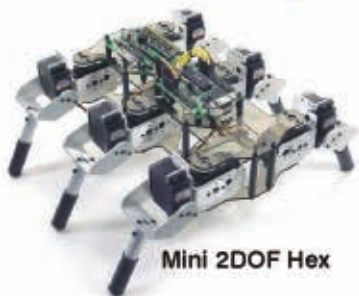
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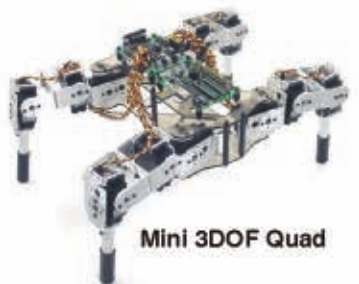
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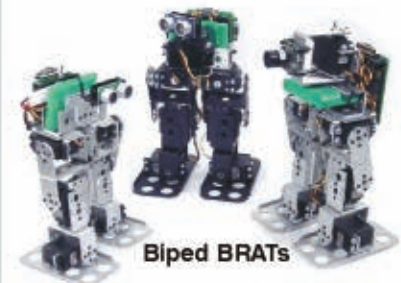
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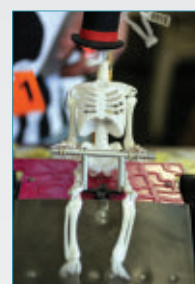
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Mind / Iron



by Bryan Bergeron, Editor

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PUBLISHER
 Larry Lemieux
publisher@servomagazine.com

**ASSOCIATE PUBLISHER/
 VP OF SALES/MARKETING**
 Robin Lemieux
display@servomagazine.com

EDITOR
 Bryan Bergeron
techedit-servo@yahoo.com

TECHNICAL EDITOR
 Dan Danknick
dan@teamdelta.com

CONTRIBUTING EDITORS

Jeff Eckert	Tom Carroll
Gordon McComb	David Geer
Dennis Clark	R. Steven Rainwater
Fred Eady	Kevin Berry
David Ward	John Blankenship
Samuel Mishal	Michael Simpson
Carlos Montesinos	Ricardo Toro
Zac O'Donnell	Blake Hooper
Nick Martin	Dave Wiley
John Frizell	William Smith

CIRCULATION DIRECTOR
 Tracy Kerley
subscribe@servomagazine.com

**MARKETING COORDINATOR
 WEBSTORE**
 Brian Kirkpatrick
sales@servomagazine.com

WEB CONTENT
 Michael Kaudze
website@servomagazine.com

PRODUCTION/GRAPHICS
 Shannon Lemieux
 AJ Cohen

ADMINISTRATIVE ASSISTANT
 Debbie Stauffacher

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Symbiotic Robots

Symbiosis — an association of mutual benefit — is a popular strategy for survival in the biological world. Think humans and normal intestinal bacteria, the common behavior of smaller fish cleaning larger fish, or even the seemingly fearless Egyptian plover bird that feeds on the leeches attached to the gums of a crocodile.

Now, imagine how symbiosis could be useful in a large, 'host' robot — say a planetary rover. Wouldn't it be useful to have relatively small, internal robots that monitor and — if needed — maintain the drive train, battery, and other physical components that can't be addressed by the basic design of the host? Think about how our white blood cells continually roam through our bodies to identify and address infections and other problems.

But how would a semi-autonomous robot working inside of another robot obtain energy? It could have an umbilical cord to the larger robot's power supply, but this would limit movement and add complexity to the construction of the host robot. It could occasionally tap into the main power of the host, something akin to a parasite latching on to an artery or vein. It could use solar energy, but this would require a transparent larger robot that works in a well-lit environment.

Another approach is to have the small internal robots scavenge energy from the vibrations of the mechanical motion of the host

robot. Although I'm not aware of such symbiotic systems, the components are nearly all commercially available. In particular, you can now purchase vibration energy scavengers, such as the Joule-Thief™ from AdaptivEnergy (www.rlpenergy.com).

As you can see from the photo, the compact, energy scavenger easily fits in the palm of my hand. You can also see the primary mechanism behind the energy scavenging: the gold, triangular beam anchored at the right and bolted to a silver weight at the left. Vibration in the environment is converted into electricity that can be used to power a transmitter or other low power device.

The current physical configuration of the Energy-joule is optimized for 60 Hz vibration. The optimum vibration frequency can be modified by changing the configuration of the triangular beam and weight. Because of this and related limitations, the Joule-thief isn't ready to drop into the battery compartment of your robot as a battery replacement, but it does demonstrate the principles of energy scavenging. As shipped by



Mind/Iron Continued

AdaptivEnergy, the Joule-Thief evaluation kit is paired with a low power transmitter that operates on a few mA at 3V.

After working with the device, it was clear to me that by incorporating a future version of Joule-thief — or something like it — practical symbiotic robotics is a possibility. I can envision repair robots that traverse spaces within the inner hull of a submarine or submersible robot, constantly looking for and repairing leaks. Again, the parallels with white blood cells that scavenge our blood for dead and dying red blood cells, foreign bacteria, and other pathogens come to mind.

An energy scavenging device is a necessary but insufficient component of our internal maintenance robot. Next, we'll need to secure highly efficient drive systems, sensors, and lightweight platforms. Imagine a robot that, say, lives under the hood of your car, constantly monitoring and — if necessary — repairing wires, connectors, and cables. I'm obviously not talking about an ordinary carpet roamer, but a robot that must navigate significant obstacles while avoiding falling to the street or into the fan belt.

Obviously, it will take some engineering and a lot of creativity to devise a robotic platform at home in an irregular, moving environment, with enough power to perform real work. If you're up to the challenge, please write in and share your designs and thoughts with other readers. **SV**



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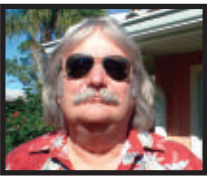
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Diggin' the ClamBot

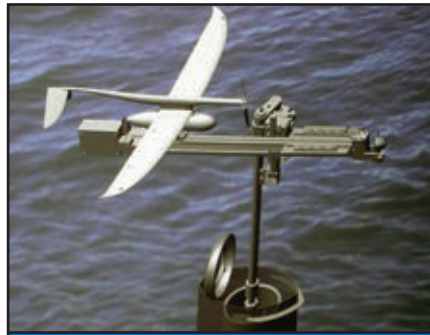


Siliqua patula — or Pacific razor clam — is the inspiration behind MIT's ClamBot design. Courtesy of Oregon Department of Fish and Wildlife.

These tasty treats are big (up to seven inches [18 cm] long), fast (able to travel 1 cm/sec through wet sand), and highly desired for food (clambers are generally limited to 15 per day and must keep the first ones caught, regardless of size or condition). But as of lately, Pacific razor clams are of particular interest to Anette "Peko" Hosoi, an associate professor in the Department of Mechanical Engineering at MIT (www.mit.edu), for different reasons. With a little support from Bluefin Robotics Corp., the Battelle Institute, and Chevron, she has assembled a bunch of pressure regulators, pistons, and other paraphernalia to create RoboClam.

RoboClam is designed to simulate the razor's ability to dig deep into the sand and lodge itself in place with 10 times the pull resistance of a standard anchor of comparable size. It seems that these guys do more than just dig a hole with their "feet" and drop into it. In fact, they also make rapid up-and-down movements and open and close their shells, thereby temporarily changing the composition of surrounding terrain into something resembling quicksand. This allows them to quickly burrow up to about 28 in (70 cm) deep. ClamBot is only entering the test stage, but the goal is to develop a robotic anchor that can be used for such things as tethering and releasing (it can operate in reverse mode) small robotic subs and safely detonating buried mines.

Subsurface-to-Air UAV



A UAV ready for launch from a submarine mast. Courtesy of EMT Penzberg.

Also aimed for sea duty is VOLANS, a system developed by German mast builder Gabler Maschinenbau (maschinenbau.gabler-luebeck.de) and mini-UAV specialist EMT Penzberg (www.emt-penzberg.de). (VOLANS is tortuously derived from "coVert Optical Airborne reconnaissance Naval adapted System," in case you're wondering.) What you get is a hoistable mast that can launch three UAVs from a submarine while completely submersed at periscope depth, typically 12 to 25 m (40 to 82 ft). The UAV then relays sensory imagery back to the sub in real time via an antenna on the boat's communications mast. The UAVs can operate at distances up to 30 km (16 nautical mi), providing vital information for special forces teams before they go ashore. The only drawback is that the spy planes cannot be recovered unless the sub surfaces; they would normally be picked up by friendly forces on land or intentionally crashed.

Multipurpose Yard Bot

Meanwhile back on land, students at the LSU Department of Computer Science (www.csc.lsu.edu) have upped the ante in automated yard care with AgBot: a



The AgBot uses an autonomous GPS system to map out your yard. Courtesy of LSU Robotics Lab.

prototype multipurpose agricultural care machine. Unlike the current single-function lawnbots (usually mowers), AgBot is a fully customizable, solar-powered machine that includes a seed dispenser, high-torque auger, and fertilizer tank. An autonomous GPS system plots out the coordinates of your yard, maps the bot's path, and stores the data for future operations.

The solar-powered machine can travel at 6 MPH for at least four hours at a time. In addition, it is fitted with a 360° night vision camera, motion sensor, and alarm system, allowing it to double as a guard dog at night.

According to LSU, "The end goal is for AgBot to be completely customizable, as if buying a computer and selecting only the desired components versus a package that includes unnecessary options. With minor adjustments, AgBot would be capable of supporting five or six different applications," hopefully including pizza delivery, rat-catching, and scaring away door to door salesmen.

Yes, I Do Windows

Also created to eliminate work around the house and render you even more useless than you already are is Winboni: a window-washing robot that won first place at last November's International Student Design Competition of the American Society of Mechanical Engineers.



MSU students Kyle Koepf and Jonathan Luckhardt stare in simulated amazement at their creation.
Photo by Jennifer Orlando.

(Its name is derived from the Zamboni, which performs a similar operation on ice.)

Measuring only 5 x 5 x 2 in (12.7 x 12.7 x 5 cm), it uses a suction fan to attach itself to a window and then moves across the surface on wheels and scrubs the pane with felt pads. Somehow it requires no solvents or paper towels to do the job.

Powered by AA batteries, it seems only a matter of time until Billy Mays starts hawking them on TV. But act now, and you'll get two Winbonis for only \$29.99, plus we'll throw in a free Doorknob Snaffler, six whoopee cushions, and the "Elvis's Favorite Polkas" CD.

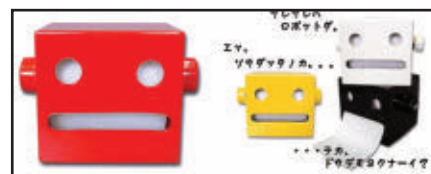


Animated version of the US Air Force's planned spy bugbot.

these sent out in a swarm," he explained. "If we know there's a possibility of bad guys in a certain building, how do we find out? This would fill that objective."

By identifying and assaulting adversaries more precisely, the robots would also help reduce or avoid civilian casualties. You don't need to put in titanium window screens anytime soon, though, as Parker doesn't expect to have bird-sized ones in operation until 2015, with insect-scale models following in 2030.

A Bot for Your Bathroom



The Airyusha Robotan toilet paper holder.

Finally, we wrap up with this month's most inane and pointless new product. It's not a robot, but it announces to your visitors (at least the ones who have sampled your Spam, bean, and haba ero pepper burritos) that you are in a robot frame of mind. Devised by Japan's Airyusha Co. Ltd. (www.airyusha.com), it may not be high-tech, but it is "made from high-quality, heavy-duty ceramic with vibrant colors," and it is said to work with all kinds of toilet paper. It even comes complete with a roll of 100% recycled paper "so you can start using Robotan right away." You can pick one up for only \$49.99 at www.audiocubes.com. **SV**

MAVs to Bug Misc. Bad Guys

Building on the old "if I could be a fly on the wall" concept, the US Air Force is working on a swarm of flying spybots that can sneak into sensitive locations, record voices, transmit video, and perhaps even fire miniature weapons. Stories of such micro aerial vehicles (MAVs) have been around for a while, but this particular application has just recently been confirmed by Greg Parker, one of the researchers at Wright-Patterson AFB. "The way we envision it, there would be a bunch of

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GEERHEAD

by David Geer

Contact the author at geercom@alltel.net

Do Robots Scream For Ice Cream?

The robotics team in the Department of Technological Studies at Ohio Northern University (ONU) has come up with yet another fun, practical, and educational use for KUKA industrial robots: free ice cream!

With two KUKA KR3 six axis arms, an ice cream machine, a PLC, touch panel interface, topping dispenser, and a camera tied to vision processing software, the ONU team erected a fully robotic ice cream machine that served up scoops and toppings to about 500 customers in three days. (Now, that is how to make a popular robot!).

The goal of this robotic construction was to demonstrate a computer vision controlled system, constituting a framework to develop strategies for bin picking, which is a popular objective in robotics. Many types of industry rely on the fast and accurate bin picking of parts for assemblies.

Adam W. Stienecker, PhD, and assistant professor of technological studies at ONU explains the details of the robotic system operation.

"Essentially, bin picking uses a camera to locate one of many randomly located parts in a bin, selecting one and then grasping it in a known orientation." In the case of the ice cream robot, the team researched bin picking by placing more than one ice cream scooper on the table for the robot to locate.

Playing Their Parts

It is important to understand each of the major components of the ice cream robot in order to understand its function as a whole. The Allen-Bradley Micrologix 1500 Programmable Logic Controller (PLC) is an industrial computer that is used to take inputs from customers via the touch panel and send commands to each of the Kuka robots. Service at the push of a button!

The touch panel — also from Allen-Bradley — is like most touch screens at mall kiosks, self-checkouts, and food ordering stations at mini-marts. Customers enter their selections on screen and, in this case, the robots go to work filling the order.

The touch panel takes requests for specific toppings on the ice cream and the PLC sets the robots in motion. The user must follow instructions on where to place the spoon and cup for the robots to successfully interact with and complete the service.

The Point Grey Research Flea2 FireWire camera serves as the eye for the robots. It takes and relays images of the spoon and its environment to the computer, which is then processed so that the robot knows how to pick it up.

Other support hardware includes a Taylor soft serve freeze and a Zevro WM300 triple toppings dispenser.

Robots sprinkle toppings on ice cream.



This article caused me to gain five pounds while writing it!



Programming

The use of two KUKA robots required two robot programs. The programs, which were mapped out using flow charts demonstrating the consecutive actions of the robots, sequence each step the robots take. "Once the flow

charts were complete, it was a matter of converting them to the robot code, KRL. This language is proprietary for the KUKA robots."

Ice cream sundae construction is split between an ice cream robot and a toppings robot. The programming of the latter enables it to select one of six different toppings based on choices made by customers at the touch panel. The same programming also enables the robot to wait for data from the camera coming through the high speed serial connection to tell it where the spoon is located.

In addition, the robot must know the orientation of the spoon so it does not place it in the cup of ice cream the wrong way. To do this, the robot relies on the VisionPro software to compute a rotation angle for the robot to use to adjust the alignment of the spoon and put it in the ice cream just right.

As for the KUKA robot assigned to the ice cream itself, this bot moves based on inputs from the touch panel, delivered through the PLC. "The code in the ice cream robot is pretty unintelligent, that is, it just moves through several points and waits while ice cream is dispensed and then returns to a position."

The VisionPro software runs on a PC. "The software receives images from the camera, analyzes them, and relays the location of the spoon to the robot by way of an RS-232 serial connection. The vision system was programmed by first giving it a training image of the spoon. That is, an image was taken of the spoon and recorded in the VisionPro software. Then, the software was told that this was what it should be looking for. It was also told at what point on the spoon it should consider its origin and at what angle the training spoon image was. Then, the program was set up to find this spoon, its location,



Robot arm releases selected toppings into cup.

and its angle of rotation, and then relay that information to the robot via the serial port."

The PLC sequencer was programmed in simple Ladder Logic, using the RSLogix500 development tool.

"It is a very simple program with only a few lines of code. It takes the topping selection button push from the touch panel, remembers it, then transmits it out to the robots when the Go button has been pushed on the touch panel."

Computer Vision

As mentioned earlier, a Flea2 Fire Wire camera fitted with a 9 mm lens was used. Optical calculations derived the ideal location for the camera in relation to the area where the spoon would be placed.

After the VisionPro was trained in spoon identification, in operation it locates a pre-determined point on the spoon within that image, down to the pixel.

Next, the computer vision software runs some vision related equations that relate the selected pixel position to the real-world measurement and placement of that part of the spoon in millimeters. The computer forwards that value to the robot via RS-232 serial communications.

"Along with that location is an angle of rotation that when applied,



A side shot of the entire robotic ice cream dispensing apparatus.



Customers making selections at interface.

yields a grasped spoon in an ideal orientation for the next task. That is, it has been relocated from a random position to a position that the robot knows." This is the core of part bin picking. The robot accomplishes this maneuver mechanically using a suction cup that attaches to the spoon at the widest point of its handle.

Design and Construction

As with any automated work cell,

From this angle, customers see the most important element of this experiment: "Free Ice Cream!"



proper location of the KUKA robots permits their work envelope to encompass the desired task – to reach the ice cream, toppings, and other elements with which they need to interact.

"The toppings dispensers also needed to be mounted within reach of the toppings robot and easily refilled by an operator. Additionally, the size of the cup into which the

toppings were dispensed dictated the vertical position of the dispensers because the robot had to be able to remove the cup with the toppings in it, so appropriate clearance had to be insured."

ONU uses KUKA KR3 robotic arms for a number of student projects and experiments, but unless next year's class does something with gelato, this is the most yummy project yet! **SV**



The KUKA manipulates an ice cream spoon.

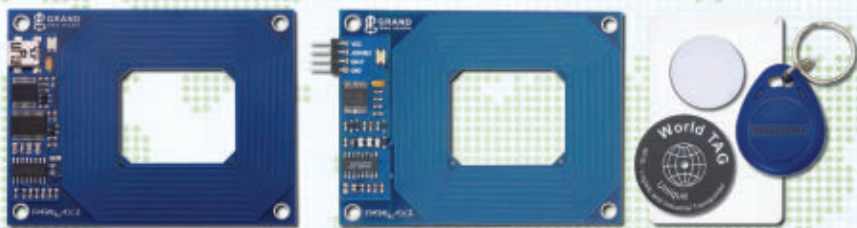
Robo Ice Cream Teamwork

Adam Stienecker and Rich Miller derived the base concept of using industrial robots to serve ice cream to University constituents. The student team leaders, together with Professor Stienecker, completed the design, followed by the implementation processes of programming, modeling, testing, and construction of the robotic apparatus.

Students were lead in five groups: the vision team, the touch panel interface team, the mechanical fabrication team, and the programming and logistics team.

Congratulations to all of the team members on a successful implementation!

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Ice cream robot video
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Department of Technological Studies
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KUKA
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RSLogix software
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Ladder Logic programming
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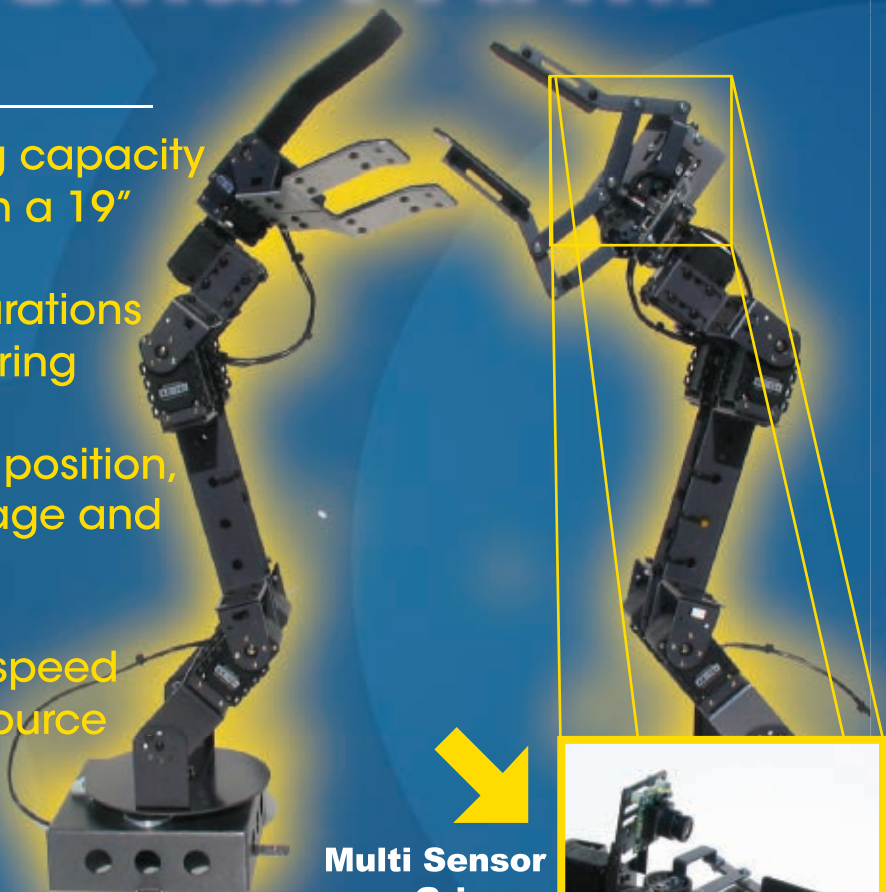
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Tap into the sum of *all human knowledge* and get your questions answered here! From software algorithms to material selection, Mr. Roboto strives to meet you where you are — and what more would you expect from a complex service droid?

ASK MR. ROBOTO

by
Dennis Clark

Reader Responses

Let it not be said that Mr. Roboto suffers the sin of hubris. Some readers responded back to me about their ideas on how to deal with a December question dealing with non-GPS robot navigation on the cheap. There are some good ideas here, to help fellow robot hobbyists!

A. This is a possible answer for Roger's problem getting a precise position for his robot. I have also been looking for a system for precise navigation for my outdoor rover robot.

Combining visual odometry and landmark recognition could be a solution. Check out www.cs.cmu.edu/~rahuls/pub/icra2005-rahuls.pdf. It might not get as close as an inch, but I think 10 cm is a reachable goal!

Needed for this approach is a computer with OpenCV software and a webcam. I built myself a Linux machine based on a cheap mini-ITX board. I'm still in an early stage, but it is promising.

—Allan Bertelsen
 Denmark

Another reader response:

A. I read your December column as usual and have a suggestion for Roger (who wanted a system for locating a robot in a 1/4 mile area).

Your idea to put a light source on the robot and cameras at the property edge is a good idea, but it can be far more effective if you *reverse* the placement.

Put a camera (or other detector) on the robot and then place any number of beacons (the number depends on the accuracy you need) throughout the intended environment. Each beacon could be coded with different colors (as you suggested) or different modulations of one color (perhaps infrared, depending on your sensor).

Alternatively, every beacon could be the same color or modulation, but give the robot the ability to remotely

turn on and off any beacon of its choice.

Either way, the robot could triangulate itself (as you mentioned) with any two beacons, giving better accuracy. It could also just move toward a certain beacon for a given distance, then turn towards another beacon, etc., letting it navigate along a path it creates (made up of movements toward beacons). This is explained in a book I wrote with my co-author (*Robot Programmer's Bonanza*, Chapter 15) where we used a series of beacons to control navigation in a suite of offices. The robot maps out a path (using algorithms similar to those in automobile GPS systems) and then turns on the required beacons one at a time, to move to the desired destination.

Probably the most powerful aspect of this algorithm is that it can be used with any type of beacon (visible light, infrared, RF, etc.) and then can be adapted to a large number of beacons in order to cover huge areas or to increase accuracy.

We wish someone would produce this product line (a sensor plus any number of beacons); the code for each beacon could be set perhaps with an eight bit DIP switch. In its simplest form, the beacons would be just bright colored light that could be controlled with an RF link from the robot. A standard webcam could be used to detect the light (just turn it off to verify proper detection). With a beacon hung at the top of each doorway (and other appropriate places of interest), navigation of a house would be easy.

RobotBASIC — the language used in the book mentioned above — provides the ability to access all ports (for controlling the beacons), commands to capture images from a webcam, and even commands for locating specific colors etc., in a user-defined grid pattern superimposed on the image.

All in all, I think a system of this type would be far less than the budget suggested in the question.

— John Blankenship
www.RobotBASIC.com

Keep those suggestions coming folks! I'm perfectly happy to provide a forum for the exchange of ideas. Now, on to our questions ...

Q . Do you remember the name of the robot that had moving eyes and mouth, and answered the phone?

— John S.
Dayton OH

A . I remember something about what you are asking, but my memory told me that it wasn't human (you didn't specify) but some kind of an animal that would signal the user that he/she had a call. If I'm not completely off base here, I believe that you are referring to Stephan Marti's Cellular Squirrel. This was a project whose design criteria was to have an inoffensive object — in this case, a stuffed squirrel — signal the cell phone owner of an incoming call. It would blink its eyes, move, and generally provide a non-offensive visual cue. You can find information about Stephan's project at <http://web.media.mit.edu/~stefanm/phd/cellularsquirrel/>

Figure 1 shows the squirrel and Figure 2 shows how much technology is stuffed inside it. (Photos courtesy of Stephan Marti.)

Q . I usually read your article first when I get my magazine. I have a question about using a voice coil actuator for small quick motions — such as eyelid blinking on a puppet or similar action — and controlling it with a PICAXE-20M. Would the outputs from the PICAXE have to be coupled through a Darlington transistor or something to protect it from possible voltage spikes from the coil? Also, how does one figure out how many windings and of what gauge wire to make the coils for five or twelve volts? I was thinking about using some small rare earth magnets due to their size and high power. Would using voice coils for some motions save space and be simpler? The article on the servos in the puppet head was very informative.

— Dale G.
New Jersey

A . Thanks for your support! I'll answer the easy question first: in no case, should you drive an inductive load directly from a microcontroller I/O line. The surge currents could quickly damage the port. You could perhaps protect against the voltage spikes, but it is far safer to use a driver transistor (Darlington not required) to drive the coil. Size your transistor to handle the current that you are expecting and use a drive resistor between the I/O pin and the base of the transistor; a 1K, 1/4 W resistor is usually fine for this task. The PICAXE is based on the Microchip PIC microcontroller whose I/O lines can typically drive 20 ma, which is more than enough to control your driver transistor. Do make sure that you guard against the

Figure 1. The Cellular Squirrel.



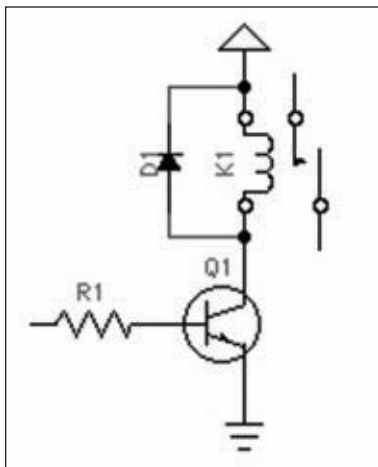
Figure 2. Servos and boards and wires!



inductive kickback (CEMF) from the solenoid with a diode just as you would for driving a relay. See Figure 3 for an example circuit.

I would not think that a speaker voice coil would have the power to do what you want. You could create your own solenoid using a magnet and winding your own coil, but really this is a topic for an entire book. Magnetics isn't exactly voodoo, but it is complex. I suggest that you use a solution that someone else has already created, especially if it is inexpensive. Use a small solenoid, for example. I found one at Jameco (www.jameco.com); PN 1919203 that

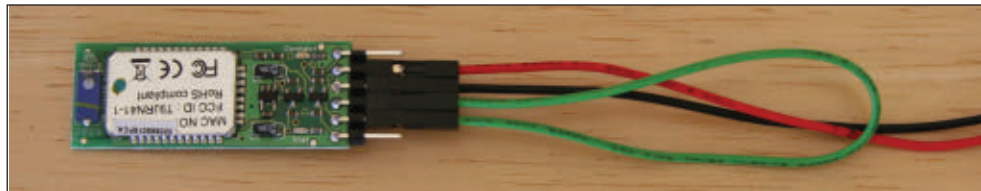
Figure 3. Relay (or any inductive coil) driver.



small, inexpensive solenoid of marginal power for your application. However, that will require experimentation to find adequate power for the throw that you might need.

should do the trick. I could not find a datasheet on it so I don't know how large it is, but the price was right (\$2.95). You could also try gutting inexpensive relays for the solenoid section; this will give you a

Figure 4. BlueSMiRF config wiring.



(which is very easy to do.) If you are having problems getting your bootloader to work, I strongly recommend that you download a free Arduino environment and use one of their files as your template for yours. Your basic problem is that buried deep in the documentation for the Atmel ATMEGA168 is the fact that the UART does not like to operate when the bit error is greater than 2%. If you look at Figure 4 (which is a table of baud rate values for various settings on an ATMEGA168 running at 8 MHz), you'll see that the fastest bit rate you can get without more than a 2% error is 38400. I walked this whole table and verified that the UART truly is that picky. Not one setting whose bit error was greater than 2% would work for me.

Table 1. ATMEGA UART baud calculations.

Baud Rate	U2Xn = 0		U2Xn = 1	
	UBRRn	Error	UBRRn	Error
2400	207	0.20%	416	-0.10%
4800	103	0.20%	207	0.20%
9600	51	0.20%	103	0.20%
14400	34	-0.80%	68	0.60%
19200	25	0.20%	51	0.20%
28800	16	2.10%	34	-0.80%
38400	12	0.20%	25	0.20%
57600	8	-3.50%	16	2.10%
76800	6	-7.00%	12	0.20%
115200	3	8.50%	8	-3.50%
230400	1	8.50%	3	8.50%
250000	1	0.00%	3	0.00%
500000	0	0.00%	1	0.00%
1000000	-	-	0	0.00%

So, the answer is simple: Modify your bootloader code and your BlueSMiRF Gold to communicate at 38400 or change your ATMEGA168 clock frequency to one that will allow a 115200 baud rate with a low error rate. There are several other clock frequencies that allow much more precise serial baud rate settings.

If you want to modify your baud rate, it isn't too hard to do. First, you'll need to know how to change it on your BlueSMiRF device. The SparkFun site has the datasheet for this Bluetooth dongle here at www.sparkfun.com/datasheets/RF/RN_BlueportII-ref-guide.pdf

Here is how to change

the baud rate on the BlueSMiRF Gold unit:

(This assumes that you have already paired your BlueSMiRF unit with your computer and you have correctly gotten all of your devices worked out. Since you have been working with this for a while, I'm assuming that you have all of this working.)

1) Set your BlueSMiRF up so that you can configure it from your computer's favorite terminal emulator. I'm on a Macintosh so I tend to use ZTerm. People have reported problems using the Windows default terminal emulator Hyperterminal, but it usually works. You can't just *talk* to the BlueSMiRF from your terminal emulator; you need to

Q I am a new subscriber to *SERVO*. I have been working on wireless re-programming for months. I have not been able to figure it out for the life of me. I am using an ATmega168 at 8 MHz, a BlueSMiRF Gold for UART, and the Bootloader from **AVRfreaks.com**. Hope you can help!

— Brandt Daniels

A Welcome to *SERVO*, Brandt! It just so happens that I have the same kind of project going, but I used the Arduino bootloader as my base and modified it

wire the BlueSMiRF Tx-O and Rx-I pins together so it will "talk to itself." See Table 1 for how to do this.

2) Put the BlueSMiRF into *command* mode by typing **\$\$\$**. It will respond with **CMD** and the red LED will start flashing quickly on the board.

3) On page 10, you'll find the **SU** command; type **SU,38<cr>** to change the power-up default baud rate to 38400. It will respond with **AOK**.

4) Now type **—<cr>** to exit command mode. It will respond with **BYE** and the red LED will stop flashing on the board. You're done! Now the unit will always work at 38400 baud until you change it to something else.

Unlike what you would expect if you tied the Rx to the Tx normally on a connected BlueSMiRF, when you are in **Command** mode the unit will NOT echo back characters. If you want to see what you are typing, then you'll need to enable local echo on your terminal emulator.

If there is any interest, I'll write about how I modified an Arduino bootloader for the ATMEGA168 to use as a generic one for my projects so that I could use a programmer button to choose between bootloader startup and jumping directly to the program code. It isn't hard, and modifying the Arduino programming environment to use it was even easier.

I'm becoming very fond of the Arduino environment as a tool to teach newcomers to robotics and embedded programming, as well as using it for a quick prototyping tool for projects! If anyone has questions pertaining to Arduino, please send them along!

Well, I've reached the end of another article, and again it was a lot of fun. As usual, if you have any questions about things robotic, please drop me a line at roboto@servomagazine.com and I'll be happy to work on it! Until next time, keep on building those robots! **SV**

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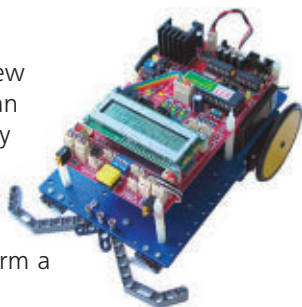
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NEW PRODUCTS

CONSUMER ROBOTS

Robot 877

Futurlec introduces their new PIC16F877 Robot 877 — an exciting, easy to use, and fully programmable robot. Based on the popular PIC16F877 microcontroller, this robot can be programmed to perform a number of instructions.



A series of microswitches are used for collision detection, together with a couple of DC motors to control robot movement. A row of optical sensors are mounted on the base for tracking operations, together with a ready-to-run LCD (2x16 characters) installed on the top for feedback and information display.

Code can be developed quickly and easily with the numerous range of programming development tools available for PIC microcontrollers. Code can be downloaded easily to the unit by a direct connection to a computer's RS-232 serial port. Users can teach the robot to perform many functions and it can provide instant feedback with the included LEDs and LCD.

Also included on the board is a provision for a DS1307 real time clock, EEPROMs, and low voltage circuit cut-out. The robot is completely mobile with a large on-board rechargeable battery that is included. There are two CD motors installed for turning and forward/reverse motion.

The robot's base is solid anodized aluminium, designed for strength and light weight. This reduces power drain on the battery and ensures the robot can withstand tough knocks combined with collisions. Additional functions such as voice, distance measuring, and infrared sensing can easily be added with the port connections and analog input connections brought out to standardized polarized connectors.

Code examples are provided in assembly language and PICBasic language for LCD operation, robot movement, analog inputs, and more.

The robot requires a standard sealed battery charger for the 6V on-board battery (not included), and is available now for USD \$129.

For further information, please contact:

Futurlec

Web: www.futurlec.com

KT-X Lite Biped Robot

The KumoTek-X (KT-X) bipedal is the latest robot to emerge from the of KumoTek Robotics and Vstone Corporation. This robot was designed to offer an entertaining, low-cost alternative to high-end robotics, while combining the latest in sleek design and cutting-edge robotics engineering.



The KT-X is a low-cost bi-pedal robot platform that (a) can be controlled using a standard wireless PS2 gamepad controller; (b) is easy to use; and (c) offers endless hours of entertainment. The robot is capable walking, running, somersaulting, and standing up from a face-up or face-down lying position. It also can be programmed to pull itself up autonomously after it has fallen over. It stands 13" tall, has 13 servo actuated joints (i.e., 13 degrees of freedom), a powerful 60 MHz HV processor with 512 kB ROM/64 kB RAM, and comes fully loaded with over 75 preprogrammed acrobatic motions.

The robot also comes with an easy to use programming interface, which is without a doubt the most user-friendly programming utility to date and requires little to no prior experience with robotics or programming. It can be purchased as a kit or fully assembled.

Software Specs:

- GUI: Slider bars for each axis; drag and drop motions for easy configuration; over 100 motion files included.
- GYRO: Provides real time gyro control, management, and feedback through RS-232 connection.
- RF and Game-pad Configuration: Easy to use RF or PS2 gamepad remote control for uploading and possible combination of motions.
- Autodemo Files: Can upload numerous external autodemo files for untethered robot demonstrations.
- Education: Great for educators and classroom environments.

For further information, please contact:

Kumotek Robotics

Web: www.kumotek.com

CONTROL BOARD

Propeller Control Board

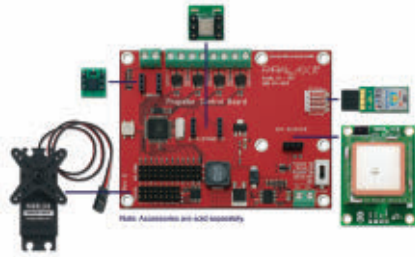
The Propeller Control Board from Parallax is a complete solution for controlling robotics platforms. This is the same control board that ships with their QuadRover robot. The Propeller Control Board is capable of interfacing directly with the Parallax GPS module, the Hitachi HM55B compass module, and the Hitachi H48C three-axis accelerometer. This provides options for tracking position, heading, and acceleration for robotic applications.

Twelve of the Propeller I/O pins are brought out for customer interfacing and control applications; eight are dedicated to servo control; and eight are dedicated to N-Channel MOSFET outputs, providing high-current drivers for solenoids or other high-current devices. There is also a magento-pickup compatible tachometer input which can be used for measuring the RPM of gasoline engines. The power switch allows you to turn all power off, only the Propeller on, or the Propeller and servos on. There is a reset button for the Propeller chip. A socketed 5 MHz crystal is included. Price is \$195.99.

For further information, please contact:

Parallax, Inc.

Web: www.parallax.com



modes of operation. In the autonomous mode, the robot explores any area which has a solid level floor (most carpets except deep pile types). In this mode, it features two IR sensors to detect objects in its path, and can back up and turn left or right. It has two white headlights that automatically come on when it's dark.

In the robot's other mode of operation, almost any brand of TV/VCR IR remote can be used to remotely control its operation. From up to 20 feet away, you can press any button on your IR remote to control the robot's response. One press and the robot goes forward. Two presses and the robot goes backwards. Three presses and it turns left. Four presses and it turns right. Five presses and the headlights turn on.

The mode at which the robot operates in (autonomous or IR remote controlled) is selected by a small moveable shunt jumper. The robot is about 4.9" x 5.1" and uses two powerful gearhead motors, a built-in piezo sounder, red, green, and white LEDs, transistors and voltage regulators. It operates from one 9V battery (not included) and comes with a CD which has the program files available for viewing.

Although not necessary for the operation of the robot, advanced students can buy an inexpensive USB programmer from Zilog distributors which allows you to create your own programs for the robot or to step through the program code that is used and watch it execute. The complete kit includes all parts, PC board, pre-programmed IC, source code used for the robot, and instructions. The skill level is set at 3.

For further information, please contact:

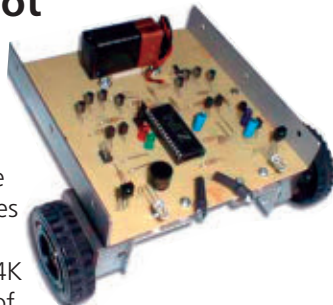
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ROBOT KITS

Microcontrolled IR Remote Controlled Robot Kit

Chaney Electronics' latest and most sophisticated Robot — the Microcontrolled IR Remote Controlled Robot Kit — uses a 28-pin Zilog Z8F0423 microcontroller featuring 4K of Flash memory and 1K of RAM. It has a program written in the industry standard C programming language using ZDS II — a freely downloadable development suite. The program code with comments can be printed out so that you can view the code that operates the robot, giving you an introduction to microcontroller program basics. The Zilog microcontroller is pre-programmed to have two separate



MOTORS

Integrated Servos Get Smarter

Animatics Corporation has taken their best selling integrated servos — the SM2315D and SM2315DT — and added power and functionality, while lowering the cost. The new products are called the SM2316D-PLS2 and the SM2316DT-PLS2.

As before, these small servomotors are actually complete servo systems with built-in controller, amplifier, and encoder. The "PLS2" versions have undergone many enhancements, some of which increase performance in coordinated motion CNC applications utilizing JenCNC and SMNC host level CNC control programs for plasma, laser, and water jet cutting, as well as CNC routing and milling.

For further information, please contact:

Animatics Corporation

Website: www.animatics.com

EVENTS CALENDAR



Send updates, new listings, corrections, complaints, and suggestions to: steve@ncc.com or FAX 972-404-0269

Know of any robot competitions I've missed? Is your local school or robot group planning a contest? Send an email to steve@ncc.com and tell me about it. Be sure to include the date and location of your contest. If you have a website with contest info, send along the URL as well, so we can tell everyone else about it.

For last-minute updates and changes, you can always find the most recent version of the Robot Competition FAQ at Robots.net: <http://robots.net/rcfaq.html>

— R. Steven Rainwater

February

12-15 Pragyan

National Institute of Technology, Trichy, India
Another Micromouse competition. Don't miss it!
www.pragyan.org/

12-15 Techkriti RoboGames

Indian Institute of Technology, Kanpur, Uttar Pradesh, India
Autonomous robots can compete in Park ur Porsche — an automated car-parking event. There's a fire-fighting robot contest and they've also thrown in some robot combat and an event called Shoot on Sight that involves robots playing a game with balls.
www.techkriti.org/#/competitions-robo-games/

15-19 APEC Micromouse Contest

Washington, DC
Micromouse maze running. Cash prizes and trophies.
<http://www.apec-conf.org/>

March

7-8 METU Robotics Days

METU Culture & Convention Center, Turkey
Events include line-following, Mini Sumo, Standard Sumo, trash hunting robots, stair climbing, slaloming, and freestyle.

www.roboticsdays.org/

13-14 AMD Jerry Sanders Creative Design Contest

University of Illinois at Urbana-Champaign, IL
JSDC is a college design contest centered around building a robot to compete in a two day engineering challenge.
<http://dc.cen.uiuc.edu/>

21 RobotChallenge

Vienna, Austria
Parallel Slalom, Slalom Enhanced, Standard Sumo, Mini Sumo, Micro Sumo. Autonomous.
www.robotchallenge.at/

21-22 Manitoba Robot Games

Tec Voc High School, Winnipeg, Manitoba, Canada
Mini Sumo, Prairie Sumo, Tractor Pull, Super Scramble, line-following, and Robo-Critters for the kids. Sumo events are held for both autonomous and tethered robots.
www.scmb.mb.ca/

21-22 Roboticon

University of Guelph, Ontario, Canada
A robot soccer event for LEGO robots.
www.collegeroyal.uoguelph.ca/

TBA Boonshoft Museum Robot Rumble

Boonshoft Museum, Dayton, OH
LEGO robot building and competition.
www.boonshoftmuseum.org/

April

16 Austrian Hexapod Championship

FH Hagenberg, Austria
Hexapod dancing and hexapod race.
www.fh-ooe.at/campus-hagenberg/aktuelles/events.html

17-19 National Robotics Challenge

Marion, OH
Autonomous robots.
www.nationalroboticschallenge.org/

The Escape Robot's built-in microprocessor enables it to "think" on its own. (KSR4) **\$29.95**



Soldering required
The robot frog moves forward when it detects sound and repeats: start (move forward) -> stop -> left turn -> stop -> right turn -> stop. (KSR2) **\$19.95**



5mm White water clear LED 3.5V 10,000 mcd (AB287) **\$0.56**

20 second voice recorder/playback module. The electret microphone is on the board. One button records, the other button is momentarily pressed to replay the message. (pre-assembled) (A96010) **\$6.60**



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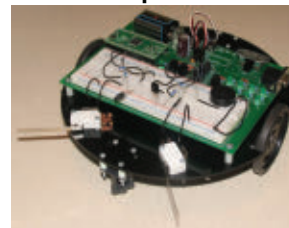
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COMBAT ZONE

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BUILD REPORT

Building a Better Battery System

● by Zac O'Donnell

My 12 pound combat robot Scurrie was designed to use two battery packs that were physically and electrically separate from each other. The pack used exclusively for the weapon consisted of twelve 1,700 mAh cells in series, while the drive pack consisted of six of the same cells. I had purchased the cells long before for a different project, so they fit my budget nicely. However, they weren't the best fit for the robot. The combined weight of the two battery packs was a pound and a half, and I

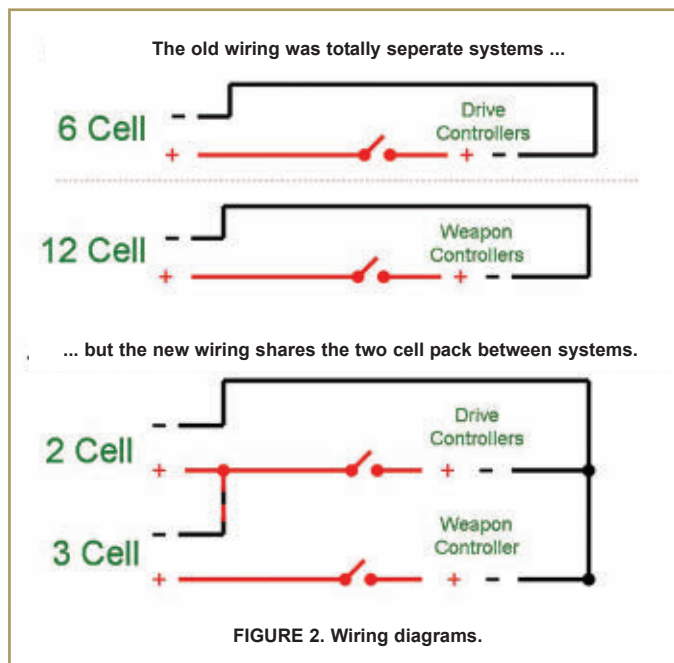
wanted to bring that down to less than a pound so that I could use the extra weight to improve the reliability of the drive system.

So, how was I going to cut my battery weight by 30 percent while maintaining the same power for the weapon and drive system in a three minute match? The answer was to find a newer technology. The 1,700 cells were nickel-metal hydride which was the most advanced technology when I bought them. However, four years turns out to be a long time in the battery industry. Two

new options had become available that offered higher energy density: lithium polymer (LiPo) and batteries from A123 Systems. I needed a battery that was rugged, could handle high charge and discharge rates, was not too expensive, and — most importantly — would meet my power and space requirements in less than a pound of total weight. A



PHOTO 1. Scurrie after taking first place at the Franklin Institute 2008 competition.



comparison of the three types of cells can be seen in Figure 1.

After balancing all of these factors, I decided to use the A123 Systems M1 batteries. This decision eliminated the possibility of having two separate battery systems in the robot because the new batteries are physically larger than the old ones, even though they weigh less. After searching for a solution for some time, I decided to ask some other builders for suggestions at <http://forums.delphiforums.com/theRFL/>. What we decided on was a split battery system that used five M1 cells because it provided the best balance of power and space for the robot's design.

When it came time to implement the battery system, there were some new problems that had to be solved. In my case, I was splitting the batteries in the robot into two separate packs: an "upper" pack that was three cells and a "lower" pack that was two cells. The packs were wired together such that the lower pack powered the drive motors *and* the

that you see in the photo is a common side-effect of ramming as much power as possible into a limited space.

One major concern when using a split battery system is overloading of the lower pack. In my case, the cells needed to be able to source enough current for the weapon *and* the drive motors so that the robot would be able to drive and spin at the same time. Because the upper pack would not be loaded as heavily, the weapon might still function for a little while after the lower pack's voltage had dropped too low. This situation causes a

weapon, but the upper pack powered only the weapon. You can see the wiring diagram for the new system in Figure 2, as well as the actual implementation of it in the robot (Photo 2).

Though splitting the battery made it worse, the "wiring spaghetti"

problem because the lower cells could have their voltages irreparably reversed. Fortunately for me (as Figure 1 shows), the M1 cells have significantly higher capacity and peak current handling abilities than the original batteries, so those boundaries were not pushed during combat in the new version of Scurrie.

Another problem was that the lower pack was being shared by two electrical systems, and great care needed to be taken to separate them. One way to separate the systems was to use a weapon controller that had optoisolation. The idea behind that is the signals coming into the controller are used to power an LED that interfaces with a phototransistor. This allows the signal to be transferred *optically* while electrically *isolating* the controller. Optoisolators are common on high voltage electronic speed controllers, but the one that I chose to use did not have one. My choices were then limited to buying a separate optoisolator or add a

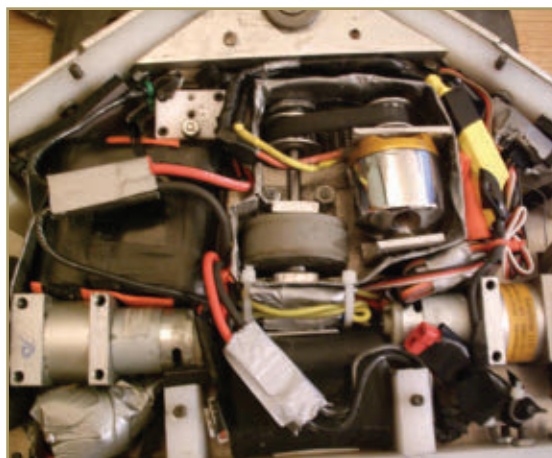


PHOTO 2. The actual wiring of the robot can get messy with so many wire ties being used.

Figure 1. Cell comparison.

	Sanyo 1,700 mAh NiMH	A123 Systems M1 Cell	Thunder 2,600 mAh LiPo
Capacity in Milliamp Hours	1,700	2,300	2,600
Continuous Discharge Rate	35A (20C)	70A (30C)	65A (25C)
Maximum Discharge Rate	50A (29C)	120A (52C)	130A (50C)
Nominal Voltage Per Cell	1.2 volts	3.3 volts	4.7 volts
Weight to Implement in the Robot	24 oz.	14.6 oz.	12.5 oz.
Estimated Cost to Implement in the Robot	\$93	\$95	\$145

Data is from the manufacturer for the A123 Systems cells and from www.robotmarketplace.com for the other two.

power switch that physically separates the power line for the weapon. I settled on the latter



PHOTO 3. The old batteries on the scale at exactly a pound and a half of total weight.

approach because switches stand up to abuse better than circuit boards and I already had an



PHOTO 4. The new batteries on the scale at just under a pound of total weight.

extra power switch.

It appears that a split battery system can be well worth the effort as long as great care is taken to avoid damaging any components. Scurrie ended up taking first place in the 12 pound class at the Franklin Cup in October 2008, and I think that the reliability that was made possible by the new batteries was the key to that success. I will be competing with this robot again in Harrisburg come February, and I was so satisfied with the new battery system that I plan to keep it exactly the same. **SV**

MANUFACTURING

Creative Armor: Sandwiching with Shock Mounts

● by Blake Hooper

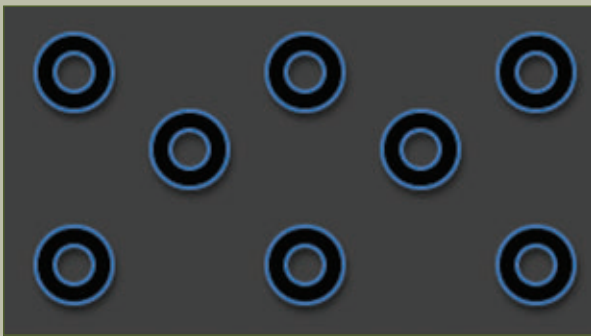


PHOTO 1. A diagonal pattern will limit the chances of your sandwich separating when you lose fasteners to damage.

When building a combat robot, one of the first problems you encounter is how to protect it. In most cases — despite creative weapon design or structural dynamics — if it cannot take a hit, you won't do well in competition. This becomes more problematic if you run into budget problems or weight issues. To this end, I will discuss the art of sandwich style and shock mounted armor.

Both sandwich style and shock

mounted armor are often confused as being one and the same. In reality, they actually complement each other.

Sandwich style armor consists of sandwiching multiple materials together to make one composite piece of armor. Shock mounted armor has less to do with the

plate you are using but rather how you are mounting your armor. The purpose of shock mounted armor is to redistribute the force from the impact in order to deal with the brunt of the blow.

Some of you might be thinking that these techniques seem silly when just using a thicker or stronger material could do the same thing with less complexity, but that is not always an option. If you are a builder on a budget, using thick

material across the whole of your robot can add up quickly, especially considering the price of the more important parts such as motors and electronics. Using one or both of these methods can greatly decrease the cost of protecting your bot. The other thing that discourages builders from using these methods is the fact that these styles of armor are more susceptible to being pulled apart. This is always a risk, but if you make sure to keep the edges of your plating protected, it won't be too much of an issue.

Let me go into the specifics of applying both methods and you can decide for yourself whether or not these styles suit you. Sandwiching armor requires you to more or less create a formula of material types. You need a core material, to offer central strength, and as the piece to take most of the damage. Following this, insulators must encase the core material, typically using some type of rubber sheeting. Once you have

selected these materials, mounting and exterior pieces must be selected. The exterior piece is the one that realistically you will have to replace most frequently, as it is the first thing to be hit by an opponent. The mounting piece needs to have secure contact at the mounting points or the entire assembly will rip off. The formula is a simple guideline to follow. How many of each type of layering you use or thickness is totally dependent on the situation your combat bot is in: its weight class, specialty, etc. In terms of the materials to use, there are several suggestions that work quite well.

The core material should be the most durable piece. Hard metals such as steel or high grade aluminum will suffice. The real dilemma in this case is cheap and thick versus thin and expensive. For insulating materials, flexibility is important. This leads to several simple options. Rubber is an excellent way to go as it is usually cheap and abundant. For mounting pieces, flexibility will still be important, but how you bracket the plating down will be the end determinant as far as dexterity is concerned.

In most situations, I would advise using thin steel as the main mounting piece. Bracketing the entire assembly together is not very complicated, since it is a matter of measuring precision. All you need to do is make sure that the holes are marked out in the same spot on each plate. After all the holes have been drilled, it is a simple matter of choosing a nut and bolt set that is the correct length and strength to tie the whole thing together. Grade 5 bolts are usually the type I use for high tolerance jobs like this; they are also a safe bet should your plating come near a spinner style opponent. With your sandwich plating assembled as one piece, it is now a matter of placing the whole thing properly on your bot's exterior. This is where shock mounting comes in.

Shock mounted armor has been done through a variety of both

simple and complex methods. The first and most common approach is essentially a more concentrated version of insulation. When bolting armor plates to their bot's frame, builders will use rubber washers over the majority of the bolt's length in order to add distance between the frame and impact, as well as distribute the force through the rubber washers. This is common and easy for those not sure about the shock mounting style. The downside to it, however, is that due to its simplicity, it is also easy to circumvent assuming the force is great enough from the impact.

Realistically, the correct method of shock mounting is more thought out than previously mentioned. The main advantage you should utilize the most is the flexibility it gives you. If done properly, an impact of significant magnitude can be distributed through the armor plating and mount itself before it gets to the actual frame of your bot. That is why you will want to design your shock mount well in advance. Some people use heavy steel springs or gas springs, but they tend to have poor reusability after competition. The real method of layering is similar to sandwich plating.

Layering for your shock mounts is far

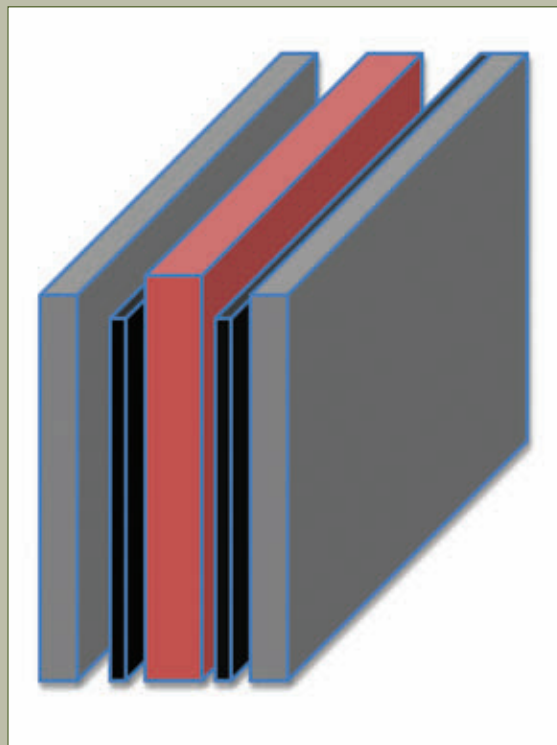


PHOTO 2. This is a side view diagram of a proper sandwich arrangement. The red represents the core material with the gray showing both the mounting and exterior plate. The interspersed black represents the insulating pieces.

simpler than layering in sandwich plating. The formula for layering these shock mounts is as follows. First, there is your core piece. Like your core material, it will act as the primary support of the mount. In many cases, this can be some form of bolt or tapped shaft. The core piece will have one end fixed to the



PHOTO 3. This is an example of grade 5 bolts with accompanying nuts.



PHOTO 4. The image on the left is an example of a bolt with a rubber insulator. When added with the spring image on the left, you get an idea of how a shock mount might look.

bot while the other end will be allowed to move inward from the impact. Once you have chosen your core piece, you then have a variety of options for the rest of your mount. Surrounding your core piece, flexibility is most important. Therefore, you will want to have three insulating layers around the core piece. The first and third layers will be on either end of the core and should be made of a rubber-like material in order to act as a shock

absorbent. Interspersed around these layers should be a larger metal washer used not only to separate the layers properly, but also as catching pieces for the material above and below. In the center, a spring-like material is required. The importance of this becomes apparent when you look at what the mount should do before and after impact. Before the impact, you want the mount to push the armor plating out as far as possible to use

distance to help displace the impact. After the blow, however, you want proper recoil of the shock mount, such as a spring-like material in the center of the layers around your core piece.

The formula mentioned above is a general guideline for putting together a sound shock mounting system; the actual specifics vary based on the situation of the builder.

However, there is one last thing to remember when trying to implement the armor setups discussed in this article: make sure to properly encase the edges. One of the most common flaws to all armor approaches is that while they will successfully take a hit, the rigging can be ripped off. Therefore, if you properly insulate the edges, any type of ripping impact will be taken by the insulation instead of your hard work flying all over the competition floor. **SV**

EVENT REPORT

Halloween

Robot Terror –

California Insect Bots

● by Dave Wiley

It's Saturday, October 25th at 9:59 AM, the last Saturday before Halloween. All is quiet at Gilroy Hobby, but not for long! As soon as the doors are unlocked at 10:00 AM, the evil known as the Halloween Robot Terror and Bot Costume contest will be unleashed on the world ... BWUHAHAHAH !!!! Well, maybe not so much evil but tons of fun, nonetheless.

This last meet marked the third



PHOTO 1. Deceptively festive costumes mask these bot's combative nature.

year for the Halloween Robot Terror and Bot Costume contest held in Gilroy, CA. The event goes from 10:00 AM till just before noon with eight one-pound ant weights, two three pound beetle weights, and two 150 gram flea weights participating. Competitors set up on their pit tables and weigh in, ready to do battle, just like many other insect class competitions. The big difference for this event is after

one round of fighting, there is a costume contest held for the robots themselves.

At high noon, the fighting started and the first weight class in the arena was the ant weights with Pooky (from Team ICE), and Wheely Wacko (from Team Science and Mayhem), matching up for the opening fight, with Pooky winning. After the ants finished their first round of brackets, the flea weights Atom Bomb (a spinning drum from Team Misfit), and Split (also from Team Misfit), fought for a best two out of three wins. The first bout was won by Atom Bomb and the second bout was won by Split, so the third bout had everyone on edge. Using most of the two minute time limit, Atom Bomb was able to kick Split off the fighting platform to take the flea weight championship.

After a short break, it was time for the beetle weights to fight for the best two out of three between Unknown Avenger (from Team ICE), and Arc Pounder (from Team Science and Mayhem). Unknown Avenger was the repeat beetle weight champion from the Gilroy Bot Gauntlets and the 2006 and 2007 Halloween Robot Terrors, and Unknown Avenger was more than ready to defend his championship by winning the first two bouts and using the third bout as a grudge match.

It was now time for the Bot Costume Contest.

The rules for the Costume Contest are simple:

- 1) The costume must go onto the fighting robot you are competing with.
- 2) The bot must be able to drive around the arena so the audience can see it.
- 3) There is no weight limit on the costume and bot combination.
- 4) No open flames.

5) The audience will vote for the best costume.

The overwhelming winner of the Bot Costume Contest was the ant weight Nibbler from Team Offbeat Robotics. Nibbler had a teddy bear over his weapon arm and when he activated the grinding disc (that was now inside the bear), Teddy started to wiggle and jiggle big time and everyone found it funny. Then, stuffing started flying out of the bear and everyone laughed harder. The teddy bear eventually fell off the weapon's arm and Nibbler took a victory lap around the arena. He accidentally drove out one of the push-outs which had everyone rolling on the floor.

The 2nd place winner was the ant weight Stumpy from Team DMV. Stumpy had a plastic skeleton sitting on him to look like it was driving Stumpy. Behind the skeleton was a hangman's gallows and from the hangman's noose was a sign that read "Vote For Me or Else." The skeleton's head turned left and right using a tiny servo; its red eyes lit up; and its arms turned handle bars when Stumpy moved left or right.

There was a tie for 3rd place between Capt. Crunch from Team Stupid Robotics, and Baron Underwear from Team Misfit. The ant weight Capt. Crunch was dressed as a pirate with a hat, stuffed parrot, and a plastic sword that was taped to its weapons.

The ant weight Baron Underwear had a two pound layer of clay over it to create a sea wave with the Greek sea god Poseidon with his trident, standing in the wave's trough.

The costume winners got their pick of monster/horror DVDs. After the Costume Contest, the ant weights got back



PHOTO 4. Pirate Capt. Crunch tied for third.



PHOTO 2. Costume contest winner Nibbler.



PHOTO 3. Second place winner Stumpy.

into battle trim and fought it out for the championship. The top four winners were:

- 1st Place - Baron Underwear, Team Misfit
- 2nd Place - Captain Crunch, Team Stupid Robotics
- 3rd Place - Rector, Team Road Rage
- 4th Place - Nibbler, Team Offbeat Robotics

The 1st, 2nd, and 3rd place winners from each weight class got their pick from a prize table. Afterwards, there were a number of ant weight no-time-limit rumbles.

There are pictures of the event and bot costumes at www.flickr.com/photos/anniecatblue/sets/72157608574338982/ or www.flickr.com/photos/21598904@N00/sets/72157608396730610. **SV**

All photos courtesy of Annie Blumenberg

EVENT REPORT 2

R_oaming R_obots UK

Winter T_our Final — 2008

● by John Frizell



Photo by Jessica Lale/Stuart Barnwell.

Pain in the Asp — a powerful six wheel drive feather — waits for its fight.

eight feathers, but many more teams came to see the finals and take part in the whiteboard fights. There were so many that they overflowed the pit area and extended it into the loading

The competition that began in Portsmouth and moved north to Nottingham finished back on the south coast in Maidstone in front of capacity audiences, with 1,200 people attending two shows. Twenty robots had made it to the finals: 12 heavyweights and

bay; late arrivals found themselves at tables in an unheated area with gusts of arctic air blowing in through the loading doors.

The random draw pitted Terrorhurtz — the most powerful of axe robots — against Thor, another axe. The one design

weakness of Terrorhurtz is that it is open at the top. As its builder commented, "Axes are rare. We'll be fine unless we meet another axe."

After a little maneuvering, the two machines squared up to each other and started pounding. Terrorhurtz was the more powerful and began to buckle Thor's armor, but Thor stood firm, despite the battering it was taking, and managed to get its axe into the open top of Terrorhurtz to the cheers of the crowd and applause from the roboteers in the pits. But Terrorhurtz kept going. Well into the fight, Thor lost power and Terrorhurtz stopped battering it and pushed it into the pit. The massive axe blows had bucked the armor enough to snap a battery wire, cutting power.

Terrorhurtz was through to the semi-finals where they faced Hydra — the only other axe machine in the competition. Hydra had blown their pneumatic system in an earlier fight, however, leaving them without a weapon so they were easily dispatched.

The featherweight final was an eight-way melee with the action swirling all around the arena. A couple of machines went out quickly but most bat-



Photo by Elizabeth van Son.

The crowd waits while Megamouse is made safe so the arena can be cleared for the next fight.

Photo by Jessica Lale/Stuart Barnwell.



Photo by Jessica Lale/Stuart Barnwell.



John Findlay — builder and driver of Ripper, the heavyweight champion.

The finalists: Ripper and Terrorhurtz.

tled on. Pain in the Asp — with its two 1 HP motors and gyro — sliced across the arena, scattering opponents, until it got stuck on the side. Aftershock, a full pressure flipper driven by 18 volt drill motors kept flipping its opponents, coming back to invert them again when they self righted and, with five machines still running at the end, became the featherweight champion by a judges decision.

The last heavyweight match saw Terrorhurtz take on Ripper, a flipper that had lost its first competition fight but won the losers melee, gained a place in the quarter finals and fought its way through to the finals. After an even battle between the two CO₂ powered weapons, with Terrorhurtz self righting after being flipped and landing numerous hammer blows, loosening the flipper's armour, Ripper flipped Terrorhurtz and propped it against a wall. With Terrorhurtz unable to self right, Ripper became the heavyweight champion of the Winter Tour. **SV**

Photo by Team Aftershock.



Aftershock — the featherweight champion — in a photo taken before the event.

ROBOT PROFILE

TOP RANKED ROBOT THIS MONTH

● by Kevin Berry

BotRank.com

Ranking Your World.

Top Ranked Combat Bots

History Score			Ranking		
Weight Class	Bot	Win/Loss	Weight Class	Bot	Win/Loss
150 grams	VD	26/7	150 grams	Micro Drive	7/2
1 pound	Dark Pounder	44/5	1 pound	Dark Pounder	13/1
1 kg	Roadbug	27/10	1 kg	Roadbug	7/4
3 pounds	3pd	48/21	3 pounds	Yeti	8/0
6 pounds	G.I.R.	17/2	6 pounds	G.I.R.	8/0
12 pounds	Solaris	42/12	12 pounds	Tourinho	13/1
15 pounds	Humdinger 2	29/2	15 pounds	Humdinger 2	29/2
30 pounds	Helios	31/8	30 pounds	Pyromancer	6/0
30 (sport)	Bounty Hunter	9/1	30 (sport)	Upheaval	8/4
60 pounds	Wedge of Doom	43/5	60 pounds	K2	14/2
120 pounds	Devil's Plunger	53/15	120 pounds	Touro	23/3
220 pounds	Biohazard	35/5	220 pounds	Original Sin	15/6
340 pounds	SHOVELHEAD	39/15	340 pounds	Ziggy	8/2
390 pounds	MidEvil	28/9	390 pounds	MidEvil	3/0

History Score is calculated by performance at all events known to BotRank
Current Ranking is calculated by performance at all known events, using data from the last 18 months

Data as of December 5, 2008

Tourinho has competed at RoboGames 2007, Winter Challenge 3 ENECA - Recife, 7 ENECA - Recife, and Winter Challenge 4 ENECA - Recife.

- Frame: 11" wide x 9.15" long aluminum chassis, 6351-T6 2" x 1/8" C-channel extrusion walls, and 2 mm 2024-T3 top and bottom.
- Drive: Two wheels powered by 16:1 36 mm Banebots gearboxes with modified last stage pins (based on the article by Nick Martin in *SERVO's* March 2008 edition), and Integrity Matrix Pro Lathe motors.
- Wheels: 3" x 7/8" Colson Performa.
- Drive ESC: Banebots BB 12-45.

- Drive batteries: Two Hextronik Lithium polymer batteries, each with 11.1V and 2,200 mAh, arranged in series to produce up to 25.2V when fully charged.
- Weapon type: 2" diameter 6351-T6 aluminum drum, with four hardened steel M12 Allen flat head bolts as teeth, spinning at more than 11,000 RPM, supported by a 1/2" 6Al-4V titanium shaft.
- Weapon power: 1.5 HP, with more than 1,000J of kinetic energy stored in the drum.
- Weapon motor: DeWalt old style, overvolted.

Tourinho – Currently Ranked #1



Historical Ranking: #11
Bot Name: Tourinho

Team Name: RioBotz
Team Location: Rio de Janeiro - Brazil
Builder(s) Name(s): Marco Meggiolaro, Eduardo Ristow, Rodrigo Estrada, Daniel Zacarias, and 11 other PUC-Rio students.
Overall Configuration: Invertible Drumbot with Two Wheel Drive

BotRank Data	Total Fights	Wins	Losses
Lifetime History	14	3	1
Events	4		

- Weapon controller: Victor HV-36, borrowed from Touro.
- Armor: The walls near the weapon shaft supports are reinforced with 3/4" thick 7050 aluminum, Kevlar, and 1/8" thick 6-4 titanium.
- Radio system: Spektrum DX6i paired with BR6000 receiver.
- Future plans: Tweak the design a little bit more and get it in good shape again for the next competition.

- Design philosophy: Tourinho was built in a few weeks since we only had less than a month between RoboGames 2006 and the hobbyweight class debut in Brazilian competitions. We aimed to scale down Touro's best features, use readily available materials and some scrap. Along the last two years, several improvements were made in order to keep it up with the competition, such as increasing the drum diameter, overvolting the weapon motor, and reinforcing the armor.

- Builder's bragging opportunity: Tourinho won all the Brazilian hobbyweight competitions since its debut in 2006. Paired up with his fellow hobbyweight wedge Puminha, Tourinho won by knock-out a fight against the

featherweight Hulk, after disabling its ICE-powered vertical spinning bar. **SV**

Tourinho vs. Agressor — Robocore Winter Challenge 2006 finals.

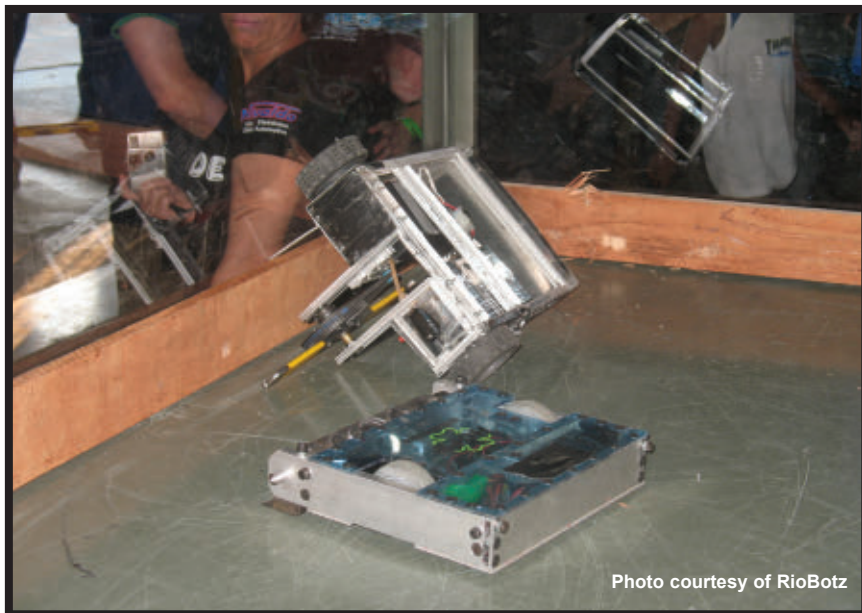


Photo courtesy of RioBotz

PARTS IS PARTS: Screwdrivers: Straight up or with a twist

● by Nick Martin

This month, we are reviewing two cordless screwdrivers from Bosch. While they share many similar features, they each have special strengths and uses.

PS20/GWR Screwdriver: Street price around \$130

This is the conventionally shaped driver of the pair. It has all the usual features of a small cordless driver and adds a few features of its own. The 10.8V pack uses lithium-ion cells, which outlast NiMh cells and appear to deliver more current too — I was

Both screwdrivers.





The PS10-2 adjustable head.

surprised at the torque from such a small tool. The small battery means the grip is quite short and slim, and will fit smaller hands well.

Just in front of the trigger is a LED light to help illuminate dark corners. It is not so useful while working on robots, but is a real bonus when you are inevitably asked to fix things around the house. If you are careful, the light can be activated before the motor starts; very handy for navigating!

The motor is a 550 sized item with an above average build quality. I was hoping for a motor more like a small DeWalt in a driver of this price, however, after a solid year of use, the motor is still going strong and if it does ever die, the replacement cost will be low. The gearbox is a typical two-stage affair with a 10 position clutch. I recommend staying away from the Lock or Drill position as it will easily strip the head out of most screws.

Rather than the usual drill



The charger and batteries.

chuck, both these drivers have a 1/4" hex socket that accepts 'power' bits. This makes the tool shorter and the bits are very quick and easy to exchange. One downside is that there is more wobble to the bits than you get with a traditional chuck, which can be annoying with long driver bits. You can easily reduce the looseness of the bits in the chuck by packing a small piece of foam into the chuck. This tool is meant for driving screws. However, you can also use it as a drill with a chuck-to-hex adaptor or special drills with a hex shank. These drills are available in sizes up to 1/4" and can be a real lifesaver in tight spaces! The 600 RPM speed makes drilling slow, but if you need the hole drilled in an awkward place, it is worth the extra time.

Bosch PS10-2/GWI Screwdriver: Street price around \$130

This screwdriver is more of a

specialty tool than the PS20 driver. The angled head is what makes this tool worthwhile; other tools have angled heads, but few are as versatile or as well made.

It comes with a soft nylon cloth carrying case; all the parts fit neatly and the case is quite protective for something made of cloth. As well as the driver, you get the same two batteries and charger as the PS20 tool.

The design of this tool results in a large grip that will not suit small hands. The trigger is quite large to make up for this and I personally don't have a problem with it. The direction control switch is fairly easy to get at and its only problem is that it will not lock in the center off position while the driver is in your toolbox.

As you can see from the photos, this tool has a rotating head that allows it to function in five positions from straight to 90° to reach the tightest spaces. In the 90° mode, you can fit in to spaces as small as 4.5" wide. The locking button is strong and easy to operate, and the chuck is the same as the PS20.

The pivoting head means that the usual mechanical torque clutch is impossible and is replaced with an electronic equivalent. This clutch mimics the mechanical one but has less start-up torque and takes some getting used to. I have found that the action will not free up tight screws as well as the PS20 model; on the other hand, it is more likely to protect the tool from abuse. The electronics provide a speed ramp-up when started that helps avoid stripped screw heads.

The angled head is what makes this tool worthwhile; other tools have angled heads, but few are as versatile or as well made.

Batteries and Charger

For your money, you get two



The PS10-2 electronic clutch.

Li-Ion packs, giving a long battery life and a quick 30 minute re-charge time. I have never been able to run a battery down before the second pack is recharged, so you will always have power ready and waiting.

The charger is a quality item with indicator lights for power and charge state. It has temperature and charge monitoring to keep the battery packs safe. The wide base ensures it will not tip over on a busy pit table and the packs are easy to insert and retrieve; it's about as good as a charger gets.

Rating

As usual, you get what you pay for. In this case, it is construction quality, unique features, and advanced batteries. Both screwdrivers work well, and look great, and the straight GWI model doubles as a light sabre handle.

The PS20/GWR model has taken a year of abuse; I can recommend it as your main power screwdriver at events and for most workshop tasks. Since buying this tool, it has become my main driver tool and other cordless tools have

either been retired or relegated to drilling.

The straight PWI model has less torque than the GWR model and poor low speed control. It is more of a specialty tool for very tight spaces and a 'last resort' tool for drilling small holes in impossible places. I do not use it as often as the PS20, however, it still gets a workout on lighter tasks and any place the PS20 will not fit. If you happen to work on computer repairs, this driver is an ideal cross-over tool and will repay itself in professional use. **SV**

EVENTS

Completed Events November 10 - December 8, 2008

Pennbots held an event in Boiling Springs, PA, on November 15th.



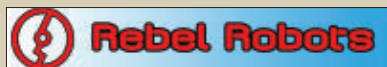
Roaming Robots held an event on November 22nd in Maidstone, England.



Robots Live held an event on November 15th and 16th in Chester, and November 22nd and 23rd in Birmingham, England.



Upcoming Events February - March 2009



by Kevin Berry

Rebel Robots 2009 Featherweight Championship will be held February 7th in Rotherham, England; www.visitmagna.co.uk.

Robots Live will hold an event in Leicester, England on February 7th and 8th, and in Burgess Hill on March 28th and 29th; www.robotslive.co.uk.

Barnsley Bash 2009 will be held in Barnsley, England on November 14th and 15th; www.roamingrobots.co.uk.

Motorama 2009 will be presented by the North East Robotics Club in Harrisburg, PA, on February 20th; www.nerc.us.



MegaCon RobotBattles will be held in Orlando, FL on March 1st; www.robotbattles.com.



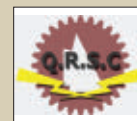
Mad Metal Machines 7 will be held in Bochum, Germany March 6th through 8th; www.dutchrobotgames.nl.



Central Illinois Bot Brawl 2009 will be held in Peoria, IL on March 7th; <http://circ.mtco.com>.



Ideas Festival 2009 will be held in Brisbane, Australia, March 27th through 30th; www.robowars.org. **SV**



BUILD THE ULTIMATE ROBOT

Part 5: Megabot

by Michael Simpson

If you are building the Megabot or a similar robot, you should have the base assembled at this point. In this article, we are going to add the electronics that will make our robot move. I'll show you the techniques I used so that you can apply them to your own design.

PROGRAM THE ACTUATOR IDS

The first thing we need to do is to connect our RX-64 actuators; before we do this, we need to assign the correct IDs to each actuator. You will use the Dynamixel Configurator software to change the default ID to the ID of your choice.

To do this, you need to do the following:

- Install the USB2Dynamixel drivers.
- Install the Dynamixel Configurator software.
- Connect the USB2Dynamixel to your PC.
- Set the selector switch on the USB2Dynamixel to RS-485.

FIGURE 1. Megabot base.



The USB2Dynamixel does not supply power to the actuator so you will need to apply the correct voltage to the power leads on the cable. You can do this by splitting some power leads from the connector that attaches the USB2Dynamixel to the actuator. You can also insert power by creating a power cable that plugs into the other connector on the actuator. The RX-64 manual goes into detail on making this connector. Since I am

only programming the ID into the actuators, I made a temporary setup using a breadboard like the one shown in Figure 2. Notice how I have marked the positive and negative power connections on the connector. For more details on the power leads, you can refer to Figure 4. If you have already mounted the RX-64 actuators, simply plug the connector into the actuator you want to program. Just make sure only a single actuator is connected to the USB2Dynamixel when you program the new ID.

FIGURE 2



FIGURE 3



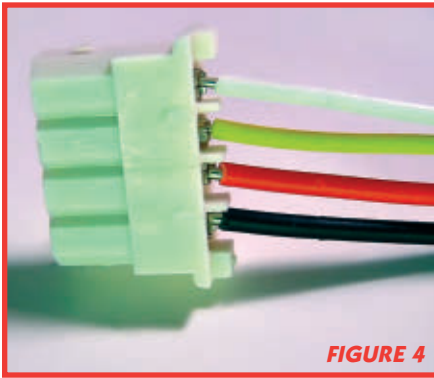


FIGURE 4

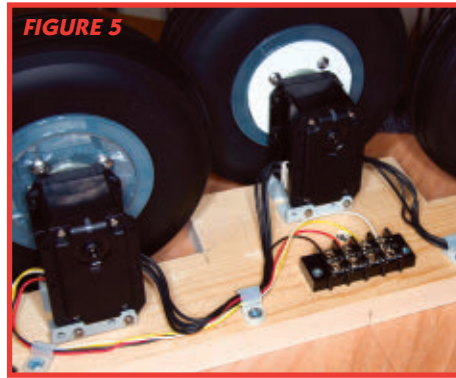


FIGURE 5



FIGURE 6

Program the IDs as shown in Figure 13. If you decide to use different IDs, you will have to make modifications to the various robot routines in order for them to work properly.

CONNECT THE ACTUATORS

Attach a four-position barrier strip in front of the center actuator as shown in Figure 3. I used two #4 wood screws. Later, we will use this to connect power to the wheel assemblies. Next, take two RX-64 connectors and connect the two end actuators to the center actuator, also shown in Figure 3. Notice how I used a cable clamp to hold the cables close to the base. This is important as you don't want the wires to snag on something and get ripped out.

To connect the actuators on this wheel assembly to the barrier strip, you need to take one of the female connectors included with the RX-64 actuator and attach four 12" wires. I recommend using the colors shown in Figure 4. I also recommend 22 gauge or larger wire. In Figure 4, the rounded side of the connector is the one facing you. I used a tool that I purchased at RadioShack to crimp the wires. You can use a pair of needle nose pliers, as well. You will need to create two of these connectors; one for each wheel assembly.

Plug the connector you just created into the wheel assembly. Plug it into actuator 3 on one assembly and 6 on the other. Add a new cable clamp and connect the wires to the barrier strip as shown in Figure 5. Use the order of black, red, yellow, and white.

INSTALL THE BATTERY

When I first started this project, the power source was the most difficult obstacle I had to overcome. In the course of this project, I have come up with two options. Both involve the use of Li-Ion batteries.



FIGURE 8



FIGURE 9

Last month, I showed you how to use the Ryobi 18V battery shown in Figure 6 to power the Firebot. You can use one for this project, as well. You can even use two — one for each wheel assembly.

To show you an alternate configuration, I decided to use a battery from a company called **atbatt.com**. It's the Amstron 133 watt universal battery shown in Figure 7. This battery has a whopping 9,000 mAh of power. Its output can be set to 16 or 19 volts and it comes with its own charger. Its low profile and weight makes it perfect for robotic applications. You can even use it to power your laptop for other projects. The battery will cost you about the same as two of the Ryobi batteries.

Place the battery on the underside of the robot between the two wheel assemblies as shown in Figure 8.



FIGURE 10



FIGURE 11

WIRE THE BATTERY

The battery comes with an output power connector and you are free to use this one. Cut off about 3" of cable and attach it to a two-position barrier strip as shown in Figure 9. Since I own several laptops, I

Place the connectors towards the front of the robot (actuators 1 and 4); make sure you leave enough room to mount a barrier strip and USB2Dynamixel controller as shown in Figure 13. There are many ways to hold the battery in place. I used some right angle connectors I had on hand, but some hook and loop straps would work just as well (choose the heavy duty kind available at most home centers). Attach the straps to the base on either side of the battery with wood and finishing washers.

wanted to keep the output cable intact, so I cut the plug off of an old AC adapter for my installation. As a last resort, you can take the battery down to your local RadioShack and they will help you find a connector.

Attach the barrier strip to the base with #4 wood screws as shown in Figure 9.

Now, take two pieces of red 18 gauge wire (12" each) and connect them together with solder. Attach the soldered end to the positive terminal on the barrier strip. Connect each of the other ends to the positive lead on the four-position barrier strip that is on each wheel assembly as shown in Figure 10. Repeat this using the two black wires. It's important that you use at least 18 gauge wires or you may not supply enough power for your actuators. Notice how I have attached everything close to the base with cable clamps.

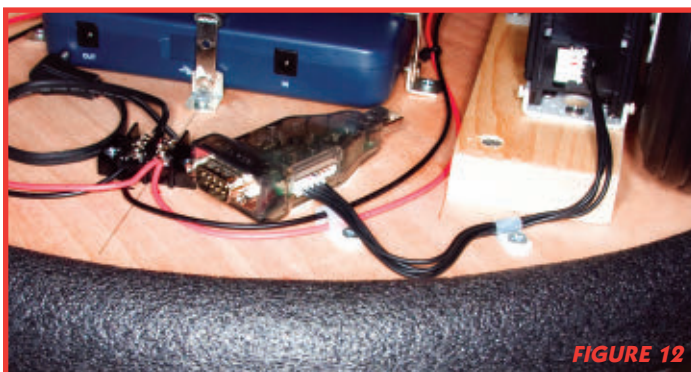


FIGURE 12

CONNECT THE DATA LEADS

Run a yellow and white wire from the barrier strips on the wheel assemblies to each other as shown in Figure 11. Make sure you route the leads through the existing cable clamps and add a couple near the rear of the battery.

MOUNT THE USB2DYNAMIXEL

Add a couple pieces of double-stick foam tape to the bottom (flat side) of the USB2Dynamixel. Make sure the USB2Dynamixel is close enough to connect to actuator 4 as shown in Figure 12. Also make sure you have room to connect a USB extension cable to the controller. Try dry fitting the controller before attaching it.

The bottom of the base is now complete. It should be laid out like Figure 13. I have included the IDs for each of the actuators.

Before we move to the top of the base, I want to say something about using two Ryobi batteries. First, I recommend mounting the batteries on the underside of the base. However, this can be problematic if the Megabot will be used on rough and rocky terrain. Connect each battery to the power leads on each wheel assembly. When connecting the data leads from wheel assembly to wheel assembly, you will also need to connect the negative terminal, as well.

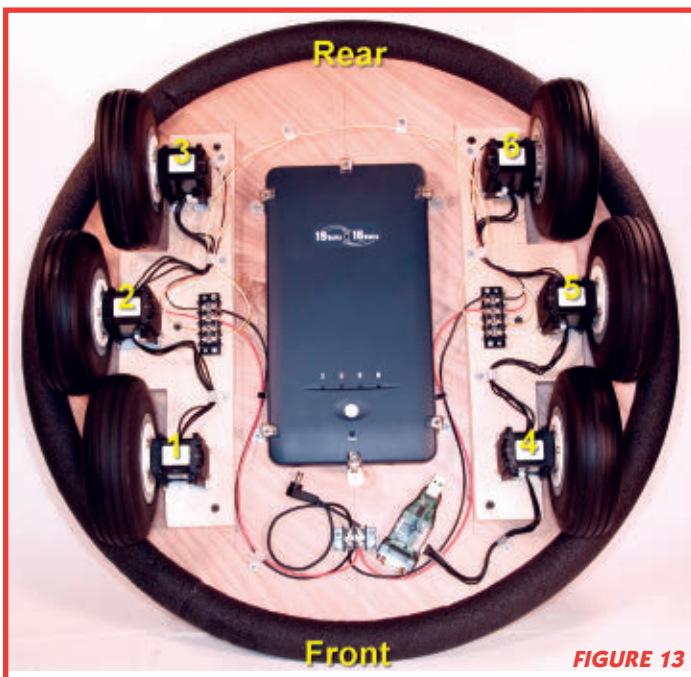


FIGURE 13



FIGURE 14

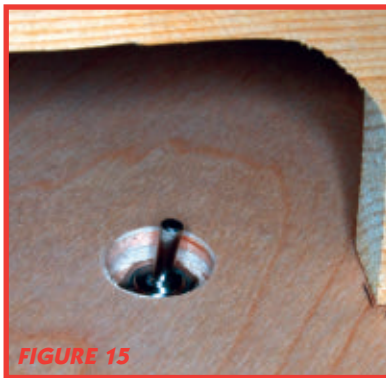


FIGURE 15



FIGURE 16

WIRE THE SWITCH (OPTIONAL)

While removing the plug from the battery works pretty well, you may want to add a switch. Just wire the switch in series with the positive lead on your plug as shown in Figure 14.

The shaft on the switch I used was not long enough to protrude through the 1/2" stock I used for the base. To solve this problem, I used a larger drill bit and created a counter-sink on the top of the base as shown in Figure 15. There are other ways you can mount the switch. If you decide to do something different, I recommend mounting the switch first. Once mounted, you can run your wires for the best fit.

BUILD A STAND

The actuators used on the Megabot are extremely powerful. If the Megabot decides to make a move unexpectedly while you are working on the program, something is going to get damaged. A stand is a must. I made the stand shown in Figure 16 out of some scrap wood. It is simply two pieces of 3" x 12" wood connected together with wood screws. It is designed so that the four outside actuators sit on top of these two pieces. I made some small notches so that the actuators sit firmly in place as shown in Figure 17. You can make your stand out of any material you have on hand. You can even use foam board or heavy cardboard. The goal of the stand is to keep the wheels off the ground. If you decide to proceed without the use of a stand, don't say I didn't warn you.

THE CRADLE

I have seen many robots — both commercial and

do-it-yourself — that lack one main component: When using a PC of any kind that has a hard drive, you have to create some sort of cradle like the one shown in Figure 18.

The cradle keeps every single bump and bounce from transferring to the laptop's internal hard drive. If you don't build a cradle, be prepared to replace the hard drive the first time you run the bot into a wall or go over a bump.

The cradle is just four pieces of wood connected together with wood screws. The cradle is large enough to give you about 1" clearance to the laptop on all sides. To actually cradle the laptop, we connect some elastic straps with some wood screws and washers.

The elastic can be purchased at most department stores that have a craft or sewing center. For the laptop I used, only four straps were needed as shown in Figure 19. The key is to support the laptop without making it too firm. If you find you need more straps, you can use thinner elastic. If the laptop wants to bottom out when it's bounced, you should add a couple more straps.

Notice that I have cut out a portion of the cradle with a jig saw. This is done to reduce weight and provide ventilation for the computer.

THE LAPTOP

Let's talk a little bit about the brain for Megabot. For most of my medium and large robots, I tend to use CE (consumer electronics) devices. The reason is that CE devices are small and are fairly rugged. They are also reasonably priced. In many cases, they can be purchased for less than \$400 new and less than \$200 used. Well, the times are changing my friends.

There is a new influx of laptops hitting the scene.



FIGURE 17



FIGURE 18

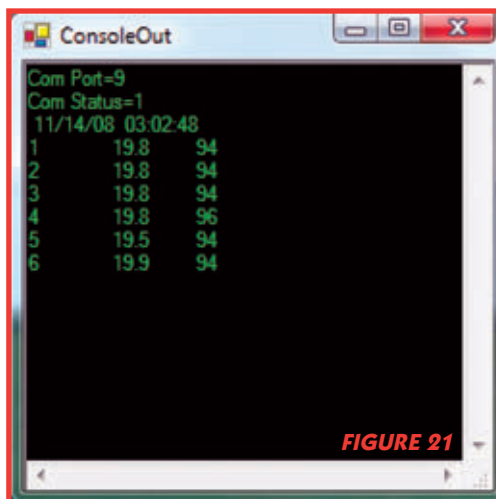


FIGURE 19



Sometimes referred to as NetBooks or ultra-portable notebooks, they are small, inexpensive, and very powerful. The device I have chosen for the Megabot brain is the **Acer Aspire One** shown in Figure 20. While the Aspire One comes in many flavors, the one I chose has the following features:

- 1.63 GHz Intel Atom processor
- 1 GB of RAM
- 120 GB hard drive
- 8.9" display
- Three USB ports
- External monitor hookup
- Built-in Wi-Fi
- Only 2.4 lb. with battery
- Windows XP home installed
- Built-in webcam



Now for the best part — it only cost me \$349. A friend just told me you can pick one up at Wal-Mart for \$339. There are other manufacturers that have similar offerings.

Using a PC over a CE device is a no-brainer for this application. It's four times faster than most CE devices and offers USB ports capable of utilizing the USB2Dynamixel

interface. The USB2Dynamixel interface allows us to communicate with both RX-64 and AX-12 actuators.

There are versions of the Acer One laptop that come with a solid-state drive that make it more suitable for robotics, but most of these come with the Linux operating system which will not work with this project.

PROGRAMMING THE BRAIN

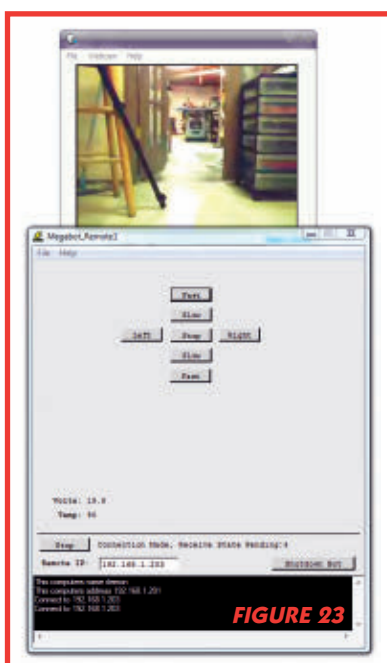
As with the Firebot in the last article, I will be using Zeus as the primary programming language for Megabot. Zeus is a very inexpensive Windows development environment for creating both Windows PC and CE applications. Let's start with a simple program that will let you test your RX-64 actuators. The program is called **Megabot_Actuator1_DT.exe**. (All programs are available at www.kronosrobotics.com.)

When you run this program on the PC that is connected to the USB2Dynamixel, it will display the output screen shown in Figure 21.

If the program is unable to communicate with the com port connected to the USB2Dynamixel interface, the Status will return a value of 0. This can happen if the software is using the wrong com port. To correct this, create a file named Port and place the number of the com port in that file.

If the individual actuators return a value of -1 and 30, this indicates that the USB2Dynamixel interface is not communicating with the actuator. This can happen if the actuators are not powered, improperly wired, or have the wrong ID.

Once you get good readings, load up the program called **Megabot_Actuator2_DT.exe**. This program will cause the wheels to move in one direction, then reverse. If you don't see the wheels



moving, go back and check your power connections.

Now it's time for some fun stuff. Normally, I like to create radio interfaces to my bots as one of the first projects. However, since I did this with the Firebot, I decided to do something a little different this time. Since the Aspire One has built-in Wi-Fi, I decided to write a little client/server program that lets me control the Megbot remotely from my desktop PC.

Load the program called **Megabot_Wifibot_DT.exe** on the laptop used in the Megabot. This is the server program shown in Figure 22. Its job is to listen on the network and wait for commands from the client program. When a command is received, it takes an action. Notice how the server program displays the IP address at the top of the form. In this case, it's 192.168.1.201. Keep this in mind, as we will need it later.

The Aspire One has a webcam. It is possible to utilize this webcam when controlling your robot from a remote location. There are a number of ways to do this. The cheapest and simplest way is to use one of the

PARTS LIST

The following are the components needed to build this portion of the project:

Jameco — www.jameco.com

- Toggle Switch — part number #76241
- Four-Pos Barrier Strip — part number #230990
- 1/4" Cable Clamps — part number #173737

CrustCrawler

- RX-64 — www.crustcrawler.com/motors/RX64/index.php?prod=67
- Treaded Wheels — www.crustcrawler.com/products/rover/wheels.php?prod=28
- Dynamixel Configurator www.crustcrawler.com/electronics/USB2Dynamixel/software/Dynamixel_Configurator

/DXCONFINST1.2.1.0.exe

Kronos Robotics

- ZeusPro Development Environment — www.krmicros.com/Development/ZeusPro/ZeusPro.htm

RadioShack

- Z4-Position Barrier Strip #274-658
- Two-Position Barrier Strip #274-656

AtBatt.COM

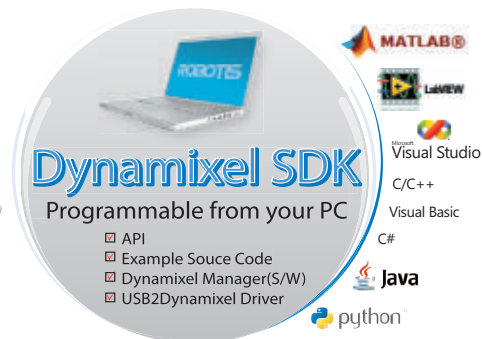
- P133 External Laptop Battery. www.atbatt.com/product/7901.asp

Messenger Software

- www.skype.com
- <http://messenger.yahoo.com/>

Why DYNAMIXEL?

High-performance networked actuators for robots



AX-12+



- Engineering plastic gear
- Control network: TTL
- Gear Ratio : 254
- Position Sensor : POT

AX-12+	at 7V	at 9.6V
Holding torque(Kgf.cm)	12.5	17.1
Speed(sec/60deg)	0.269	0.196
Weight(g)	53.5	

RX-28 / DX-117



- Full metallic gear
- Control network: RS-485
- Gear Ratio : 193
- Position Sensor : POT

	RX-28		DX-117	
	at 12V	at 16V	at 12V	at 16V
Holding torque(Kgf.cm)	28.3	37.7	28.3	37.7
Speed(sec/60deg)	0.167	0.126	0.167	0.126
Weight(g)	72		67	

RX-64



- Full metallic gear
- Control network: RS-485
- Gear Ratio : 200
- Position Sensor : POT

RX-64	at 15V	at 18V
Holding torque(Kgf.cm)	64.4	77.2
Speed(sec/60deg)	0.188	0.157
Weight(g)	125	

EX-106



NEW

- Full metallic gear
- Control network: RS-485
- Gear Ratio : 164
- Position Sensor : Encoder

EX-106	at 14.8V	at 18.5V
Holding torque(Kgf.cm)	84	106
Speed(sec/60deg)	0.182	0.143
Weight(g)	155	

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peer-to-peer programs like *Skype* or *Yahoo! Messenger*. All you need to do is create an account for your desktop machine and one for your laptop. Before proceeding, you may want to set up these accounts and get the display working on your laptop. It's easy and best of all, it's free.

Now load up the program called **Megabot_Remote1_DT.exe**. This program is the client and it will connect to the server. It's best to overlap the Megabot_Remote1 form on the remote webcam display like I did in Figure 23.

Before doing anything fancy, I really recommend that you test everything out using a stand. Also be advised that the lid on the laptop needs to be up when using the built-in webcam. This means you may need to add a couple more elastic straps to balance things.

Step 1

Take the IP address shown on the top of the server (bot) form and place it in the Remote IP: field on the client (remote1) form.

Step 2

Click the **Start** button at the bottom of the form. This will connect the client to the server. Once connected, both forms will enter state 4; this is Receive State Pending.

In this state, you can issue commands.

Step 3

Hit the buttons to issue the various commands. Try each to make sure they are all working. Once you are satisfied, you can place the Megabot on the floor without the stand. Give yourself plenty of room and start off simple.

WHAT'S NEXT

Originally, I was going to add an interface to the GPS Data Logger I did in a previous article. I wanted to try and experiment to determine if a GPS unit with a WAAS enabled receiver is accurate enough to keep a robot on a 5' driveway that twists and turns down the front of my property. However, since I have been experimenting with the Wi-Fi remote I want to take it one step further and add a robot arm and a couple of sensors to help automate the remote control system.

Next month, we are going to add an upper base and attach a robotic arm. I will also show you how to connect some AX-S1 sensors to the system.

Be sure to check out the Kronos Robotics website for updates to this project at www.kronosrobotics.com/Projects/megabot.shtml. **SV**

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AN INTRODUCTION TO PNEUMATIC SYSTEMS

Part 2: Adding Electronic and Computer Control to a Pneumatic Circuit

by David A. Ward

Now that the basic pneumatic system is operating manually, let's add some electronics to control it. If we add an electromechanical relay and two limit switches, we can make the piston travel back and forth continuously; check out the electrical schematic in Figure 1. The limit switches being used are momentary SPDT miniature switches (Photo 1) and the electromechanical relay is a DPDT 24 VDC coil relay with a socket (see Photo 2). By adding electrical limit switches at the fully retracted position, the circuit will turn the control valve and relay on in this position and keep it on with the electromechanical relay contacts until the cylinder rod fully extends.

Photo 3 shows a normally open contact. Photo 4 shows it in the fully extended position. When the extended limit switch is hit — which de-energizes the valve solenoid, as well as the electromechanical relay coil — the piston and rod will go back to the retracted position and the cycle will start

over again. The speed of the cylinder travel can be controlled with the flow control valves.

As a final control demonstration, we'll show how the pneumatic system can be controlled by a PLC (programmable logic controller) or industrial computer. I realize that most people's access to PLCs outside of industrial settings is pretty limited, however, there is a two-part series of articles entitled *Getting Started With PLCs*, published in the

PHOTO 1

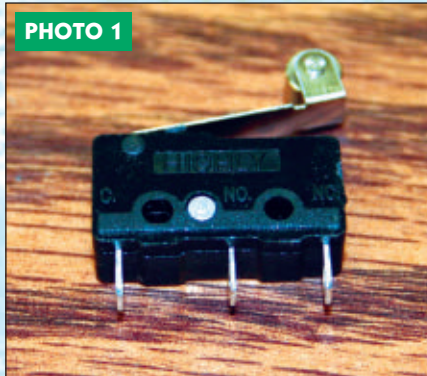


PHOTO 2

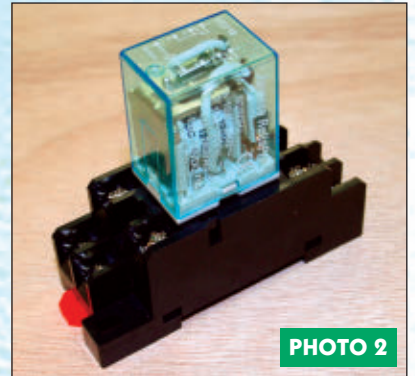
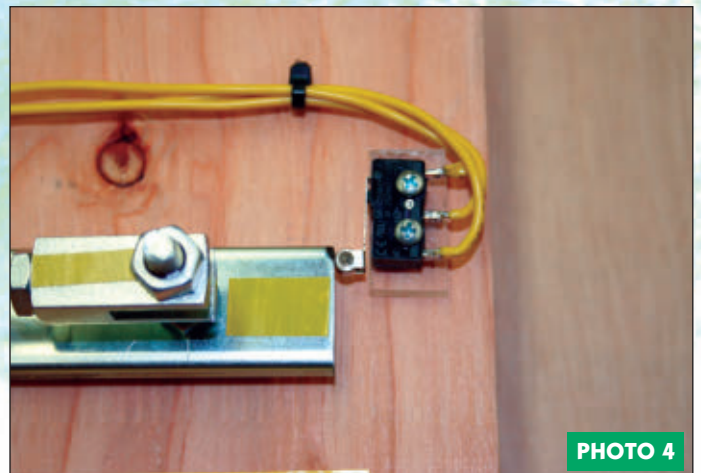
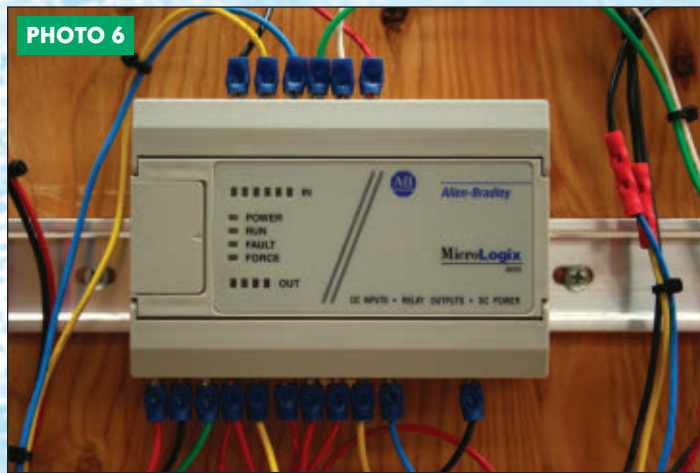
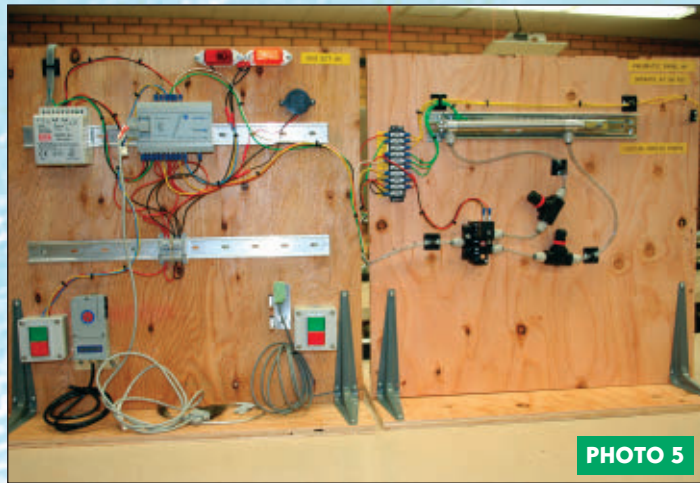


PHOTO 3



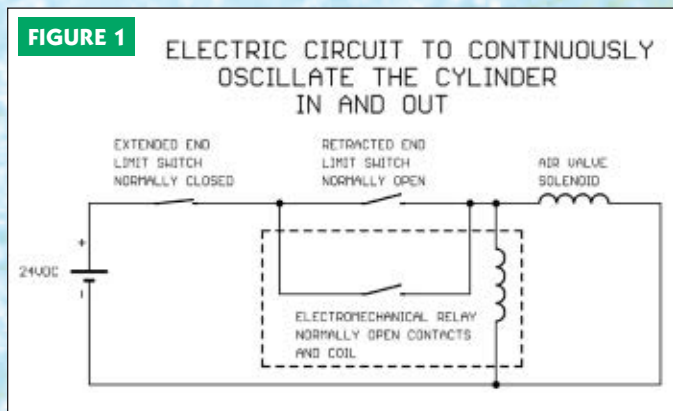
PHOTO 4





February and March 2005 issues of *Nuts & Volts* that can get you started.

Figure 2 shows the connections necessary from the pneumatic circuit to the PLC. Photo 5 shows the PLC connected to the pneumatic board. Figure 3 shows the complete ladder logic program that is being run in the PLC to control the pneumatic circuit. This program was written in RSLogix Starter for 10 point Micrologix 1000 software, and can be downloaded from <http://ab.com>. The PLC used in this demonstration is from Allen-Bradley (see Photo 6).

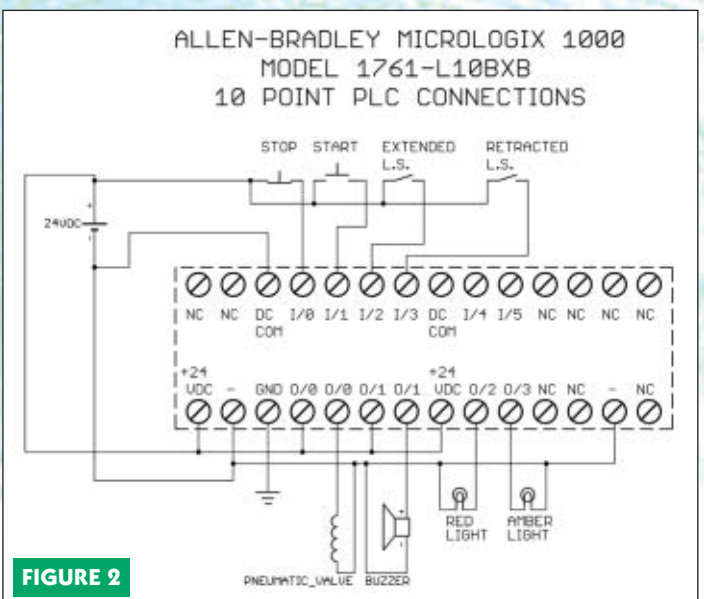


The PLC Program

The ladder logic program has several features in it that I'll use to illustrate industrial types of situations that may be encountered, in an industrial setting. The system operates as follows: First, the operator must press and hold the green start button for three seconds to get the system started. While holding the start button in, a buzzer sounds and an amber light comes on. The purpose of this is to warn people that the system is about to move. If the start button is released before the three second delay has completed, the timer will reset and the button must be held in for a full three seconds again. After this warning delay, the PLC piston output will be on as well as a red light; the piston will extend until it hits the extended limit switch. At this point, the output from the PLC to the cylinder will turn off and the cylinder will return to its retracted position. When the retracting piston hits the retracted limit switch, the PLC piston output will be turned on and continue the cycle over again. However, there is a counter keeping track of every time the extended limit switch is hit (rung #5 in the program). When the counter's accumulated value equals the preset value (currently 5), the system stops, resets the counter, and goes back to its original position waiting for an operator to hold in the start button again. Pressing the red stop button at any time will turn off all PLC outputs, reset the counter, and put the PLC back into its original state waiting for a three second start button press.

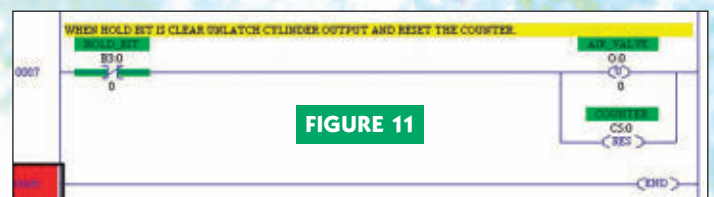
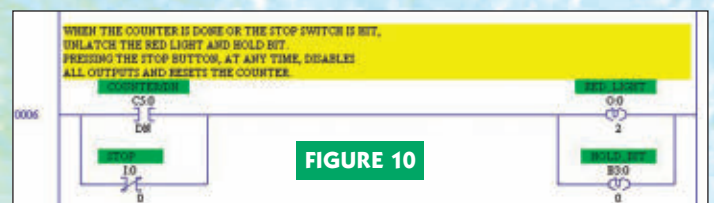
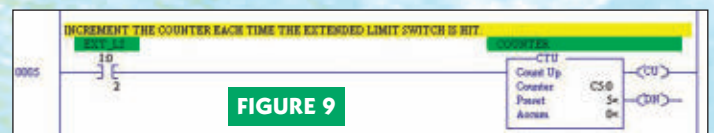
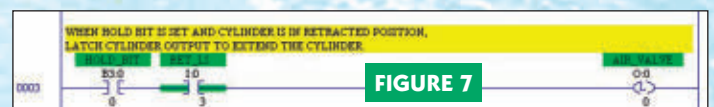
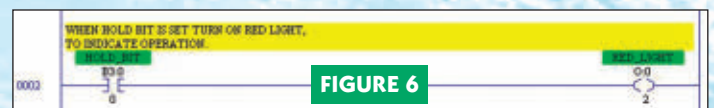
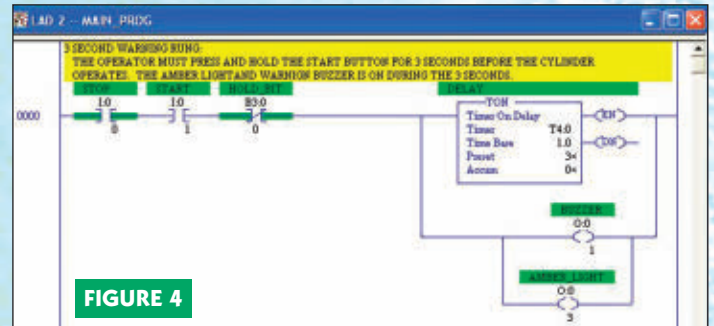
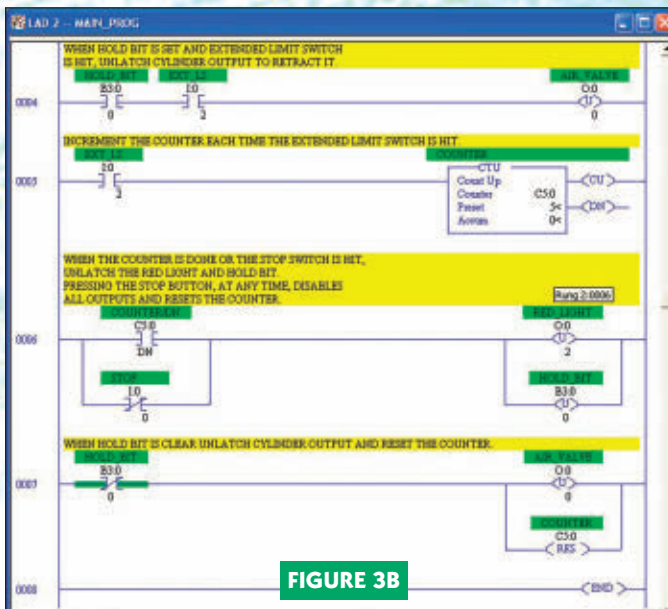
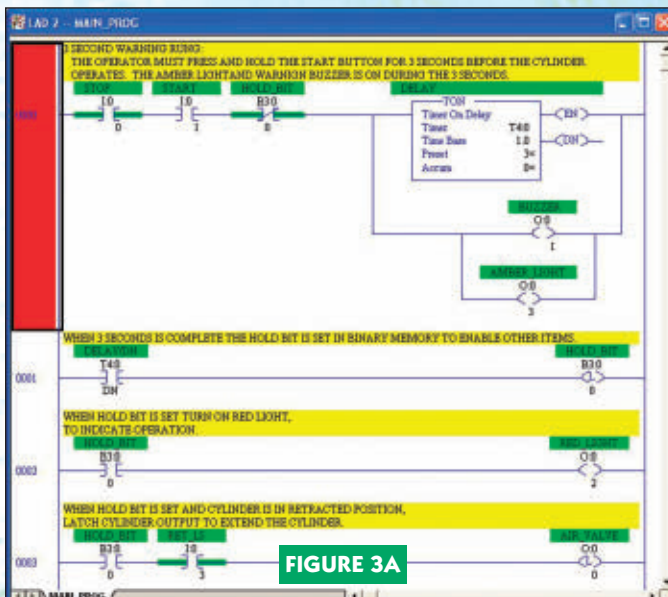
Industrial Control 101

Let's go through the ladder logic program one rung at a time. Rung 0 (see Figure 4) waits for three seconds while the start button is pressed. It turns on the buzzer and amber light. Rung 1 (see Figure 5) sets a bit in memory when the three second delay has completed. This bit in



memory is named "HOLD_BIT" and is used to control other logic in the program. Basically, if the HOLD_BIT is set the system should operate; if it is cleared, the system goes into its original power-on state waiting for an operator to press the start button. Rung 2 (see Figure 6) turns on the red light when the system is in operation mode. Rung 3 (see Figure 7) turns on the pneumatic valve output (latches it on) if the HOLD_BIT is set and the retracted limit switch is closed. This occurs when the piston is at its fully retracted position. Rung 4 (see Figure 8) turns the pneumatic valve output off (unlatches it) if the HOLD_BIT is set and the extended limit switch is closed. This occurs when the piston has reached its fully extended position. Rung 5 (see Figure 9) increments counter C5:0 each time the extended limit switch is hit. The DN or done bit of this counter will be true when the preset value of 5 equals the accumulated value. To change how many times the

piston cycles, simply edit this rung and change the preset value of counter C5:0. By the way, if the preset value is never set or remains at zero, the piston will not oscillate at all. In Rung 6 (see Figure 10) the COUNTER/DN bit will be true when the counter in the previous rung preset and accumulated values are equal (5 at this time). It also turns off the red light and clears the HOLD_BIT (unlatches both). Clearing the HOLD_BIT puts the program back into its beginning power-on state. This rung will also be true if the stop button is pressed. Rung 7 (see Figure 11) will turn off the pneumatic valve output and reset





If your pneumatic system needs to be portable and cannot be tied to a bulky compressor, there is a CO₂ regulator kit available from Lowe's Home Improvement Stores. These kits can regulate high pressure CO₂ cylinders — such as those used with paintball guns — down to a range of from 0 PSI up to 120 PSI (see Photo 7). The Kobalt portable compressed CO₂ regulator kit sells for approximately \$85. The kit allows you to

connect your normal compressed air fittings to the compressed CO₂ cylinder throughout the regulator. It is made to clip onto a person's belt for portable use of nail guns. Lowe's provides a cylinder exchange system for these CO₂ cylinders, similar to the LP gas cylinder exchanges for home BBQs. You must purchase your first full cylinder for \$24 for a nine ounce cylinder and \$36 for a 20 ounce cylinder. When they are empty, you can exchange them for a full cylinder for about \$5. Be sure to read all of the instructions, precautions, and safety notes when using this regulator.

PARTS LIST

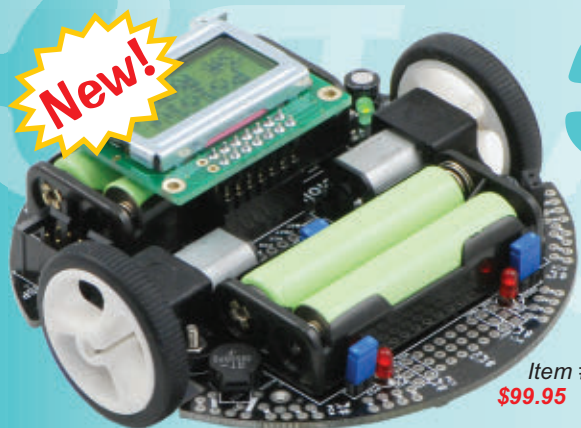
Part #	Description
Jameco #159599	Miniature limit switches
Jameco #175565	24 VDC DPDT electromechanical relays
Jameco #275531	Relay socket

the counter when the HOLD_BIT is clear. The HOLD_BIT can be cleared either by the system having completed all of its oscillations or by pressing the stop button at any time.

Granted, this PLC demonstration program may not appear to be terribly practical, but its purpose is to use several of the programming features of the PLC and interface with a pneumatic system.

It has been my experience that pneumatic components can add some real muscle to projects along with electronic controls as the brains. It would be pretty difficult — and probably very expensive — to find an equivalent linear electronic actuator capable of producing up to 66 pounds of force (at 150 PSI) while travelling a full six inches as quickly as the pneumatic components shown in this article will.

I hope that you found this information helpful and it will help to get you started with pneumatic components and systems. **SV**



Item #975
\$99.95

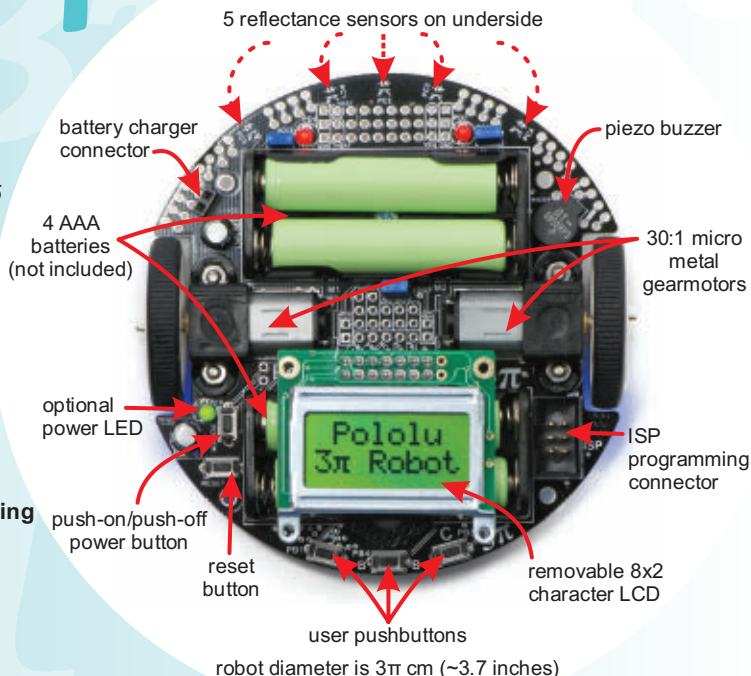
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BLDC HARDWARE PART 3

by Fred Eady

It's time to get cerebral with our BLDC motor project. However, before we transition our fingers from spinning the BLDC motor shaft to tapping out some BLDC motor code, I have a confession to make. Before I relate my experiences, though, let's have a moment of silence for the fallen Microchip TC4428 and Fairchild NDS9952A parts that are littered all over my workbench under a cloud of "magic smoke."

My hot air soldering station has been very busy removing "exhausted" chips from the pair of prototype BLDC motor controller printed circuit boards I'm using to support our motor controller project. I built up a second PCB without the chips, but with the intention of writing and testing as much of the motor controller firmware as I could without having to worry about accidentally smoking the three phase circuitry. I figured that everything was going as planned. In my mind, I was on the road to success with only a small bug to root out and fix.

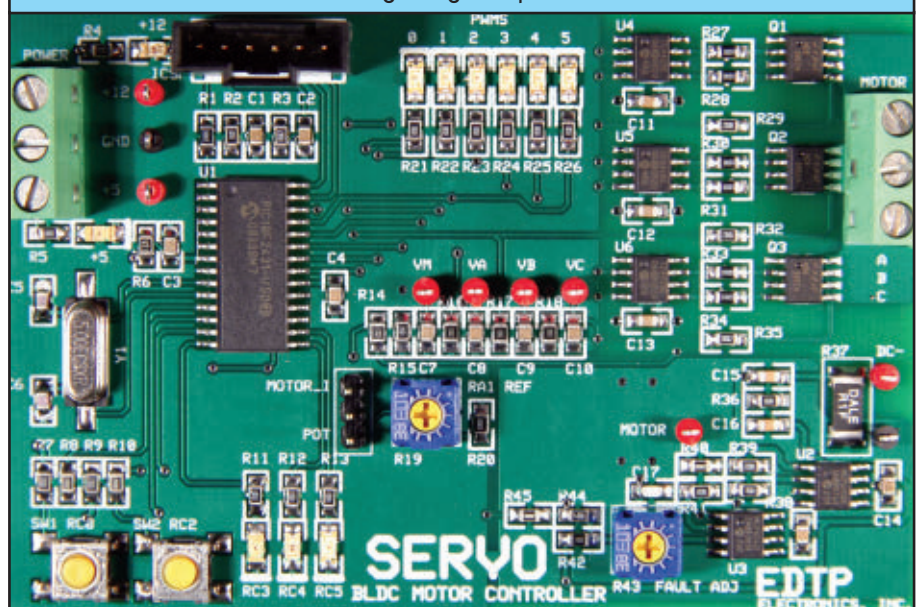
I was having a problem with PWM0 that was presenting itself to me visually via the PWM0 activity LED. After some head scratching, I finally found that I had configured PWM0 and PWM1 as *complementary* PWM outputs instead of *independent* PWM outputs. The PWM0 LED no longer acted funny after I switched the PWM0 and PWM1 output pair to independent mode. Without the PWM0 LED indicator, I would have not caught the bug until later in the firmware testing process since at the time I was not looking at the output of the PWM pairs simultaneously using an oscilloscope.

The LEDs saved the day yet again when I noticed that PWM4 and PWM5 on the PCB didn't match the definitions in the test code I was executing. After

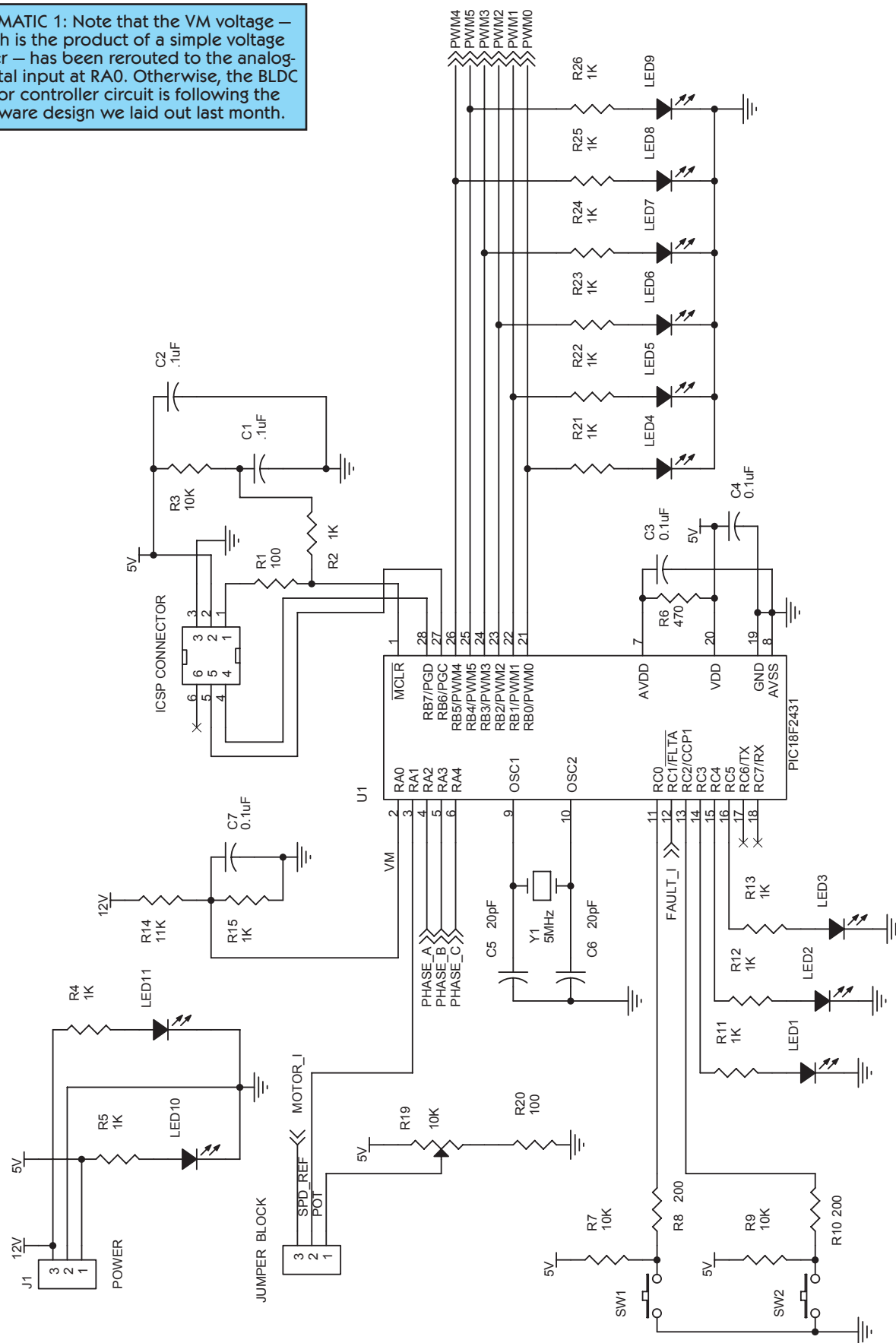
getting down off my donkey, I found that I had "assumed" that the PWM signals were laid out in numerical order on the PORTB pins of the PIC18F2431. In fact, if you take a look at the PIC datasheet, you'll see that PWM4 is presented on RB5 instead of RB4 as I "assumed." A couple of pieces of wirewrap wire and two precise moto tool-assisted trace removals fixed that problem. At this point, I'm looking very closely for more "assumption bugs" before moving on with my firmware testing.

Satisfied that I had conquered all of the problems, I mounted the TC4428s and NDS9952As on the prototype PCBs. I figured I'd play it safe and not electrically attach the BLDC motor to the controller electronics at this time. The absence of the BLDC motor coils would eliminate the possibility of passing any harmful overcurrent levels through the NDS9952A output pairs. To add some extra "no smoke"

PHOTO 1. No major changes were made to the updated BLDC motor controller hardware. I simply fixed some trace boo-boos and tied the VM voltage output to a dedicated analog-to-digital input on the PIC18F2431.



SCHEMATIC 1: Note that the VM voltage – which is the product of a simple voltage divider – has been rerouted to the analog-to-digital input at RA0. Otherwise, the BLDC motor controller circuit is following the hardware design we laid out last month.



security, I didn't apply power to the motor controller's +12 VDC power input. Things seemed to go well without the BLDC motor and its power source. I tested with this configuration without incident.

After coding what I thought was enough to try to spin the BLDC motor shaft, the time had come to apply power to the three phase driver. My goal was to simply nudge the motor shaft by forcing it to assume the final (sixth) position in a BLDC motor electrical cycle. Recall from our previous discussion that the PWM outputs are selected using the PIC18F2431's OVDCOND and OVDCONS registers as follows:

```
PDC0H = 0x01;           //PWM0 and PWM1
PDC0L = 0xFF;
PDC1H = 0x01;           //PWM2 and PWM3
PDC1L = 0xFF;
PDC2H = 0x01;           //PWM4 and PWM5
PDC2L = 0xFF;
OVDCOND = 0b00100000;   //PMW5 active only
OVDCONS = 0b00000000;
```

Upon applying +12 VDC BLDC motor power along with the +5 VDC logic power, I noticed that the +12 VDC power indicator LED dimmed and started to pulsate. After quickly powering everything down, I noticed that the TC4428s were very warm. I simply noted this as "odd" and moved on. I again applied full power and observed the pulsating LEDs which seemed to be in rhythm with a jerking of the BLDC motor shaft. After whiffing the aroma of "magic smoke," I decided to "finger test" the warm parts. The TC4428s were hot enough to put a blister on my now-temperature-calibrated thumb. Something was wrong in a bad way. I figured I was somehow pushing too much current through the motor coils. Following some deep meditation, I came to the conclusion that the pulsating I was seeing was the wall wart switching power supply attempting to save itself from an overcurrent condition by discontinuing power to the load. As I'm trying to keep the nasty thought of "bad design" out of my mind, I'm burning up TC4428s at a fanatical rate. It came down to the fact that simply applying full power to the motor controller smoked the TC4428s. My feelings, my pride, and my thumb were hurt. For the next few days, I was clueless as to what was going on in a bad way with this project. Everything that had to do with polarity of power supply voltages and PWM signals looked fine on my CleverScope display and the NDS9952A MOSFETs were never getting warm. How could I be frying the TC4428s and not the MOSFET pairs?

More meditation led me to the conclusion that since I was only releasing the "magic smoke" when +12 VDC was present, I was somehow overloading the TC4428's push-pull outputs. To kill the outputs meant that I had to be punching clean through the gates of the MOSFET pairs causing them to present a short to the TC4428 outputs. I whispered to myself, "No way."

Once again, I got down off of my donkey. The problem had to lie in the way I was driving the TC4428 inputs with the firmware. After a couple of days of intense datasheet scrutiny, I found that the problem was embedded in this set of firmware commutation patterns:

```
/**      COMMUTATION PATTERNS
/**
/**#define pos0 0b00111001 ;PWM1-PWM2 A-B (C)
/**#define pos1 0b00101101 ;PWM1-PWM4 A-C (B)
/**#define pos2 0b00100111 ;PWM3-PWM4 B-C (A)
/**#define pos3 0b00110110 ;PWM3-PWM0 B-A (C)
/**#define pos4 0b00011110 ;PWM5-PWM0 C-A (B)
/**#define pos5 0b00011011 ;PWM5-PWM2 C-B (A)
```

Do you see the problem? If you don't see the problem, don't feel bad. It took me two days to discover the source of my hardware pain. If you're still under the spell that was cast upon me, look back at the OVDCOND/OVDCONS code snippet I pulled from last month's discussion. Each and every commutation pattern has not one, not two, but four MOSFETs activated. I was lucky not to set fire to the bench! Here's the commutation pattern set that removed the spell and led me out of the magic-smoke-filled forest:

```
/**      NEW COMMUTATION PATTERNS
/**
/**#define pos0 0b00000110 ;PWM1-PWM2 A-B (C)
/**#define pos1 0b00010010 ;PWM1-PWM4 A-C (B)
/**#define pos2 0b00011000 ;PWM3-PWM4 B-C (A)
```

etc.

That's more like it! Now, only two PWMs are activated per commutation position. One of the PWM signals is always a +12 VDC enabler and the other PWM is always pulsing the ground return MOSFET. I am happy to report that the +12 VDC indicator LED no longer dances to the beat of an overloaded wall wart and the TC4428s are no longer burning holes in my thumb.

The Big Picture

The updated motor controller hardware is saying "cheese" in Photo 1. During the firmware test period, I discovered that the VM voltage output needed its own dedicated analog-to-digital (A-to-D) input. So, I moved the MOTOR_I voltage output point to the jumper block point that was occupied by the VM voltage output. For the purposes of a fault condition, there's no reason to keep up with the motor current as the current op-amp/comparator monitor circuit can be used to detect and report an overcurrent condition to the PIC18F2431. The output of the MCP6542 comparator feeds the PIC's active-low FLTA pin. When an overcurrent condition is sensed by the MCP6542, its output falls to a logical low level, which kicks in the PIC's automatic fault handler. Upon seeing a logical low at the FLTA pin, the hardware PWM outputs are driven to an inactive state until the fault condition is removed. With some simple register programming, we can choose to remain in a fault mode until the firmware clears the fault indicators or exits fault mode automatically when the overcurrent condition is removed. If you wish to monitor the current with the A-to-D module, you'll be pleased to find that the voltage in millivolts presented at the MOTOR_I node is identical in value to the current being drawn. For instance, if you read 37 mV at the MOTOR_I node, you're pulling 37 mA of current through the three phase driver

phase pairs. When I was burning up TC4428 parts, I was attempting to perform a BLDC motor shaft alignment operation. Since we are not using Hall sensors to report the rotational position of the motor shaft, we must begin the BLDC motor spin-up process from a known electrical position. Look back at the corrected commutation pattern code snippet and you'll see that there are six rotational positions in every electrical cycle. The number of electrical cycles needed to perform a single 360° physical rotation of the BLDC motor shaft depends on the number of poles contained within the motor. For the BLY171S-24V-4000

motor, the pole count is eight. One electrical cycle per pole pair is required to perform one full 360° shaft rotation. That equates to four six-position electrical cycles per revolution for the BLY171S-24V-4000, or 24 commutations.

Once the motor shaft has been aligned, the next thing we must do is apply our commutation table to the PIC18F2431's PWM outputs just as if the BLDC motor was a stepper motor. If we flow through the commutation table slowly while ramping up the PWM voltage, the motor shaft will begin to spin. The idea is to begin spinning the shaft slowly so it can easily follow the forced commutation

pattern. Once it is spinning, we gradually increase the speed in which we flow through the commutation table, which increases the rotational speed of the motor shaft. As the rotational speed increases, so does the magnitude of the BLDC motor's BEMF. The motor must spin up to about 800 RPM to get enough BEMF for a valid A-to-D measurement. Once we can see BEMF electronically, the trick is to switch as seamlessly as possible from forced to sensorless commutation mode.

Once we jump to sensorless regime, we come under the control of the closed-loop system formed by the PIC's BLDC motor controller firmware and its timers. The voltage divider networks we set up in hardware can now come into play. The VM voltage is the voltage applied to the three phase drivers divided by 12. Thus, with +12 VDC as our applied motor supply voltage, we will measure one volt at the VM node. Each PHASE_X (PHASE_A, PHASE_B, PHASE_C) voltage divider node is fed by a divide-by-6 voltage divider, which is coupled with a 0.47 µF low pass filter capacitor. The maximum voltage at any PHASE_X node would be two volts with a +12 VDC motor input supply voltage. Ideally, we want to see 50% of the motor input voltage level at the midpoint of the BEMF signal we are measuring. That equates to a point in time that is 30° after commutation begins. So, with a +12 VDC motor input voltage, ideally we want six volts of BEMF at a point in time that lags 30° behind the initiation of a

FIGURE 1: The vertical dotted line is the 50% or 30° point of the BEMF waveform. This figure depicts an ideal situation in which the BLDC motor's speed versus the applied voltage is perfect.

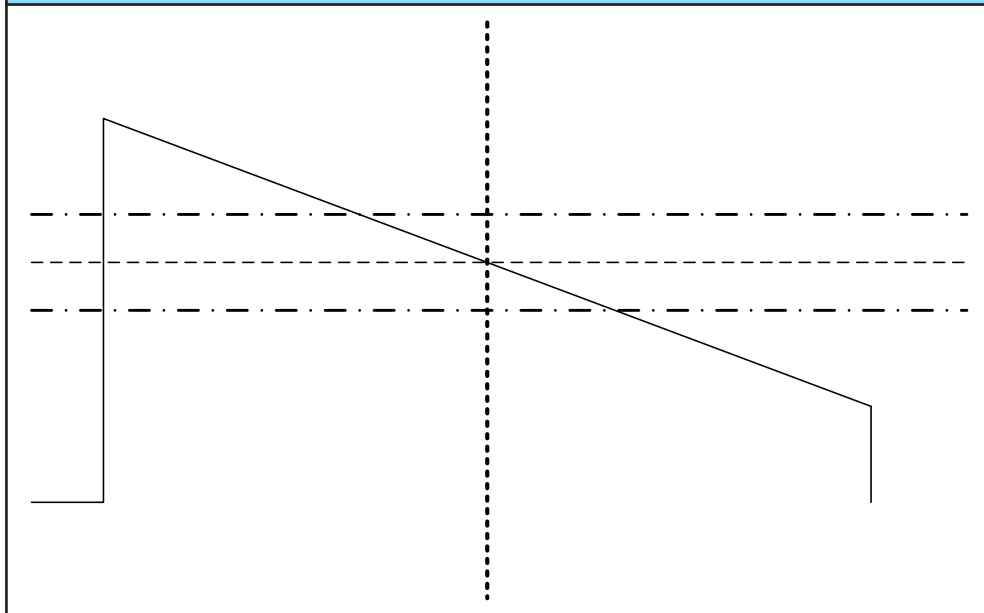
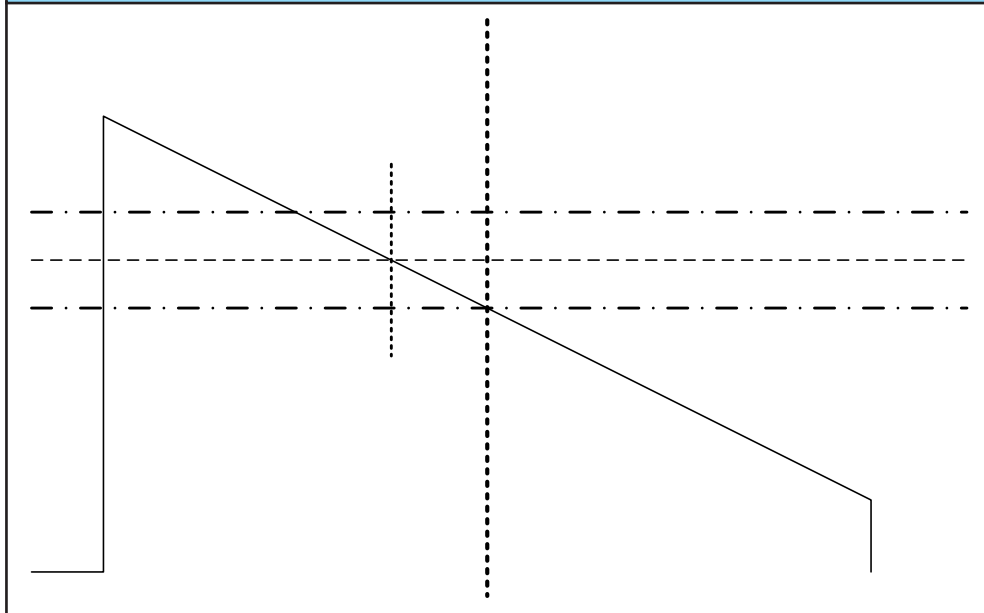


FIGURE 2: This graphic shows us that the BEMF voltage level will register below the 50% motor voltage rail target when the motor speed is too slow or the voltage applied to the motor for that speed is too high.



commutation period. This ideal situation is shown graphically in Figure 1; commutation occurs at the far left of the figure. The resulting ramp-like BEMF waveform you see is generated. The vertical dotted line in Figure 1 is the 50% or 30° commutation center point where we want to see 50% of the motor supply rail. Figure 1 is reality when the speed of commutation and the voltage applied to the BLDC motor by the PWM source are causing the BEMF ramp to center itself at one half of the motor supply rail at the midpoint of the commutation cycle.

What if the motor speed is too high and the voltage applied by the PWM signals for that speed is too low? The BEMF ramp will shift down and to the left as shown in Figure 2. Thus, the measured commutation midpoint voltage will come in lower than 50% of the motor voltage rail. Yep, Figure 3 shows us what happens to the BEMF voltage ramp when the motor speed is too high relative to the motor's applied PWM voltage. In Figure 3, you can easily see that the BEMF voltage ramp has shifted up and to the right within the commutation period. The result of this measurement is a voltage higher than half of the motor voltage supply rail.

Judging from what we see in Figures 1, 2, and 3, our firmware BLDC motor control loop must regulate the commutation interval versus the pulse width. The goal of the control loop firmware is to put the middle of the BEMF ramp in the center of the commutation period. Let's determine how we know where the 50% voltage points are. If 12 volts is two volts at the PHASE_X nodes, then half of the motor voltage supply rail (12 volts) is equivalent to one volt at the PHASE_X nodes. The VM voltage divider is a divide-by-12 circuit. So, 12 volts is one volt at the VM node. Thus, we can reference our PHASE_X voltages to the VM voltage. Basically, if the PHASE_X voltage is equal to the VM voltage, the BEMF ramp is centered within the commutation period. If we consider the motor speed as a constant, then we can control the BEMF ramp position by throttling the PWM signals we are presenting to the three phase motor driver.

Aligning A BLDC Motor Shaft

One can easily fall into a pit of math vipers when working with BLDC motor control algorithms. I don't like snakes of any kind, especially math vipers. So, in snake country I watch my step and keep things as simple as possible. With that, let's approach BLDC motor alignment from an intuitive angle.

We know that we want to force the BLY1715-24V-4000's shaft into the final position of a BLDC motor electrical cycle. Let's take another look at our commutation pattern definitions and add a vehicle that will allow us to

load the OVDCOND with the binary commutation patterns:

```
/**      NEW COMMUTATION PATTERNS
**      PHASES:  USED  OPEN
**#define pos0 0b00000110 ;PWM1-PWM2 A-B (C)
**#define pos1 0b00010010 ;PWM1-PWM4 A-C (B)
**#define pos2 0b00011000 ;PMW3-PWM4 B-C (A)
** etc.
char comm_sequence[6] =
    {0x06,0x12,0x18,0x09,0x21,0x24};
```

By placing the commutation patterns into an array, we can easily place a commutation pattern into the OVDCOND like this:

```
OVDCOND = comm_sequence[5];
```

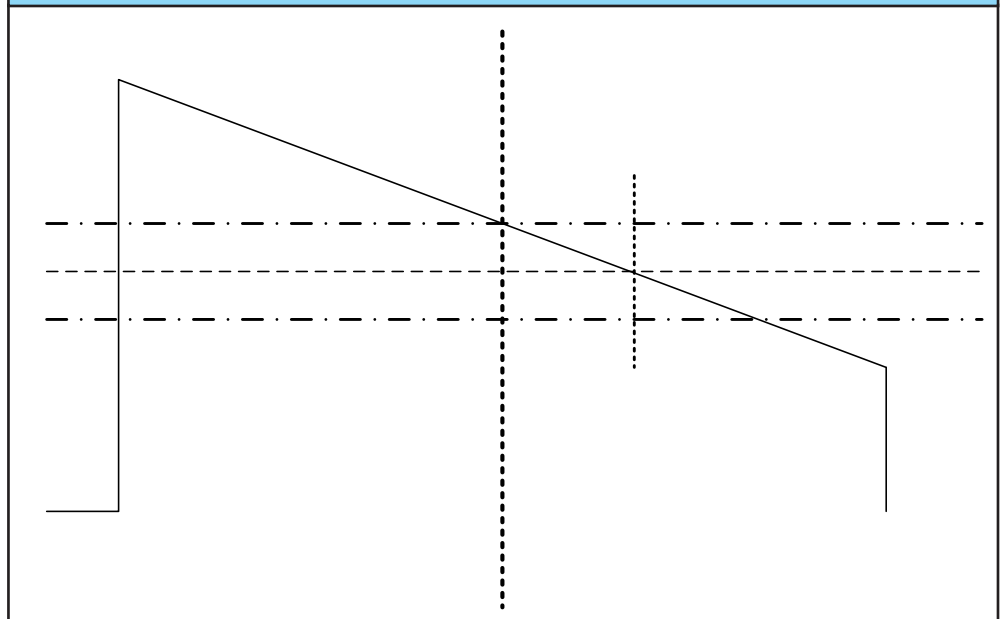
One good line of code leads to another. The commutation function is built around loading OVDCOND:

```
/** *****
** COMMUTATE
** *****
void commutate(void)
{
    OVDCOND = comm_sequence[comm_ptr++];
    if(comm_ptr > 5)
    {
        comm_ptr = 0;
        LED1 ^= 1;
    }
}
```

The *commutate* function spins continuously through the six electrical cycles. However, to align the motor shaft, we need to load only OVDCOND with the sixth electrical cycle's binary commutation pattern and spin it in place.

The *main* function of the motor controller firmware driver contains routines that define pushbutton SW1 as the Start button and SW2 as the STOP button. Upon power-up, the motor controller enters STOP mode and spins, looking

FIGURE 3: As you can see in this graphic, the midpoint commutation voltage is too high when the motor speed is fast relative to the applied PWM voltage.



for a START button depression and release. The START button code drops the firmware into ALIGN mode. Here's the ALIGN mode entry code:

```
case ALIGNMODE:
    mode = ALIGNMODE;           //set the operating mode
    pwm_control(OFF);           //disable the PWMs
    disable_TMR5int;            //no interrupts for TMR5
    TIMER5OFF;                  //TMR5 OFF
    TMR5H = 0x00;               //TMR5 counts up to PR5
    TMR5L = 0x00;
    PR5H = 0x13;                //load TMR5 for 1ms
    PR5L = 0x88;
    voltage = 0;                //PWM voltage = 0
    delay = 0;                  //init align delay ctr
    tenms_counter = 0;          //10ms counter
    LED2 = ON;                  //for us humans
    OVDCOND = comm_sequence[5]; //set commutation
                                // state
    comm_slice = 1;             //tell TMR5 interrupt
                                // not to commute
    pwm_control(ON);            //enable the PWMs
    enable_TMR5int;             //kick off TMR5 1ms
                                // intervals
    TIMER5ON;                   //TMR5 ON
    break;
```

The ALIGN mode operational code shares space with the STOP, START, and RUN mode algorithms in the *mode_service* function, which is called by the TMR5 interrupt handler. The *ALIGNMODE* case statements I just revealed also share program memory area with the entry code that supports the STOP, START, and RUN operational service routines. Each entry mode routine prepares the environment to execute the newly selected operational mode that is about to go into service.

As you can see in the *ALIGNMODE* entry case statements, TMR5 is shut down to allow it to be loaded with a 1 ms period value. TMR5 is unique in that it counts up towards a match with the PR5 register pair. Once the TMR5 registers have incremented their way to the same value as the PR5 registers, the TMR5 registers are reset to zero, the TMR5 interrupt flag is set, and the upwards count begins anew.

The alignment process ramps up the PWM voltage for approximately 1 second and holds it there for about two seconds. The idea is to supply just enough PWM voltage to nudge the BLDC motor shaft into the desired electrical position. In our case, the desired electrical position is 6. The code to accomplish the nudge goes like this:

```
// align time in 10ms increments
#define ALIGNMODE_LOOPS 255

//*****
//** ALIGN MODE
//*****
case ALIGNMODE:

    //has 10ms passed?
    if(++tenms_counter >= 10)
    {
        if(voltage <= 560) //ramp voltage to 50%+
        {
            voltage += 5; //inc 5 PWM counts/sec
                        //load new PWM values
            PDC0H = make8(voltage,1);
```

```
            PDC0L = make8(voltage,0);
            PDC1H = make8(voltage,1);
            PDC1L = make8(voltage,0);
            PDC2H = make8(voltage,1);
            PDC2L = make8(voltage,0);
        }
        else if (delay < ALIGNMODE_LOOPS)
            //2 second align delay
        {
            ++delay;
        }
        else
        {
            enter_mode(STARTMODE); //enter next state
        }
        tenms_counter = 0;          //clr the 10ms
                                    //counter
    }
    break;
```

The 50% PWM voltage mark is represented by the value 512 loaded in the PDCX duty cycle registers. You'll find that the BLY171S-24V-4000 will not spin its shaft for any PWM voltages that are specified as less than 512. We want to be sure that the BLDC motor shaft locks into the nearest selected electrical cycle position. So, we add a boost factor to the 50% PWM value of 512. If you're wondering how I came up with the PWM voltage value of 560, it was by experimentation. I increased the PWM voltage value until I could actually feel the motor shaft move into the alignment position. Note also that I applied a liberal fudge factor to the alignment delay value. After feeling that little "thump" of the BLDC motor shaft, I let out a big YEEE HAAAA!

Spinning Up

Here we go. The BLDC motor shaft is aligned at electrical position 6 and START mode has been entered. Like the ALIGN mode, the START mode is also preceded by START mode entry code:

```
case STARTMODE:
    mode = STARTMODE;
    comm_slice = 0; //commutate on next TMR5
    interrupt
        LED2 = OFF; //exiting alignment routine
        voltage = 0; //reset PWM voltage value
        tmr5val = 0xA018; // 305 RPM value
                        //kill TMR5 and its interrupt
    disable_TMR5int;
    TIMER5OFF;

    TMR5H = 0x00; //init TMR5 up counter
    TMR5L = 0x00;

                                //TMR5 counts up to PR5 value
    PR5H = make8(tmr5val,1);
    PR5L = make8(tmr5val,0);
                                //enable TMR5 and its interrupt
    enable_TMR5int;
    TIMER5ON;
    break;
```

The *tmr5val* variable is derived from a basic BLDC motor equation:

$$\text{tmr5val} = 60 / \text{RPM} * \text{Pole Pairs} * 6$$

Where:

RPM = Desired motor shaft speed

Pole Pairs = Number of motor poles divided by 2

The code says we need to spin the BLY171S-24V-4000 motor shaft at 305 RPM:

$\text{tmr5val} = 60 / 305 * 4 * 6 = 8.2$ milliseconds

We can't readily load the raw 8.3 ms into our PIC18F2431 TMR5 registers. So, we'll have to do a bit of conversion. We simply need to convert 8.3 ms into TMR5 ticks. Here's what we need to load into the PR5 registers:

$\text{tmr5val} = 8.2 \text{ ms} * (\text{FOSC} / 4) = 8.2 * 5000000 = 40984\text{h} = 0\text{x}A018$

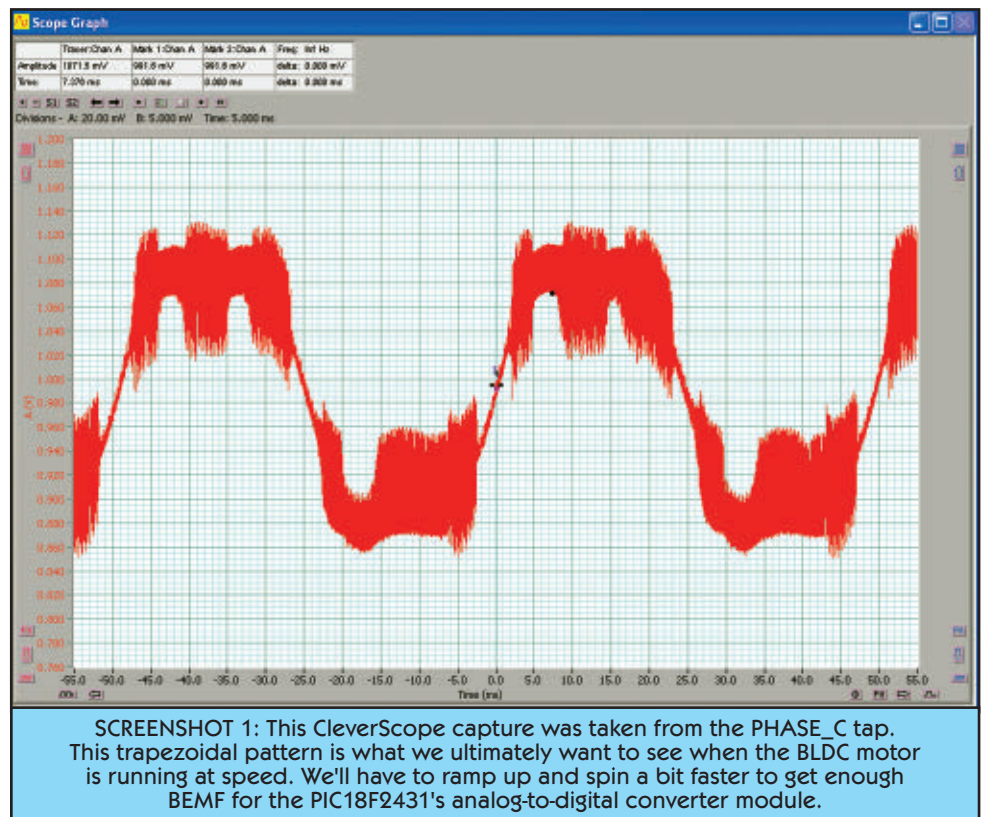
The stage is now set to run this code:

```
//*****  
// * START MODE  
//*****  
case STARTMODE:  
    if (voltage < 570)  
    {  
        ++voltage;  
        PDC0H = make8(voltage,1);  
        PDC0L = make8(voltage,0);  
        PDC1H = make8(voltage,1);  
        PDC1L = make8(voltage,0);  
        PDC2H = make8(voltage,1);  
        PDC2L = make8(voltage,0);  
    }  
    break;
```

Again, I used physical touch coupled with a minimal current draw to come up with the 570 PWM voltage value. Right now, all I want to do is smoothly spin the BLDC motor shaft. So, I dialed in the PWM voltage value for optimal symmetry of the waveform you see in Screenshot 1. The initial START mode voltage ramp is controlled by the *tmr5val* as the STARTMODE function is called with every TMR5 interrupt. The rotational speed is very slow and there isn't enough BEMF being generated to be recognized above the noise by the PIC18F2431's A-to-D module. However, my CleverScope picked up the beginnings of the trademark BLDC motor trapezoidal commutation waveform in Screenshot 1.

Spinning Out

The BLDC motor controller hardware is verified and the BLDC motor driver firmware is well along the way to being



viable. The next step is to transition from the START mode to the RUN mode. I think you have enough BLDC motor theory in your battery compartment to follow along in the complete version of the BLDC motor controller code that I'll put in the download package on the *SERVO* website (www.servomagazine.com) along with the ExpressPCB file. **SV**

Fred Eady can be reached via email at fred@edtp.com.

SOURCES

CleverScope
Saelig
www.saelig.com
A Microchip ICD3 was employed as the debugging device.

ICD3 Programmer/Debugger
PIC18F2431
MCP6002
MCP6542
TC4428
Microchip
www.microchip.com

NDS9952A
Fairchild Semiconductor
www.fairchildsemi.com

BLY171S-24V-4000 BLDC Motor
Anaheim Automation
www.anaheimautomation.com
The code for the BLDC motor driver is compiled using HI-TECH PRO for the PIC18 Family.

HI-TECH Software
www.htsoft.com

Implementing a Digital PID Controller on the PenAeroBot

by Carlos Montesinos and Ricardo Toro

Our philosophy at the Control Systems Laboratory at the University of Illinois is to teach students through interacting and experimenting with their own designs. In our lab, we continuously develop experiments to implement the theoretical concepts learned in class. One of the projects we are currently developing is the Pendulum Aeronautic Robot (PenAeroBot). This system is an inverted pendulum and the idea is to control its angle of inclination, propelled by two motors. The robot uses a digital Proportional-Integral-Derivative (PID) controller to levitate until it reaches the given set point. Figure 1 shows the first prototype and a block diagram of the robot's architecture. Throughout this article, we describe how we built the investigational PenAeroBot and how to implement a digital PID controller. We also show experimental results of the controller's performance and the design of a more flexible PenAeroBot platform for teaching controls systems theory.

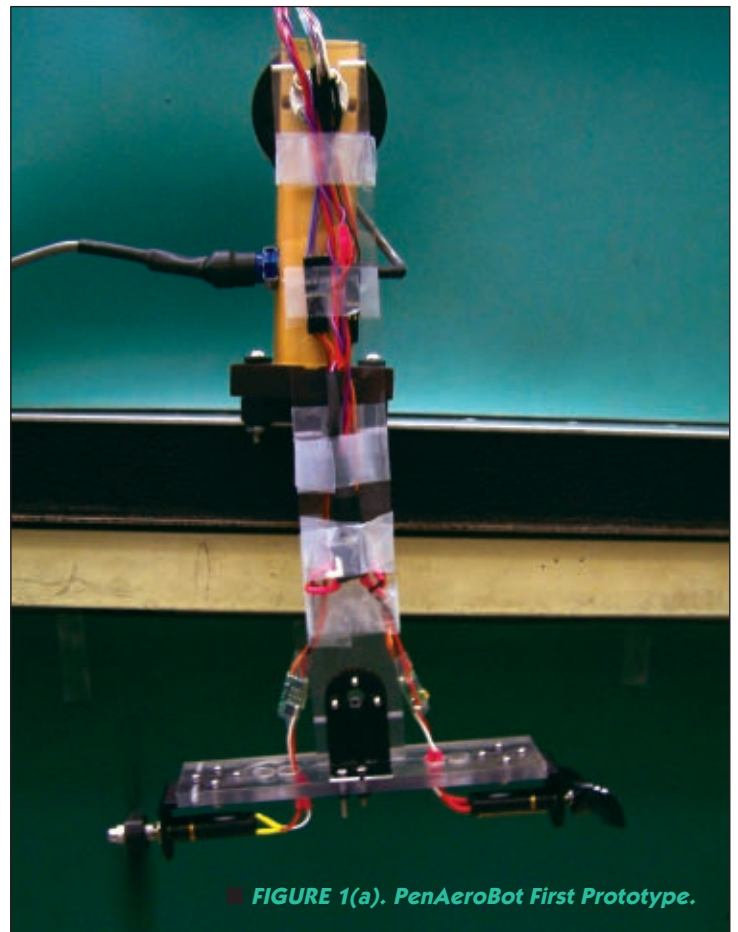
The Robot's Structure

The PenAeroBot was developed using standard hobby parts. Table 1 shows a list of all the parts needed to build it. As shown in **Figure 1**, the robot consists of a digital signal processor (DSP), two DC motors, and two encoders. The DSP is in charge of sampling the encoders, running the PID controller, and manipulating the actuators. The motors are oriented 90 and -90 degrees from the body of the pendulum to maximize the torque exerted on the system. The encoder mounted on the pendulum's base provides inclination feedback while the reference encoder is used to input the desired inclination set point.

Assembling The Robot

Figure 2 shows how to mount the robot. Components (1) and (3) are custom-made parts from acrylic. The "L" connectors (2) and (4) are standard servo motor brackets. The material and parts may be purchased from US Plastics (www.usplastic.com) and Lynxmotion (www.lynxmotion.com), respectively. The brackets have a hole in the center allowing the DC motors to screw in, providing enough support as shown in **Figure 3**. We used two AstroFlight motors since they are small, light, and powerful. We also used these motors because they are controlled using a custom velocity servo circuit.

The velocity controller takes standard PWM servo motor signals and translates them into velocity



■ **FIGURE 1(a).** PenAeroBot First Prototype.



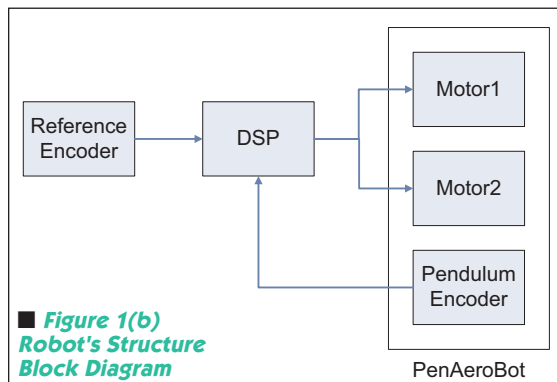
■ **FIGURE 4.**
Base of the
system.

commands. The motor and its velocity control can be purchased at AstroFlight (www.astroflight.com). To measure the inclination of the pendulum, a two-channel quadrature encoder was attached to a custom-made aluminum base as shown in **Figure 4**. The sensor we used was produced by DRC (www.drc.com). Unfortunately, they do not produce it anymore. However, US Digital (www.usdigital.com) produces similar and more modern products.

We powered the motors using a 12V, 1A power supply. The sensors and the motor controllers are connected to a TMS320C6713 DSP Starter Kit board. This project could be implemented using more simple and less expensive processing units, as long as they are capable of reading data from encoders and producing PWM signals.

Digital PID Controllers

A controller is run by an algorithm which makes decisions on how to manage the behavior of a plant or system, e.g., a robot. The control system theory is divided into classic controls and modern controls. Classic techniques may be implemented using analog or digital computers. However, modern techniques may only be implemented in digital computers. Classic techniques use basic mathematical operations like addition, subtraction, multiplication,



■ **Figure 1(b)**
Robot's Structure
Block Diagram

Table 1 - Parts List

Pendulum

Item	Part#	Qty	Source	Website
Optical Encoder	M21AAS0DB03	1	DRC	www.drc.com
Encoder Base	-	1	-	-
Hut	HUB-02	1	Lynxmotion	www.lynxmotion.com
Propeller 5x3	LP05030	2	APC Propellers	www.apcprop.com
Motor	799C	2	Astroflight	www.astroflight.com
Acrylic Rect. Bar 5.5" x 1.4" x 0.225"	44350	1	U.S.Plastic	www.usplastic.com
Acrylic Rect. Bar 11" x 1.4" x 0.225"	44350	1	U.S. Plastic	www.usplastic.com
Alum. L Connector Bracket	ASB-06	4	Lynxmotion	www.lynxmotion.com
7/8-9 UNC Screw 1/2" Long	-	15	Any store	-
7/8-9 Nut	-	15	Any store	-

Electronics and Power Supply

DSP	TMS320C6713 DSP Starter Kit	1	Texas Instruments	www.ti.com
Power Supply	E3631A	2	Agilent	www.home.agilent.com

integration, and derivation. Controllers using modern techniques make use of linear algebra and mathematical models of the robots being controlled. Analog computers are built with operational amplifier circuits, which makes it extremely hard to operate with vectors and matrices. In the following paragraphs, we describe how we implemented a classic digital PID controller on a DSP.

Controlling The Robot Using One Motor

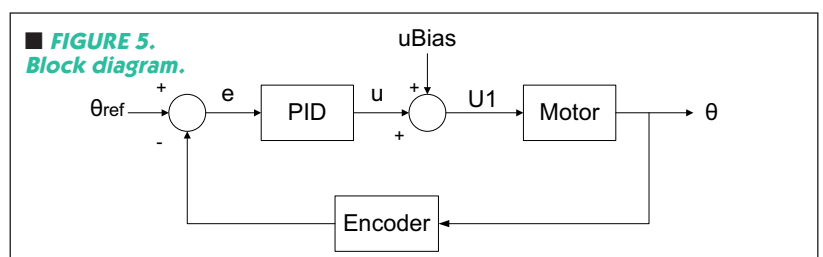
As mentioned earlier, the objective of the system is to reach a given inclination set point angle. We want to achieve this as quickly as possible and with the least number of oscillations. To simplify and learn more about the behavior of the system, we first operated it with one motor.

Figure 5 shows the control system's negative feedback loop we implemented. Later on, we'll describe how we added a second motor to try to increase the robot's performance.

The first signal we need to create is the error signal e defined in Equation 1:

$$e(t) = \theta_{ref} - \theta$$

The error signal tells us how far we are from the desired inclination angle. Assume we want the robot



■ **FIGURE 5.**
Block diagram.

[illegible]

The way the PID block processes the error signal to command the motor is described in Equation 2. The controller takes the error signal and produces the motor's control effort u :

$$u = K_p \times e(t) + K_i \times \int e(t) dt + K_d \times \frac{d}{dt} e(t)$$

■ **FIGURE 3.** DC motors and the velocity control circuitry.

■ **FIGURE 3. DC motors and the velocity control circuitry.**

Software Architecture

The main function runs only once and initializes the devices connected to the DSP board, its internal registers, and its memory. The *runAeroPendubotController()* function runs the control algorithm and it is called every 1 ms.

Implementing The Controller's Code

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error signal. The function returns the control effort for later use. The PID controller's code looks as follows:

```
float acPidController(float var_k, float theta_,
                    float kp_, float ki_, float kd_,
                    float dt_)
{
    // Implement controller
    derivativeValue = acDerivate(var_k, dt_);
    integralValue = acIntegrate(var_k, dt_);
    controlEffort = kp_*var_k +
        ki_*integralValue +
        kd_*derivativeValue;

    return controlEffort;
}
```

This function is pretty much returning $u = K_p * e + K_i * \text{acIntegrate}(e, 0.001) + K_d * \text{acDerivate}(e, 0.001)$, where $K_p * e$ is the proportional controller. Next, we have the integral controller and its code looks as follows:

```
float acIntegrate(float var_k, float dt_)
{
    static float area = 0; // Signal area
    float delta_area = 0;

    // Sample at time k-1
    static float var_k_1 = 0;

    // Use Tustin's rule to integrate
    delta_area = ( var_k_1 + var_k ) * dt_ / 2.0;
    area += delta_area;

    // Update samples
    var_k_1 = var_k;
    return area;
}
```

This function integrates the error signal using Tustin's rule, a numeric integration technique. The idea behind it is to find the area under the error's curve. Assuming we have an error signal as shown in Figure 7, the area at time t is calculated using Equation 3:

$$A_t = A_{t-1} + \Delta A_t$$

Delta area may be estimated using the area of a trapezoid given by Equation 4, where T is the sampling period:

$$\Delta A_t = \frac{(B+b) \times h}{2} = \frac{(e_{t-1} + e_t) \times T}{2}$$

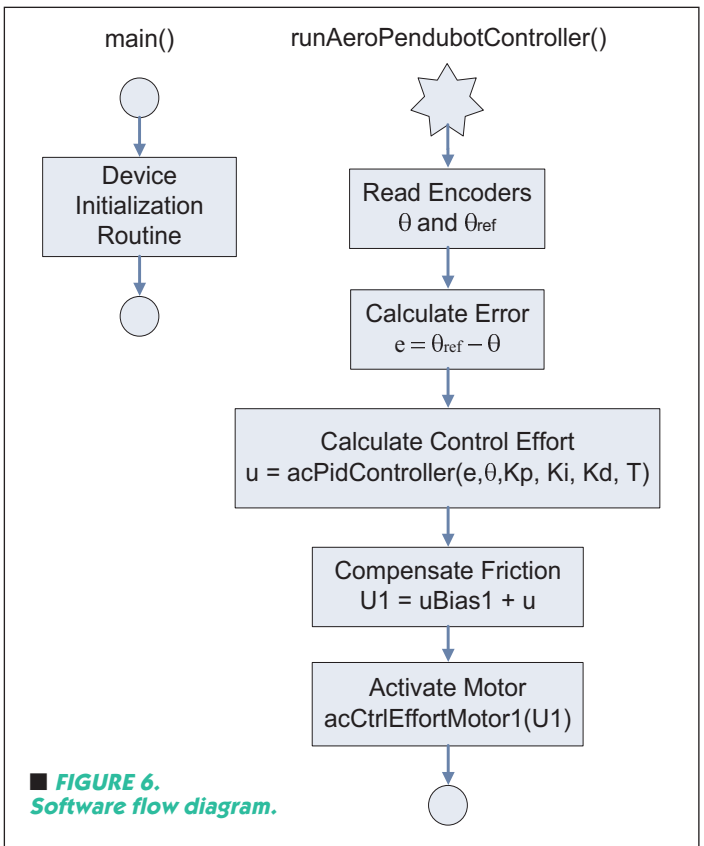
The final term in the PID is the derivate controller. Its code looks as follows:

```
float acDerivate(float var_k, float dt_)
{
    float dV_dt = 0;
    static float var_k_1 = 0; // Sample at time k-1

    dV_dt = (var_k - var_k_1) / dt_;

    return dV_dt;
}
```

This function takes the derivative of the error signal using the discrete approximation shown in Equation 5.



Here, we approximate the rate of change of the error within period T :

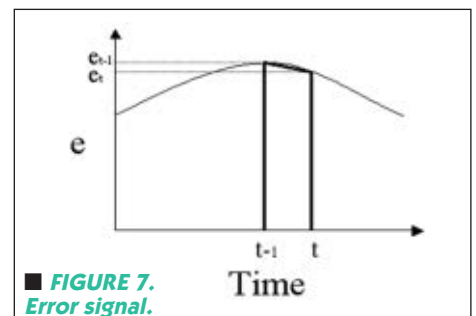
$$\frac{d}{dt} e(t) \approx \frac{e_t - e_{t-1}}{T}$$

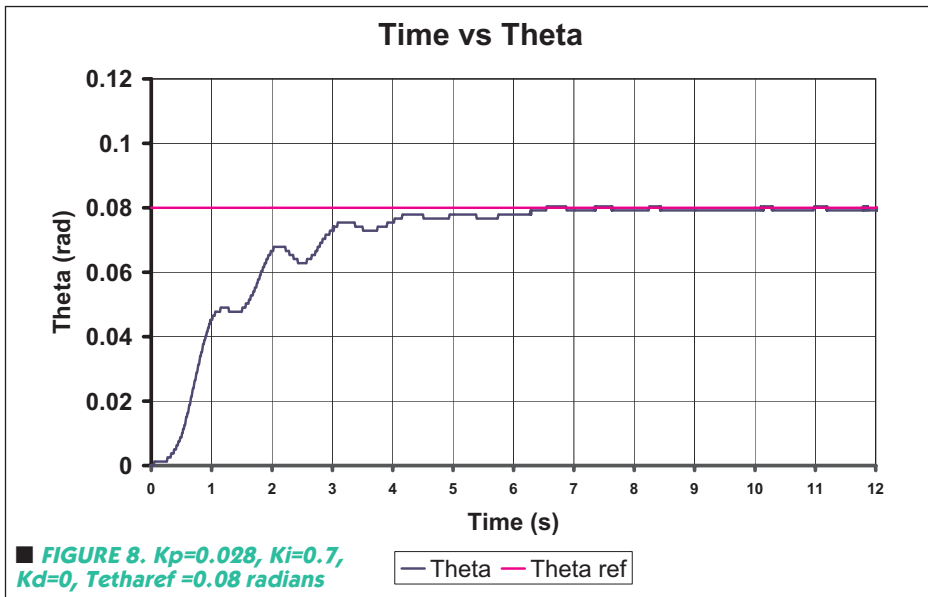
Tuning The Controller

Once the code is up and running, we need to find the right values for the control effort bias and the K_p , K_i , and K_d gains. There are many methods to determine good gain values. We used Zigler and Nichols method as a guideline. The first gain we tuned was K_p . Once we got a steady response, we increased K_i to eliminate the steady state error. Our final results — for a set point of 0.08 radians — are shown in Figure 8.

The controller we tuned responds pretty well for a set point of 0.08, however, it becomes unstable for other set points. **Figure 9** shows a stability graph where unstable set points are represented by zeros and the stable set points are represented by ones.

Tuning servo controllers is an art. When tuning a PID controller, you must know the system really





Implementing Two Motors

Motors with propellers have a lot of inertia. If a motor is spinning in one direction and it's suddenly commanded to turn in the opposite direction, it will take a lot of time to stop and start spinning in reverse. The implementation with one motor relies on gravity to push the pendulum in the opposite direction when needed. By adding a second motor, we avoid the inertia problem and we avoid relying on gravity to stabilize the system.

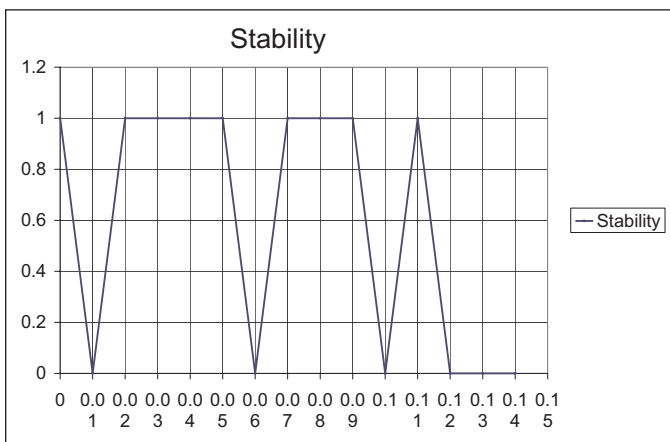
Controlling The Robot Using Two Motors

well. You should also have in mind how the proportional, integral, and derivative controllers affect the behavior of the system. From a theoretical point of view, one can never achieve a given set point with just a proportional controller. Even if it were possible, just a proportional controller by itself would not suffice.

Any system will wear out with time, requiring the controller to apply different control efforts, depending on those changes. In order to avoid this problem, one must use an integral controller. Integral controllers scale the integrated (accumulated) error in time. This means that even if the error is zero, the control effort will not necessarily be zero. This behavior could be thought of as the system building up memory as time passes by. It has a drawback, though. If friction is huge or the system gets stuck for any reason, the control effort will keep building up until it saturates. This could possibly damage the controller or the actuators.

The last control element is the derivative which slows down the response of the system. This controller scales the rate of change of the error, decreasing the control effort if the system reacts too fast. This is useful to avoid overshoot. However, it may slow the system's response too much.

■ **FIGURE 9.** System stability.



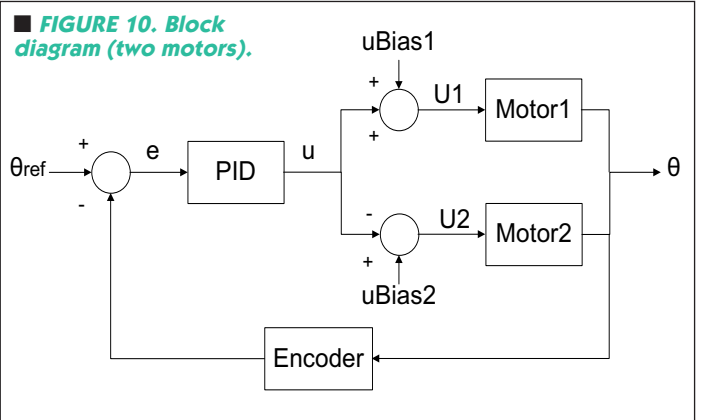
In this case, we used the same PID we designed previously. The only difference is that the control effort is split to activate each motor individually. The second motor takes a negative control effort such that if one motor needs to speed up to push the pendulum in one direction, the other slows down. **Figure 10** shows the block diagram for the new setup.

Software Architecture For Two Motors

The architecture for two motors is pretty much the same as the one for a single motor. The main difference is the addition of a calibration routine. We did this to avoid determining the bias for each motor individually. The bias is approximately 50% of the maximum control effort each motor can take. This allows the controller to speed up or slow down the motors within their operation region while keeping the system in equilibrium. **Figure 11** shows the new flow diagram.

Controller's Code For Two Motors

The only addition to the previous code is the calibration function `acCalibMotorBias()`. The goal of the function is to



turn on the first motor gradually until it hits 50% of its total power. Then, the second motor will be activated until the system is in equilibrium as close as possible to 0 radians. The function is only called until the bias values are determined for both motors. Once these values are determined, the PID controller routine is called.

The code for the calibration routine looks as follows:

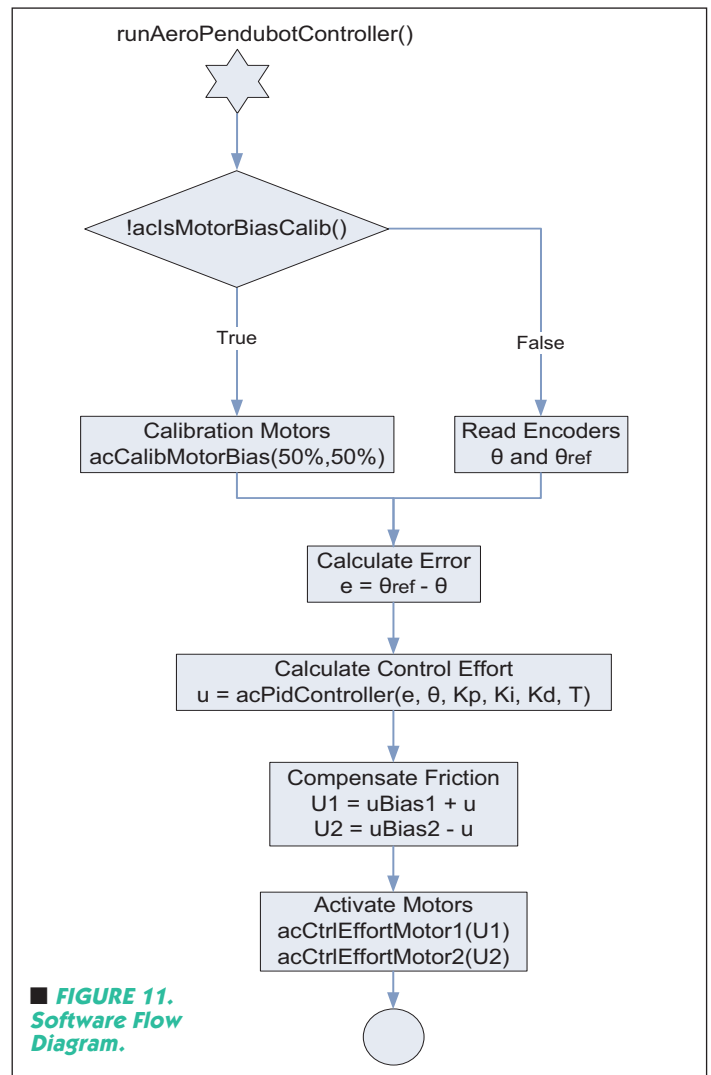
```
void acCalibMotorBias(float half1, float half2)
{
    static int calibState = 0;
    static int conta = 0;
    static int flag = 0;
    static double ramp1 = MIN_CTRL_EFFORT1;
    static double ramp2 = MIN_CTRL_EFFORT2;

    // Calibration state machine
    switch( calibState )
    {
        case 0:
            if( ramp1 < half1 )
            {
                // Ramp the first motor to 50% of the max
                power possible:
                ramp1 += DELTA_RAMP;
                acCtrlEffortMotor1(ramp1);
            }
            else
            {
                // Set the motor #1 bias -> motor1Bias
                motor1Bias = ramp1;
                calibState = 1;
            }
            break;

        case 1:
            if(half2 > 0 && flag == 0){
                // Ramp the first motor to 50% of the max
                power possible:
                ramp2 += DELTA_RAMP;
                acCtrlEffortMotor2(ramp2);
            }
            else
            {
                // Set the motor #2 bias -> motor1Bias
                motor2Bias = ramp2;
                flag = 1;
                if ( conta == 3000 )
                {
                    calibState = 2;
                    biasCalibFlag = 1;    // Indicate the
                    calibration is over
                }
                conta++;
            }
        }
    }
}
```

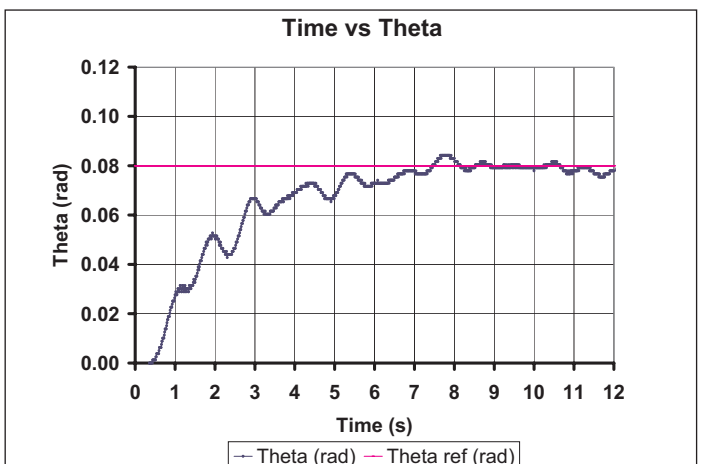
Tuning The Controller For Two Motors

Once again, we used Zigler and Nichols' method as a guideline to tune the system with two motors. We first tuned Kp and then Ki. **Figure 12** shows the results for the tuned system.

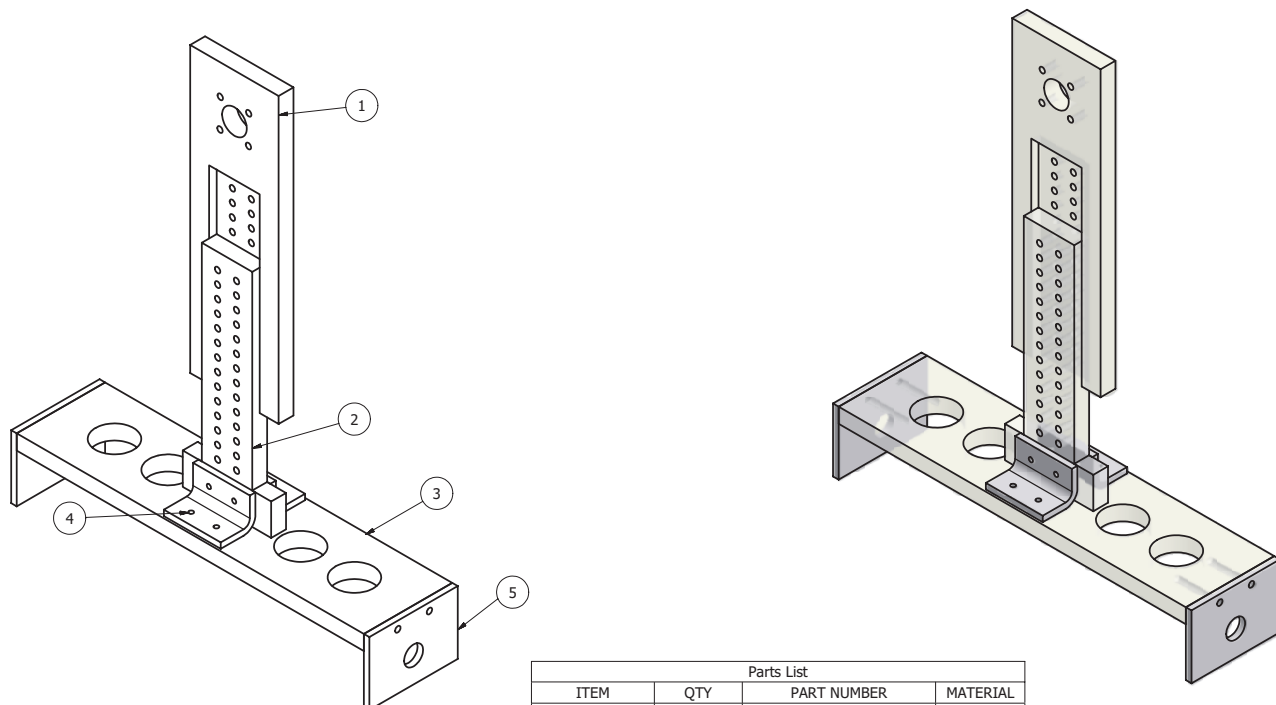


Conclusions

The first prototype of the PenAeroBot has proven to be very useful. This platform has allowed us to explore the implementation of a digital PID controller and it has also taught us a lot about the robot's properties. The



■ **FIGURE 12.** Kp=0.01, Ki=0.3, Kd=0, Tetharef=0.08 (rad).



■ **FIGURE 13.** New PenAeroBot design.

Parts List			
ITEM	QTY	PART NUMBER	MATERIAL
5	2	MOTOR LEVER	ALUMINUM
4	2	BRACKETS	ALUMINUM
3	1	LEVER H	ACRYLIC
2	1	LEVER DOWN	ACRYLIC
1	1	LEVER UP	ACRYLIC

implementation with one motor works relatively fast and it can be used to introduce students of all levels to the theory behind automatic controls. The implementation with two motors still needs to be refined.

As shown in **Figure 12**, it takes the system more time to reach a steady state with two motors than with a single motor. We think this is because the first prototype was not

thoroughly built. Asymmetries in the structure introduce undesired oscillations. Also, the fact that we are using the same PID controller for two different motors could be a cause for poor performance.

We are currently working on a new design, shown in **Figure 13**. In the future, we'll use modern control theory, including state space design and/or gravity feed forward controllers. We are confident that by using these techniques, we'll be able to achieve better results compared to PenAeroBot with one motor. **SV**

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References

Astroflight
www.astroflight.com

Static Thrust Calculator
www.mmaa-modelairplanes.org/staticthrustcalculator.htm

Motor-Propeller Damped Compound Pendulum
<http://prism2.mem.drexel.edu/~paul/thrustTester/thrustTester.html>

Digital Control of Dynamic Systems
 by G. F. Franklin, J. D. Powell,
 and M. Workman

Building projects involving electronics and robotics is much easier today than in the past because many of the commonly needed subsystems are available as ready-to-use modules (we will call them HMs [helper modules]). For example, there are HMs for controlling nearly any type of motor (DC, stepper, servo) and HMs to collect environmental data (humidity, temperature, distance to objects). You can find switching power supplies, battery chargers, Bluetooth modems, and dozens more just by looking through the ads in this magazine.

A Practical Quadrature Encoder

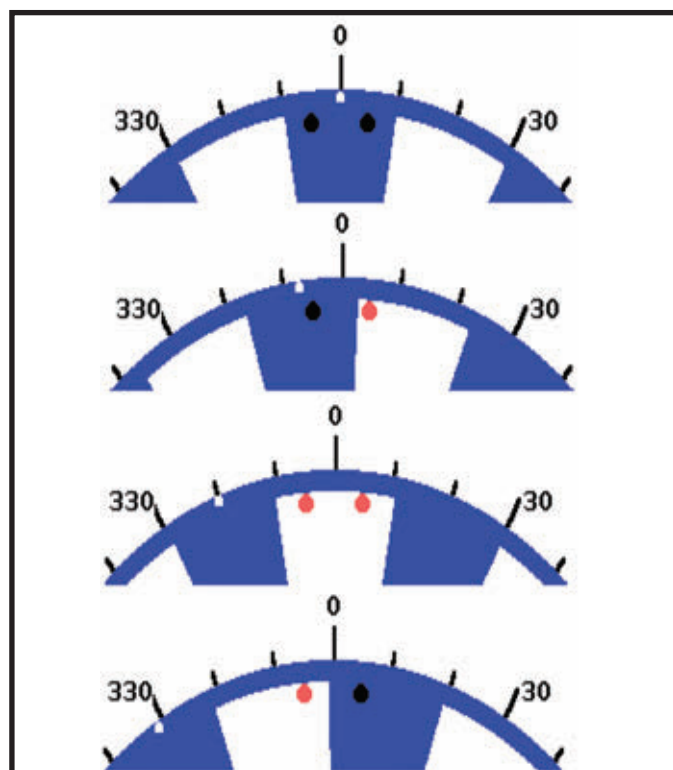
By John Blankenship and Samuel Mishal

Many HMs communicate using a serial protocol (SPI, I²C, etc.) allowing a microcontroller (μ C) to do a lot more work than would be possible with the limited number of I/O pins it has. Another major advantage provided by HMs is concurrent processing. To control multiple servomotors, for example, a μ C would have to constantly generate pulses to keep each motor correctly positioned against a load. A μ C that has interrupt timers can achieve this, but the necessary programming is more complicated. Remember too, that the more tasks the μ C has to perform, the less time it has to dedicate to each process. HMs provide a solution for the above. Once they are told what to do, they do it in the background *concurrently*, freeing the μ C to be able to manage multiple tasks that it may not be able to achieve (simultaneously) by itself. The HMs themselves are often a μ C wired together with some additional circuitry in a small package.

In a recent project, we built a scale model of a space station to experiment with attitude control (using digital PID methodology). An accurate measurement of the heading of the model station was needed. To do this, we needed to measure the amount a disc (modeling the station) rotated (clockwise or counter-clockwise). A very accurate method for achieving this is quadrature encoding. There are numerous websites that sell ready-made quadrature encoders for industrial and other purposes. Unfortunately, besides being very expensive, many of these devices are bulky and are not suitable for most hobby projects. We are going to show a way to create a cost-effective, practical, and versatile Quadrature Encoder helper module.

receiving infrared light from an infrared LED, thus creating a data pair of [00]. If the wheel turns counter-clockwise (CCW), the right sensor will be uncovered (on=1) while the

FIGURE 1. Moving CCW generates these sensor states.



Principles Of Quadrature Encoding

The image in Figure 1 represents a wheel composed of equal sized spokes and openings. The two dots on the vertical spoke represent two infrared sensors. In the current position, both sensors are blocked (off=0) from

TABLE 1. Quadrature state transition sequence.

Moving CCW (left)		Moving CW (Right)	
Both blocked	(00 = 0)	Both blocked	(00 = 0)
Left blocked, Right clear	(01 = 1)	Left clear, Right blocked	(10 = 2)
Both clear	(11 = 3)	Both clear	(11 = 3)
Left clear, Right blocked	(10 = 2).	Left blocked, Right clear	(01 = 1).
Then repeats		Then repeats	

FIGURE 2. The quadrature encoder simulator.

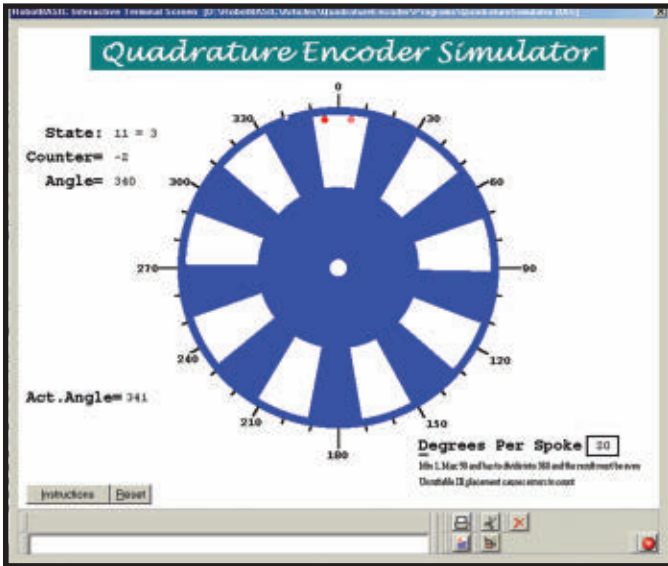


FIGURE 3. A quadrature encoder helper module.

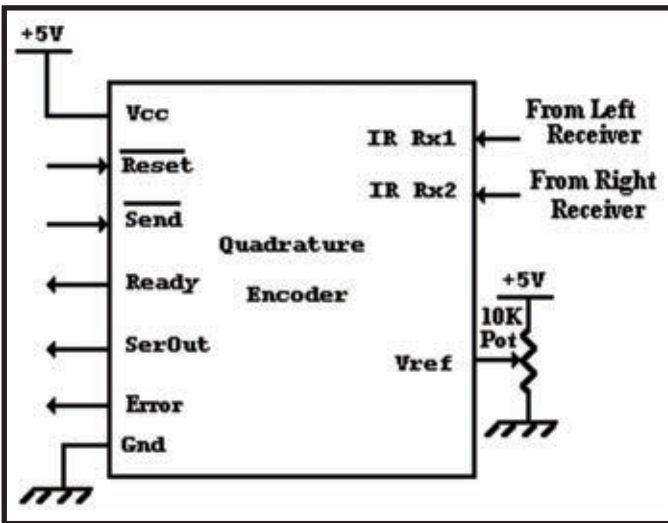
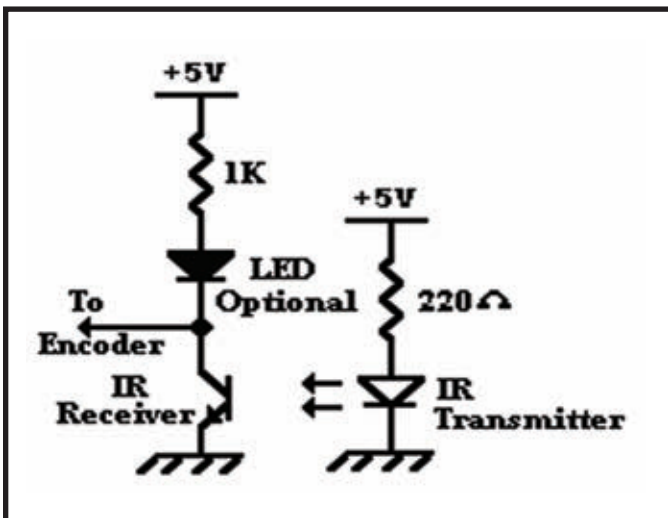


FIGURE 4. Infrared receiver and transmitter circuit.



left one will still be covered by the spoke. So the data pair becomes [01], then [11] as the wheel continues to turn, and both sensors are now under an opening. As the wheel turns further, the next spoke will move to cover the right sensor while the left one is still uncovered [10]. Finally, the two sensors will both be covered again repeating the cycle of four states. These four patterns (states) repeat as long as the wheel is turning CCW. If the wheel turns clockwise (CW), we will get the following four states: 00, 10, 11, and 01 (see Table 1).

To help in visualizing and understanding quadrature encoding, we wrote a simulation using RobotBASIC that makes it easy to see exactly how the patterns are generated in real time (see Figure 2). You can download a free copy of RobotBASIC and the Quadrature_Simulator.BAS program (and many other programs) from www.RobotBASIC.com.

It is relatively easy to program a μC to monitor the pulses from the two infrared sensors to maintain a count of how many state transitions have occurred (positive is CW and negative is CCW). This count can be used to calculate (see later) the wheel's position relative to its initial starting point. However, if the μC is busy doing other tasks like sending serial commands to a motor-control HM, it might miss some state transitions, especially if the wheel is turning fast. If we had an HM that counted the transitions all by itself, the μC would communicate with it whenever it needed to know the count and thus no transition will be missed.

The encoder only returns a count of the movement and direction (negative or positive values). It is up to the system that utilizes the encoder to convert the count into an absolute value of movement meaningful for the situation at hand. For example, in Figure 1, since we know the size of the spokes, we can calculate how far the wheel moves. Since the movement from one spoke to another (a spoke and opening) produces four states (i.e., a count of 4) then:

$$\text{Amount_Of_Movement} = \text{Count_Of_State_Changes} \times (2 \times \text{Spoke_Size}) \div 4$$

In Figure 1, the spoke size is 20° , so for every state transition we know the wheel has moved 10° ($1 \times 2 \times 20 \div 4$). If the wheel rotates to create 33 state transitions, we can infer that the wheel has moved 330° . If the movement is CCW, the value would be -330° (-33×10). To maintain the value within 0° to 360° , (as in a compass heading), we need to add 360° resulting in a heading of 30° . This is all performed by the μC , not the HM. The HM only returns the count of state transitions. With a 20° spoke and correctly placed sensors, the **resolution** of the system is 10° ($2 \times 20 \div 4$), giving us $\pm 5^\circ$ of precision. You can see how this works in real time using the simulator. You can observe the precision limit by comparing the calculated angle value (per above logic) and the actual angle of the wheel as you move it. You will notice that the actual angle may be within $\pm 5^\circ$ of the calculated angle.

For a finer resolution, a smaller spoke size should be used. The simulator mentioned above allows you to

experiment with different sizes. However, changing the size may cause the positioning of the sensors to become inappropriate and you will have to reposition them. Additionally, the smaller the spoke, the closer the sensors will have to be (the sensors need to be separated by 1/2 the size of a spoke for optimum results). In real life situations, if the sensors are too close they may interfere with each other. Also, it may be necessary for your project that you position the sensors some distance apart due to other considerations. The simulator will allow you to experiment with various placements to see how they work. As long as the relationship of the sensors to the spokes remains the same, the sensors can be placed on separate spokes if you need to separate them to prevent interference.

If you position the sensors incorrectly, the wrong sequence of state transitions will occur. The simulator indicates this by lighting a simulated LED, and you can observe the value of the state that caused the error and how it affects the count. Some arrangements of sensors may not cause an error but will cause a count in the wrong direction. The simulator can help you understand how misplaced sensors can cause these types of problem and this, in turn, helps you place your real sensors correctly.

Designing The Quadrature Encoder HM

The HM we would like to have is shown in Figure 3. When a μC wishes to read the current quadrature count, it sets the *Send* pin to 0 (active low) and monitors the *Ready* (active high) pin. When the *Ready* pin goes high, the μC sets the *Send* pin high (to prevent another send later) and waits for the asynchronous serial data (the state transition count) to come on the *SerOut* pin from the device. The *Error* pin (active high) can be monitored to determine if an illegal state transition has occurred, although generally this can be ignored, as errors should be rare for reasonable wheel speeds. The *Reset* pin (active low) can be set to low whenever resynchronization is required (setting the count back to zero).

The device also has two additional pins that receive analog voltage levels from infrared receivers (pins *IR Rx1* and *IR Rx2*). These voltage levels will be compared to a reference voltage established using a rheostat tied to the *Vref* pin. This will be used to determine when a receiver is blocked and when it is active (see later for how this is done).

The IR receiver and transmitter circuits are shown in Figure 4. The IR receiver acts as a variable resistor that varies with the amount of infrared light falling on it. The circuit behaves as a voltage divider that gives a variable voltage depending on the

```
' {$STAMP BS2}
' {$PBASIC 2.5}
'=====
ResetPin    PIN 0
Interrupt   PIN 1
ReadyPin    PIN 2
SendPin     PIN 3
ErrorPin    PIN 4
LeftIR      PIN 5
RightIR     PIN 6
OldIRS      VAR Nib
IRS         VAR Nib
dA          VAR Nib
dAngle      VAR Word
HdAngle     VAR dAngle.HIGHBYTE
LdAngle     VAR dAngle.LOWBYTE
'=====
Main:
  GOSUB Initialize
  DO
    GOSUB Quadrature
    IF ResetPin = 0 THEN GOTO Main
    IF Interrupt = 0 THEN
      HIGH ReadyPin
      GOSUB Quadrature
      LOW ReadyPin
      SEROUT SendPin,84, [HdAngle,LdAngle]
    ENDIF
  LOOP
END
'=====
Initialize:
  OldIRS.BIT0 = RightIR
  OldIRS.BIT1 = LeftIR
  dAngle = 0
  LOW ReadyPin
  LOW ErrorPin
RETURN
'=====
Quadrature:
  LOW ErrorPin
  dA = 0
  IRS.BIT0 = RightIR
  IRS.BIT1 = LeftIR
  IF IRS <> OldIRS THEN
    dA = 3
    SELECT OldIRS
      CASE 0
        IF IRS = 2 THEN dA = 1
        IF IRS = 1 THEN dA = 2
      CASE 1
        IF IRS = 0 THEN dA = 1
        IF IRS = 3 THEN dA = 2
      CASE 2
        IF IRS = 3 THEN dA = 1
        IF IRS = 0 THEN dA = 2
      CASE 3
        IF IRS = 1 THEN dA = 1
        IF IRS = 2 THEN dA = 2
    ENDSELECT
    OldIRS = IRS
  ENDIF
  SELECT dA
    CASE 1
      dAngle = dAngle-1
    CASE 2
      dAngle = dAngle+1
    CASE 3
      High ErrorPin
  ENDSELECT
RETURN
```

LISTING 1. Quadrature control.

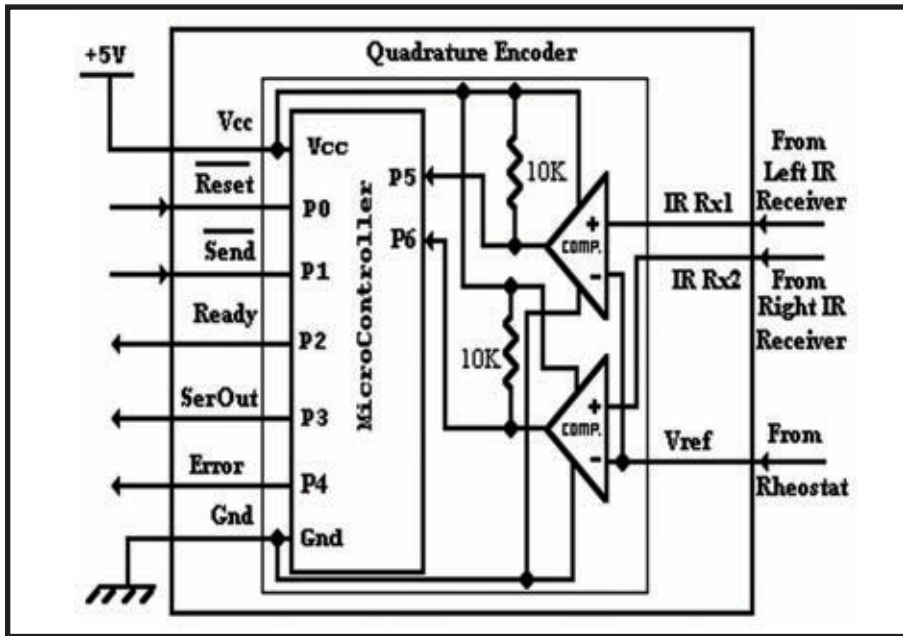


FIGURE 5. Detailed design of a quadrature encoder helper module.

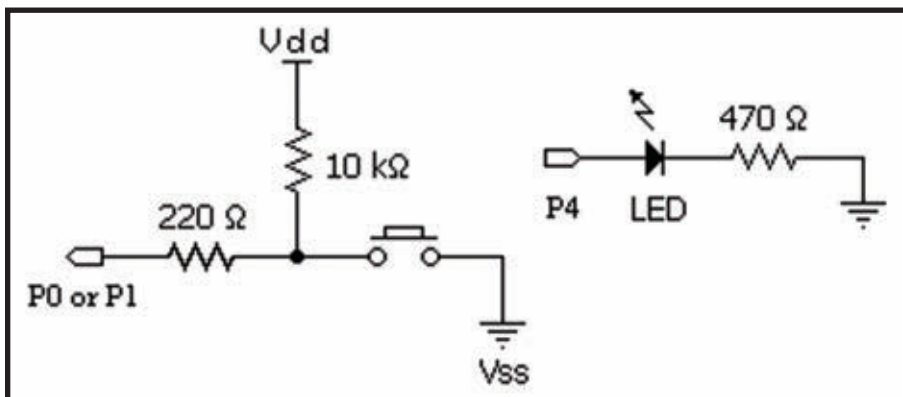


FIGURE 6. Wiring P0, P1, and P4 for manual testing of the HM.

amount of IR light. The reference voltage serves to establish a cutoff point for when the receiver will be considered as blocked and when it is active (active is low resistance, i.e., lower voltage). The optional LED is not necessary for the operation of the circuit but will serve as a visual indicator during debugging.

Building The Quadrature Encoder HM

Figure 5 shows the detailed design of the desired HM. The comparators (e.g., LM339) are used to compare the voltage from the IR sensors with the voltage reference to generate a TTL level digital signal for on/off. As the system is designed, the signal reaching the μC will be low (0V) if the sensor is not blocked and high (+5V) if it is.

Instead of a microcontroller as shown, we could use an FPGA (clocked counters, flip flops, shift registers, and so on). However, powerful and cheap microcontrollers that can be programmed to do serial communications make it very

simple to achieve the desired operations mentioned above with a minimal effort.

The ideal microcontroller for the job should be able to perform interrupt driven serial communications. It should also have an external interrupt capability to serve as the *Send* pin (this way we would not need a *Ready* pin). The speed of the controller (MIPS) should ideally be high. An example of a very suitable controller is the SX available from Parallax, Inc. Besides being very cheap, the SX has all the desired characteristics, and there are C and Basic compilers available for it (though for this job assembly is more suitable).

For the purposes of this article, however, we will use the Basic Stamp 2 (BS2) available from Parallax (www.parallax.com) because it is very popular, well supported, and easy to use. Besides, it has many advantages during the research and development stage. It is easy to program and has very nice debugging features that can be invaluable when you are testing new ideas. Once the details of the design have been honed and tested, the BS2 can be easily replaced with any other (more powerful and cheaper) μC like the SX.

Programming The Microcontroller

The PBasic program in Listing 1 keeps track of the quadrature states and performs the logic described earlier. The count is sent on the serial line high byte (MSB) first using a communication rate of 9600 baud with eight bits, one stop bit, and no parity. The BS2 keeps a count of the transitions in a two-byte register. This effectively gives a possible count range of -32513 to +32512. You should take this into account when using the controller and issue a Reset if the count is about to be exceeded. The negative values are in 2's complement.

Since the BS2 performs 4,000 instructions per second and since the two bytes (16+4=20 bits @ 9600 baud with one stop bit and one start bit) will take 2.1 ms to transmit, The maximum states transition rate must not exceed 1 per ≈ 2.5 ms. If we assume 5° spokes for good precision, (2.5° resolution and $\pm 1.25^\circ$ precision), then to be able to use the BS2 based HM the wheel must not rotate any faster than 2.5° per 2.5 ms. This is 1,000° per second, or 2.8 revolutions per second. As long as the wheel speed does not exceed this rate, the BS2 based HM will be very suitable.

Obviously, if a μC with interrupt driven serial communication and high clock speed is used the allowable

revolutions per second can be much higher. Also, if you select a coarser resolution the revolutions can be proportionally faster. Another option is to use the BS2SX module, which is considerably faster than the standard BS2 (instructions per second, as well as serial communication rate) and runs the exact same programs.

Testing The HM Alone (without a controller)

Here is where the convenience of using the BS2 outweighs its shortcomings. To test the final design, you do not need to use a μ C to control the device and view the count. Just replace the line: `SEROUT SendPin, 84, [HdAngle, LdAngle]` with the line: `DEBUG Dec dAngle, CR`. This will display the states transition count on the debug window of the BS2 development IDE whenever the *Send* pin goes low, instead of sending it on the *SerOut* pin (remember that negative numbers are in 2's complement format). To manually control the device and to view the status of the *Error* pin, wire the P0, P1, and P4 pins of the BS2 as shown in Figure 6.

To create a quadrature wheel, you can print the graphic in the simulation program and glue it on a piece of cardboard and cut out the white areas and around the wheel rim; IR transmitter/receiver pairs can be purchased from many electronics stores.

Final Thoughts

The quadrature encoder helper module we have designed is easy to use and is a nice addition to the family of helper modules available on the market today. Our HM performs only one quadrature encoding (one pair of sensor inputs) and the count is only two bytes long (16 bits). A commercial module may have multiple encoders and keep a 32 bit (or 64 bit) count for each. Another possible addition that can be of great utility is to calculate the speed and acceleration along with the count. A μ C using our HM can differentiate (digitally) the count once to get speed and

twice to get acceleration, but why not have the HM do this? If the circuitry is fast, there would be no time constraints in doing the necessary calculations and sending the information to the user μ C. We used infrared sensors, but Hall-Effect transducers could be used instead, in exactly the same manner. Also, we illustrated the principles of quadrature encoding using a wheel (rotational movement), but the principle applies equally for linear movement.

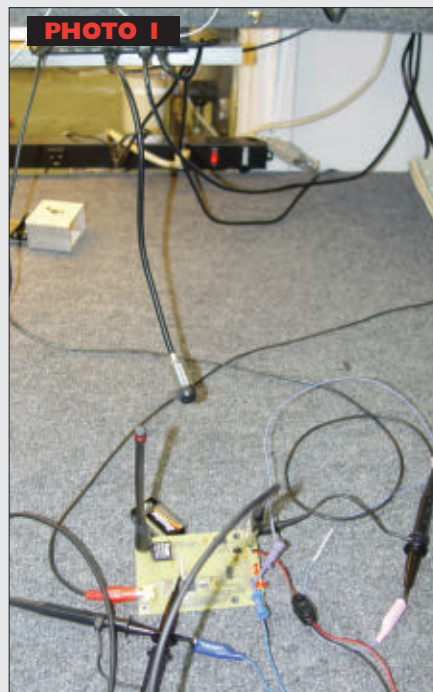
The important thing is that helper modules — whether you buy them or build your own — provide distributed processing capacity that enables reduced programming complexities, faster completion times, and projects that might otherwise be impossible with just one controller or processor. **SV**

A Shop Trick

By Jim Miller

The other day, I was digging through a box of “good stuff to keep but don’t know why.” I ran across a USB fan on a flexible goose neck. It had been given to me as a Christmas present years ago. I used it for a while but its utility and novelty wore thin and it eventually exited my computer bag as clutter. Anyway, I pulled it out of the box to toss it and set it on my electronics bench. I have a powered USB hub there so I plugged it in just one more time. The fan actually worked great, so I left it there spinning away. It has soft blades so if you stick your hand in it there is no injury. Later that same day, it cleared the solder smoke out of my face. Last night, I had a voltage regulator out of its case and it was getting a bit warm so I just swung the fan around and it fixed the problem. I’m sure it will come in handy for other things too, but I was excited that I saved it from the land fill and think it’s found a new life on the workbench.

It’s kind of challenging to pick it out in the photos from all of the test leads, but in Photo 1 you can see it plugged into the hub. Photo 2 is just a bit closer to the action in the cooling mode.

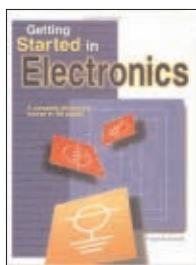


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ROBOTICS

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by Grant Imahara

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Forbidden LEGO

by Ulrik Pilegaard / Mike Dooley

Build the Models Your Parents Warned You Against.

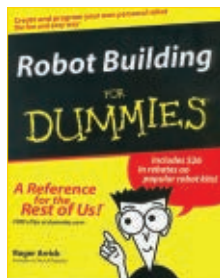
Forbidden LEGO introduces you to the type of free-style building that LEGOs master builders do for fun in the back room. Using LEGO bricks in combination with common household materials (from rubber bands and glue to plastic spoons and ping-pong balls) along with some very unorthodox building techniques, you'll learn to create working models that LEGO would never endorse. **\$24.95**



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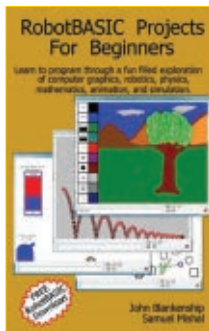
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by John Blankenship, Samuel Mishal

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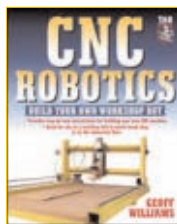
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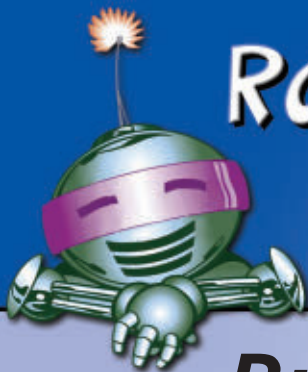
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ROBOTICS RESOURCES

Tune in each month for a heads-up on where to get all of your "robotics resources" for the best prices!



Building a Robot? Don't Forget LEGO!

Take the Wayback Machine to about 2001 and you'll see thousands of eBay sales, hundreds of websites, and dozens of books on LEGO robotics. While the giddy heyday of the LEGO robot has come and gone, it's still a viable — and fun! — platform for learning all about desktop automatronics.

What's more, publishers keep coming out with new and improved books on the subject, and the websites devoted to robotics and LEGO continue to be an active, if not mature, community of like-minded enthusiasts.

In this month's column, we'll look at the current state-of-the-art in LEGO robotics — why you might want to consider using a "toy" to construct your next automaton, what you need to get started, and where to find some great deals.

At the Core: The Mindstorms Controller

There are many different ways to construct a robot using LEGO parts. One is to merely cobble together the bricks, plates, beams, and other common LEGO components and combine it with some gear or servo motor you found in your junk bin. Add a BASIC Stamp or some other microcontroller board, mix in some batteries, and there's your bot.

For the purposes of this column, I'm talking about LEGO robots built around the LEGO Mindstorms controller: an electronic device that is

sold as part of the LEGO Mindstorms set. The set is available for purchase at many toy stores, electronics outlets, Amazon, and other online retailers.

There have been several iterations of LEGO Mindstorms since its introduction 10 years ago. The first was the ubiquitous yellow programmable "brick" controller known as the RCX. The RCX was part of a now-discontinued product — the LEGO Mindstorms Robotics Invention System (or RIS). Today, LEGO sells a second generation package called the LEGO Mindstorms NXT. This kit of parts sports an updated programmable brick (the NXT). The basic idea behind the NXT brick and its older RCX brother is the same: Create a program on your PC, download it to the brick, then hit the Run button. The controller plays your program and operates your robot. Both the RCX and NXT connect to motors to move the robot around and sensors to detect the machine's immediate environment. The sensors in the NXT package are more sophisticated and varied than those that shipped with the earlier RIS, but you can make perfectly functional bots with either Mindstorms version.

As noted above, you develop your Mindstorms programs on your personal computer. Once created, your program must be compiled and transferred to the Mindstorms brick. On the RCX, this was accomplished using an infrared "tower." The first RCX units came with a tower that required a serial connection to the

host PC. Later versions worked with a USB connection, making them more universal with the latest computers, which lack the nine-pin serial connection common in PCs of the 80s and 90s. The NXT eschews the infrared tower and works with a USB cable that connects the computer to the brick. Though the RCX is no longer made, it's worthy of mention because it's still available for resale on sites such as eBay, and usually at considerable savings. You can often find the RCX brick sold by itself for under \$40 and complete Robotics Invention System packages (including the RCX) for less than \$100. List price for a new Mindstorms NXT is \$199.

Motors and Sensors and Other Goodies

While a robot is not absolutely required to be mobile or even have moving parts, the LEGO Mindstorms is set up for building rolling machines. Both the RCX and NXT controllers support up to three motors. Two are included with the older RCX-based sets; three come with the NXT. You can buy more if you need them. The motors for the NXT are more sophisticated and incorporate rotation sensors to determine distance of motion (rotation sensors are extra for RCX-based robots and are not part of the RCX motors).

Sensors make the robot, and this applies to LEGO Mindstorms as well. The sensors packed with RCX kits were rudimentary: just touch (contact



switch) and light. You could buy additional sensors such as the motor rotation types mentioned above, and the simplicity of the RCX allowed you to readily make your own. The NXT's sensors include the touch and light switches as with the RCX, but also have a microphone for detecting sound and ultrasonic, for measurement, and determining nearby movement.

Various third parties sell enhanced sensors for the NXT. For example, HiTechnic Products (www.hitechnic.com) offers an electro optical proximity detector similar in function to the popular Sharp IR units commonly used in homebrew robotics. They also provide color sensors, compasses, accelerometer and tilt sensors, gyroscopes, and a multiplexer that lets you connect up to four NXT touch sensor switches to one input port.

Construction Parts

Both the RCX- and NXT-based robotics kits come with hundreds of LEGO Technic parts. The exact assortment varies between the two packages and, of course, you are free to combine parts from other standard LEGO and LEGO Technic kits, such as the low-cost Bionicle line. You can also purchase specific parts sets from retailers, including LEGO's own online store. The basic building block is the LEGO brick. Bricks are composed of one or more raised "studs" or "nubs" on the top and corresponding sockets on the bottom. The studs and sockets mate, allowing you to stack bricks one on top of another. Plates are like bricks, but are a third the height. Like bricks, plates have one or more studs on the top and corresponding sockets on the bottom, and are made to be stacked together. And finally, beams are specific to the Technic brand of LEGO parts, included with the Mindstorms sets. Beams are bricks with one row of studs and have holes down their sides. Beam variations include L- and hooked-shaped versions which can be used for things such as building robotic grippers.

Assembly parts hold things

together such as attaching one beam to another or a wheel to a beam. Connector pegs are typically used with the LEGO Technic line to attach beams together. Connectors pegs are either 1/2, 3/4, or full – the difference lies in the length of the connector on one side of its middle collar. Some connector pegs are half connector and half axle (see below for more information on axles). Available in a wide variety of sizes and shapes, friction connectors are used with both LEGO Technic and standard parts for such things as connecting beams and wheels.

Other assembly parts include the bushing and the cross axle. For turning parts, there are wheel hubs provided in a number of different sizes; pulleys which are used with string, rubber bands, or O-rings to create a kind of belt; and gears which are used to construct more sophisticated motion transfer mechanisms. Rounding out the parts list are toggle joints, sloped beams, catches, cross blocks, axle connectors, cams, and a raft of others. Check out the LEGO website for more information on LEGO parts. For some good books on LEGO, go to the *SERVO* webstore at

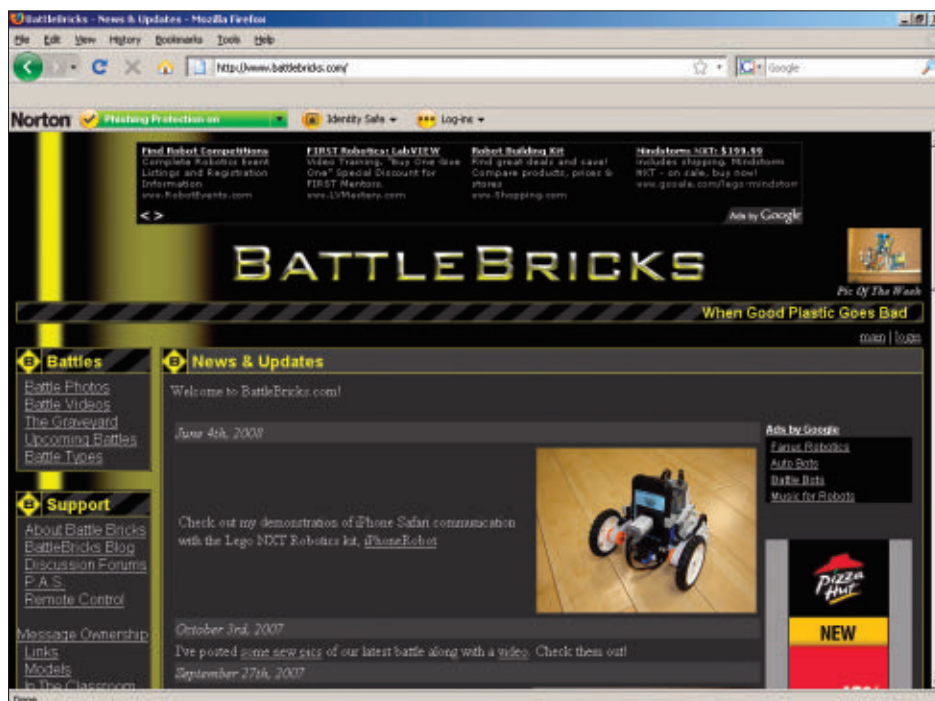
(<http://store.servomagazine.com>).

All these parts are made to snap together. When properly constructed, the snap-together system provides for relatively strong joints. In fact, most "purists" frown on building LEGO mechanisms in any other way, including using fasteners or glues. Still, depending on the type and size of robot you're constructing, you might consider cementing non-movable pieces using ordinary household cement (such as Duco), epoxy, contact, or CA (cyanoacrylate) glues. If you want a very permanent bond, use clear ABS plastic solvent (LEGO parts are made of ABS plastic). ABS solvent cement is available at most home improvement stores, as well as many hobby stores. Another approach is to use hot melt glue. You can use either the standard or low-temperature variety.

Temporary constructions can be taped together using black electrical tape or even strapping tape used for box packaging. Do note that most tapes leave a sticky residue. Acetone can be used to remove this.

Tie wraps are used to bundle wires and other loose items. They consist of a single piece of plastic with

Battlebricks extends the sport of combat robotics to LEGO Mindstorms creations.





a catch on one end. You thread the loose end of the tie through the catch; a ratchet in the catch keeps the tie in place. Most ties have a one-way ratchet that cannot be undone; others have a catch "release" so you can re-use the tie. Tie wraps are available in a variety of lengths, thicknesses, and colors. Get an assortment and use the size that best suits the job. You can use small hardware to attach LEGO pieces together. In most cases, you'll need to drill through the plastic for the screw or bolt. You can use self-tapping metal screws if the plastic material is thick. When using self-tapping screws, be sure to use a drill size slightly smaller than the screw.

You can attach non-LEGO things to LEGO pieces with hook and loop fastener. Get the kind with self-stick adhesive backing. Put the hook material on one piece, and the loop on the other. Press them together for a very strong — but not permanent — bond. Double-sided tape also performs well here, though some versions are made to set up after a day or two, making disassembly difficult. If you want to be able to take your contraptions apart, be sure to use temporary tape.

Programming Languages and Options

LEGO Mindstorms comes with a graphical programming environment that is suitable for both students and adults. With graphical programming, you assemble blocks of programs by selecting actions, connecting them in a way that the robot does something functional like follow a line, retreat when it encounters a bright light source, or reverse direction when it hits an object.

Over the years, a number of alternative programming languages have been developed for the RCX and NXT bricks. For instance, NQC (for Not Quite C) is a text-based programming language specifically designed for the LEGO RCX brick, and allows the construction of fairly elaborate event-based programs (in event-based programming, actions are taken when a certain event occurs such as triggering a touch switch or even the passage of a certain amount of time).

Similarly, ROBOTC and RoboRealm allow you to program the NXT (as well as the RCX and some other popular

educational platforms) and offer a laundry list of useful functions. Some programming alternatives will allow for interactive play. For instance, NXT++ allows you to control the NXT brick using the C++ program, connecting to the NXT with USB or Bluetooth.

Sources

Here are some sources to get you started in the world of LEGO Mindstorms robotics. Because of the breadth and depth of LEGO, this list is necessarily selective. You can find many more online sources for products, add-ons, how-tos, and information by using your favorite Internet search engine.

Amazon.com

www.amazon.com

Best-known as a book seller, Amazon sells books about LEGO Mindstorms, as well as the Mindstorms sets themselves.

Battlebricks

www.battlebricks.com

Combat robots, LEGO style. Includes forums, photos, videos, and more.

FIRST LEGO League

<http://firstlegoleague.org>

Home page of the FIRST LEGO League — an international consortium that promotes engineering (through building LEGO robots) among young students.

HiTechnic Products

www.hitechnic.com

Specialists in third-party sensors for the NXT. Includes optical, gyro, and accelerometer, and color sensors, plus modules for building your own custom sensors.

Hobby Engineering

www.hobbyengineering.com

Full service retailer of LEGO products and books on LEGO Mindstorms.

LDraw

www.ldraw.org

LDraw is an open standard for

LEGO Mindstorms' main page is your first port of call for anything related.





LEGO CAD programs that allow the user to create virtual LEGO models and scenes.

LEGO.com
www.lego.com

The first stop in anything LEGO related. Also see **mindstorms.lego.com** — the specific portal for the Mindstorms product.

LEGO Mindstorms Internals
www.crynwr.com/lego-robotics

Informational page about the inner workings of the RCX brick. Lots of links to external sites that offer additional help.

LEGO Shop
http://shop.lego.com

Online outlet for LEGO products, including spare parts (when available).

LUGNET User Group
www.lugnet.com

Huge user-to-user group on all things LEGO. The site hosts newsgroup messaging, events, LEGO sources, and more.

Michael Gasperi's LEGO Mindstorms NXT/RCX Sensor Input Page
www.extremenxt.com/lego.htm

Support page for popular LEGO Mindstorms book author Michael Gasperi. Includes projects and downloadable program blocks.

Nuts & Volts Store
http://store.nutsvolts.com

Go here for books on LEGO such as the popular *The LEGO MINDSTORMS NXT Idea Book*.

NXT Central
www.nxtcentral.com

Blog and forums all about LEGO Mindstorms NXT.

NXT Motor Internals
www.philohome.com/nxtmotor/nxtmotor.htm

Informational page on how the special NXT servo motors operate. Includes nice cut-away drawings,

torque charts, and step-by-step disassembly pictures.

NXT Programs
www.nxtprograms.com
Resource site of NXT-based projects, including programming code.

NXT Programming Software
www.teamhassenplug.org/NXT/NXTSoftware.html

Informational site provides a handy table comparing various programming languages and products for the LEGO Mindstorms NXT.

Nxtasy.org
www.nxtasy.org

Web community specifically for the LEGO NXT enthusiast. Forums, blogs, and links.

Roboteq
http://roboteq.sourceforge.net
Graphical programming environment for LEGO Mindstorms. Open source.

Target
www.target.com
Retail stores and online site. Both offer great deals. Make it a habit of

regularly checking the website for clearance items.

Techno-Stuff
www.techno-stuff.com

Makers and sellers of RCX- and NXT-compatible third-party sensors. Products include an IR flame sensor, UV flame sensor, and passive infrared motion sensor.

Vernier and LEGO Mindstorms
www.vernier.com/nxt

Product page for Vernier software (such as Logger Pro) and interface products used for testing, measurement, and education, with specific how-to and advice on integrating with the LEGO Mindstorms NXT.

Wal-Mart
www.walmart.com

Retail stores as well as online, Wal-Mart is arguably the world's largest seller of toys. Be sure to check out their clearance items. **SV**

CONTACT THE AUTHOR

Gordon McComb can be reached via email at robots@robotoid.com

HiTechnic Products offers numerous enhanced sensors for your NXT-based robot.



Beginner ELECTRONICS

by William Smith

Introducing AtomNano

Many years ago, Basic Micro, Inc., introduced the Basic Atom modules to compete with other popular modules. Basic Micro still offers the Basic Atom modules and the interpreter chips they're based on. Now, they are introducing the AtomNano which is similar to the PICAXE and it has many additional features. I've only seen a beta version of the chips; they are very similar to the Atom interpreter chips but at a lower cost. Let's take a quick look at the new AtomNano.

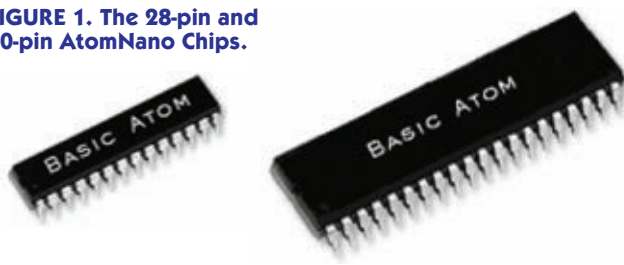
them to opt instead for the lower cost PICAXE chips. This forced the previous Atom user to learn all the new commands the other system required. The AtomNano was developed to answer the call for a lower cost chip without having to leave the Atom development environment. Plus, it runs faster than 4 Mhz. Basic Micro has maintained the free download of the Basic Compiler software. All of this should be available by the time you read this article so look for the AtomNano at **BasicMicro.com**.

The reasons why I like these chips and why I recommend you take a look at them is they offer more program space, RAM space, and easy access to the internal features of the PIC they are based on. The AtomNano chips in Figure 1 represent the 16F887 (40-pin) and the 16F886 (28-pin). The AtomNano chips rely on the internal 8 MHz oscillator offered in these chips so they will be a little slower than the Atom modules. Speed can be important when you are trying to drive servos and read sensors. In other cases though, the

For those not familiar with the Basic Atom, it is based on the Microchip PIC development system that has been around for a while. Basic Micro has created some new development boards for the AtomNano, which will be offered in a 28-pin and 40-pin version just like the Basic Atom interpreter chips were. However, the AtomNano will include an 18-pin version. This comes at a really good time because I see many robotic users struggling with large programs. If you want a full featured Basic compiler and low cost chip that doesn't need a lot of extra hardware to program it, then you should look at the AtomNano chips.

As mentioned, the Basic Atom comes in module form. (In fact, they have Atom modules that are pin for pin compatible with the BASIC Stamp). At some point though, you may want to use more than one microcontroller in a design and paying \$50 per module can get expensive. (This is where the interpreter chips offered at \$20 each come in handy). To some people this was still too expensive, causing

FIGURE 1. The 28-pin and 40-pin AtomNano Chips.



effectiveness of the command set is important so the Atom Basic compiler has a huge list of commands including one for driving an LCD module. The smaller Atom modules that are compatible with the Parallax BASIC Stamps are shown in Figure 2. Because of this compatibility, these modules can be used in numerous development boards originally designed for the Basic Stamp 2.

Both offer analog-to-digital converters (ADCs): the 24-pin version has them as extra pads underneath that you need to solder to, but the 28-pin version brings these connections out to leads you can easily access in a breadboard setup or development board. Having an ADC is a great resource that gets carried over to the AtomNano chips and the Atom Basic compiler has a single command to handle the ADC measurement. The AtomNano makes it very easy to write and run code to, for example, read a Sharp infrared distance sensor with analog output using an ADC channel.

Programming

Programming the Atom modules simply requires a serial connection to your PC. A USB-to-serial adapter will also work. Programming the AtomNano chips takes a little more effort. The PC serial port signals at +12V to -12V, or +5V to -5V on many laptops. To convert this type of signal to a TTL or 0V to 5V digital signal (so the AtomNano chips can read them) requires a level shifter. You can do that with a transistor circuit as shown in Figure 3 or you can use an RS-232 level converter chip. The smaller Atom modules have this transistor circuitry built in while larger versions (that use leaded devices instead of surface mount) use the RS-232 converter chips. To support these new AtomNano chips, Basic Micro developed their own USB-based programming adapter instead of using RS-232. The board is designed to plug into a breadboard for communications and power. Just connect a matching USB cable from your PC to the board and you have a programming connection. Figure 4 shows the Basic Micro USB programmer board next to an RS-232 programmer.

Software

As mentioned, the Atom Basic compiler is free to

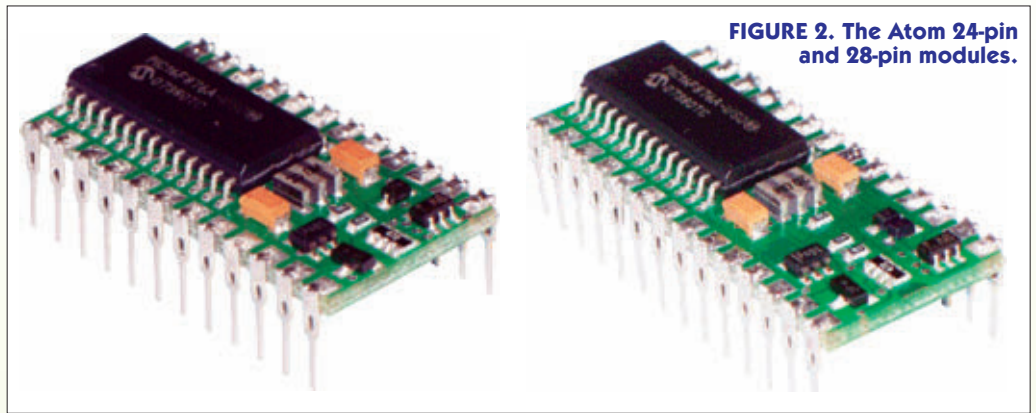


FIGURE 2. The Atom 24-pin and 28-pin modules.

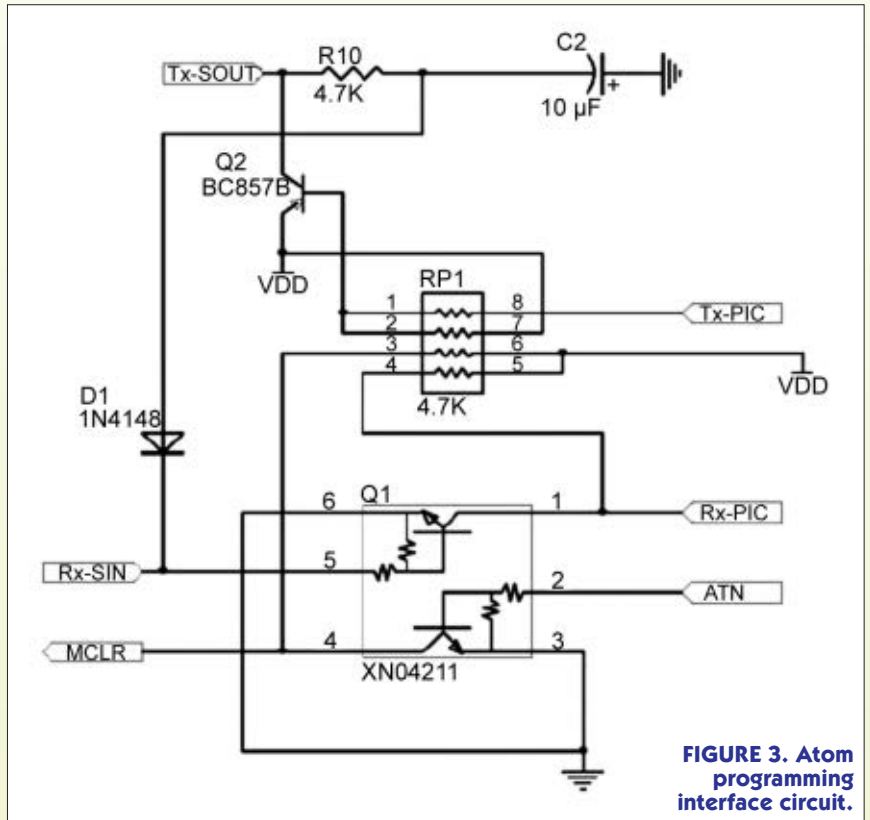


FIGURE 3. Atom programming interface circuit.

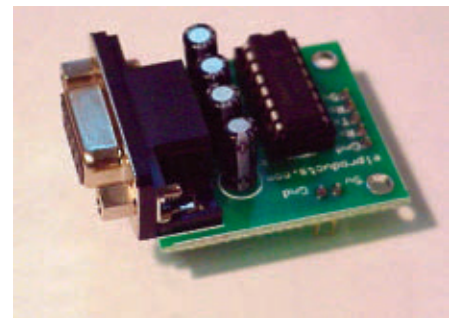
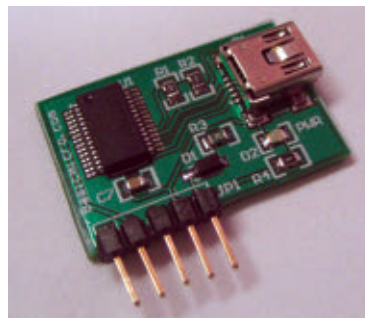


FIGURE 4. USB and RS-232 programming adapters.

download from the Basic Micro website. When you install it, you get a full IDE interface (Figure 5). Also included is the debugger software that lets you step through code and watch your variables change as the program runs. The program is actually running in the chip — not being

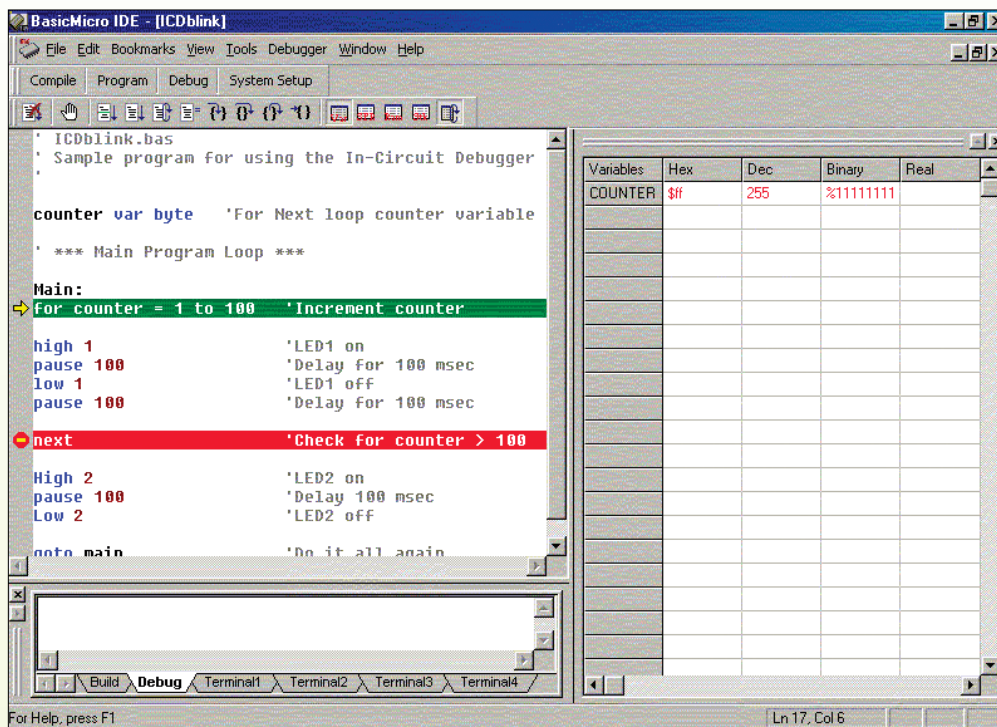


FIGURE 5. The Basic Atom IDE.

simulated — so this slows down operation a little. However, this is priceless for debugging code. Because the commands are similar between the Basic Stamp, PICAXE, and Basic Atom, no matter what path you take you end up with a lot of sample code to choose from — as long as you don't mind investing a little time modifying it to work for your application.

I like the fact I can debug the code in the actual chip rather than a simulation, although they both have their advantages. The programming is also handled through the IDE. Similar to the Stamp setup, the Basic Atom has a one-click compile and program ability. This makes it great for anybody just getting started. It also saves money because you don't have to buy a separate programmer or other special hardware to program and debug. The Atom debugger lets you look at the internal

registers of the PIC, as well. (This is more of an advanced topic but having that extra capability is a very nice feature when you get into those tougher projects.)

Books

Having access to support is very important, so check out the user forums to get answers for your questions. I also like to have a book or two for my own reference. The Basic Atom has several online manuals available in PDF format. One book in particular, titled *"Programming the Basic Atom Microcontroller"* by Chuck Hellebuyck, is written for the beginner and takes you through several projects that teach you easy things like flashing LEDs

and then moves on to harder topics like driving LCD modules and using interrupts or timers. This book is available from the *SERVO* online bookstore at <http://store.servomagazine.com>.

Another book I recommend is *Programming the Microchip PIC with MBASIC* by Dick Smith (no relation). This book is written about the Basic Micro MBASIC compiler for Microchip PICs, but the Basic Atom software was developed from this compiler so all the information applies. Dick has numerous projects and explains them well, though some go beyond the capabilities of the Basic Atom.

Development Boards

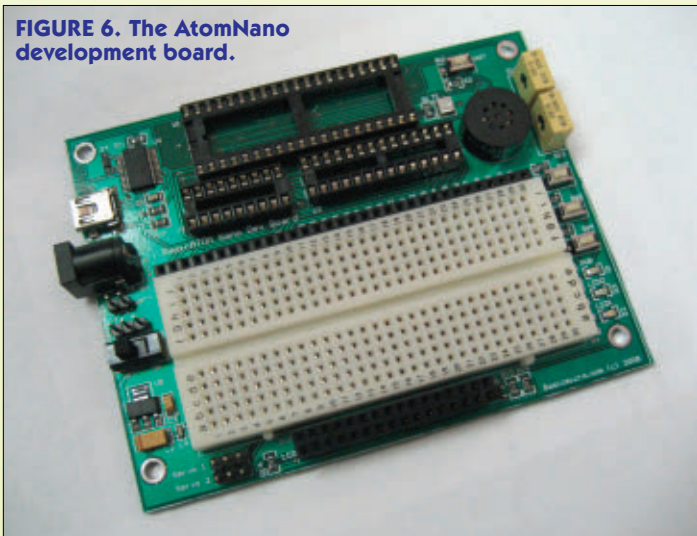
Basic Micro is developing some new boards for the AtomNano. Figure 6 shows an early picture of one that supports all three chips. The board has a couple switches, LEDs, potentiometers, a speaker and a socket for plugging in a parallel LCD module. The board has two servo connectors which should make it a great robotics board. There is a large breadboard area, as well.

The board has the USB programming adapter built in so all you need to do is supply power and plug the cable in to program your AtomNano chip. I'll be using a few of these boards for some projects in future columns.

Conclusion

If you've been thinking about getting into robot programming, the AtomNano is your new low cost path. It offers all of the higher level commands you need, easily expressed through a Basic language and supported by a great development environment. If you can't wait for my future columns, visit the Basic Micro website, place an order and get some code running at home! **SV**

FIGURE 6. The AtomNano development board.



Mobile Robot Driven by Miniature Onboard Motors for Cardiac Intervention

A paper presented at the 34th Annual Northeast Bioengineering Conference written by P.Allen, N. Patronik, and C. Riviere of The Robotics Institute at Carnegie Mellon University, and M. Zenati of the Division of Cardiac Surgery at the University of Pittsburgh describes the development and construction of a mobile robot driven by miniature ultrasonic piezoelectric motors (SQUIGGLE motors) for minimally invasive cardiac therapy. The robot design extends upon previous prototypes of the HeartLander miniature mobile robot that moves in an inchworm-like fashion.

The HeartLander OMNI (Onboard Motor Navigational Instrument) has been developed to reduce tether stiffness by utilizing small onboard SQUIGGLE motors which would result in more efficient turning capability by eliminating the drive-wire mechanism from the tether of the robot. The development of the robot allows for increased turning capability and higher traction during locomotion, and represents the first step in designing a wireless mobile robot for cardiac therapy. The lifetime specification for the SQL-I.8 SQUIGGLE motor has been increased to >1 million cycles.

The robotic design was developed as a proof of concept to demonstrate mobility on the cardiac surface. Construction of the system included motor selection, body design, and development of the control system. The paper presents the design of the robotic platform and preliminary testing results in vitro.

A copy of the paper is available for download (with permission from CMU) at www.newscaletech.com/doc_downloads/CMU-Heartlander-NEbioengConf2008.pdf.

Robotic Grape Vine Pruner

Just recently, in the agriculturally rich central valley of California, Vision Robotics planned to test a prototype robotic device for the pruning of grape vines. "It's a highly skilled job," says CEO Derek Morikawa. "The manner in which you prune the vines affects the quantity and quality of the grapes you get in the coming season."

Vision Robotics already builds a number of robots that harvest apples and oranges, but this is its first foray into an area that requires significantly more precision and mechatronic expertise. The \$150,000 robot combines computer software, machine vision, and robotics to perform the work.

The two-ton prototype is 15 ft long, 10 ft tall, and 9 ft wide. It covers the length of a six foot vine in about 45 seconds (Morikawa estimates the production version will move three times as fast). Its cameras take pictures of the vine and (in real time) an on-board computer converts that information into a graphical representation of it. The computer model then transmits instructions to the robotic pruners. "We developed an engine that incorporates different rules about pruning the vine," says Morikawa. "Different growers have different rules based on age of the vines and the varietal of grape."

The prototype has to be as tall as it is because some trellis

methods and stakes can be as high as eight feet tall, Morikawa explains. The time it takes to do the imaging dictates the length of the prototype. "The cameras are six feet ahead of the robotic arms," says Morikawa, "so we can see an entire vine before we make a decision about pruning." The size is typical for agricultural equipment, he says.

Tests are scheduled to take place in Lodi, CA in a number of vineyards with flat ground. Morikawa says his company will work on a version for pruning grape vines on steep hills such as those in Tuscany in the future.



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Then and NOW

ROBOTS: FROM NANO TO TINY TO SMALL

by Tom Carroll

Electro-Biological Intelligence

I take a lot of magazines — most to keep abreast of a broad range of technical subjects — but I rely on *Forbes* for business and financial articles, at least until the November 12th issue. An article in the issue, "Interview With A Cyborg" by Courtney Myers caught my eye. I had heard bits and pieces about Kevin Warwick's experimentation with cybernetics and the implantation of a 100 electrode array that he had in his arm back in 2002. (Figure 1 shows

Warwick with his 'bionic' hand controlled by the electrode implant.) As the "world's only cyborg," he has been able to control an artificial hand and an electric wheelchair from brain impulses picked up by the electrodes and transmitted through a computer to these devices. What interested me most was his little robot, Gordon, that he and fellow university professor, Dr. Ben Whalley, head of the School of Pharmacy unveiled that past August. This robot is nothing short of amazing with a most unique brain (see Figure 2).

Gordon's brain is not a RAM or ROM coupled with a microcontroller/processor, but fetal rat neurons in

what is similar to a very small Petri dish, connected to an array of 60 electrodes sprouting out the edge. As in the electrode array Warwick has in his arm, these cells — immersed in a nutrient-rich and oxygenated saline solution — emit faint electrical 'signals' that can be used to control the robot. The cell culture is remote from the actual mobile robot and is interconnected by an RF link. The cells need to be temperature controlled and cannot withstand the jostling of a mobile robot. No computer intelligence is used; the learning and control is strictly by the rat neurons.

These cells mature in a week or so, and begin to actually divide and grow from 50,000 to

FIGURE 1. Warwick and hand.



FIGURE 2. Whalley examining the rat neuron "brain."



150,000 cells or more. The neurons begin to form interconnections within 20 minutes and establish an elaborate network within a week after the growth starts.

Gordon has a suite of four pairs of ultrasonic transducers to detect obstacles — the same Devantech SRF08 ultrasonic rangefinder modules many robot experimenters use on typical robots. The modules are pinged every 250 ms and map the room in much the same way as maze-running robots.

What is more unique is that the cell mass acts as a simple brain and is able to learn the boundaries of its pen and not repeatedly bang into the walls. Yes, the sonar units locate walls and other boundaries, but the brain actually decides what is possible to traverse and what is not, or where the walls are and are not. This learning process is accomplished with just a small mass of neuronal cells.

Starting out with a new batch of cells, the robot may bang into the enclosure's walls every minute or so, but after a few weeks, Gordon avoids the walls entirely. After two hours of controlling Gordon, the brain must be placed back in a warm incubator for a while; the lessons learned are still retained. The brains live for about three months before becoming inactive through aging, as the needed voltage stimulation becomes too high to be effective. Warwick hopes to add speech recognition inputs so the robot can go right or left on spoken command.

Powering Small Robots

Most robotics researchers have not used living neurons and brain tissue as did Warwick with his little Gordon, but have gone the route of microprocessors and microcontrollers for control of their experiments. Microprocessors may



FIGURE 3. Fraunhofer-Institut camera pill.

not have the number of active neurons of animal brains, but they have over a hundred million transistors. Simple Flash drives the size of your fingernail have the equivalent of many billions of transistors. The complexity of these chips would have been unthinkable in the 1960's. Processing power of this magnitude would have been a robot designer's dream. Even today's less complex microcontrollers have astounding 'power' compared with earlier robot circuits and they control 95% of experimental robots built these days.

Nanobots are the dreams of doctors and experimenters worldwide. They are perceived to be so small that their very bodies are not much larger than individual memory cells in chips or sub-nanometer discrete transistors. With this in mind, future nanobots must rely on even more microscopic circuitry than is available today to have any sort of intelligence, and even this does not allow for propulsion and steering motors, sensors, manipulators, programming,

and vision systems for the little robots to be useful for any tasks.

The Need for Smaller Robots

Back in 1966, the premise of the film *Fantastic Voyage* was that the only way to save an important physicist with an inoperable blood clot in his brain was to miniaturize a submarine and its crew, and send it in to do the work.

Fantastic Voyage may have seemed a bit silly to anyone with any sort of scientific background, but it did highlight the desire the medical community had to want to see inside a human body. X-rays, endoscopes, and later CT scanners, MRI machines, and PET scanners do allow amazing non-invasive views of the interior of a patient's body, but do not satisfy a medical doctor's needs to see the very tiniest sections of a body such as the vascular system or the interior of various organs.

Newer surgical instruments such as the da Vinci Surgical Robot allow surgeons to perform delicate surgery with minimal invasiveness, but the need is still there for something or some method to allow imaging and manipulation in very inaccessible areas of the body. Crawling inside a patient's body is about as possible as sending a nano-sub with a doctor aboard down a vein to treat a patient's ailment.

Mini Surgical Assistant Robots

Pills that contain what can be described as a "miniature color television station" have been used by doctors for several years. These pills contain a color TV camera, a series of LED lights, and a transmitter to send photos to a recorder outside the body. The devices can photograph the entire



FIGURE 4. Robot Minidoc.

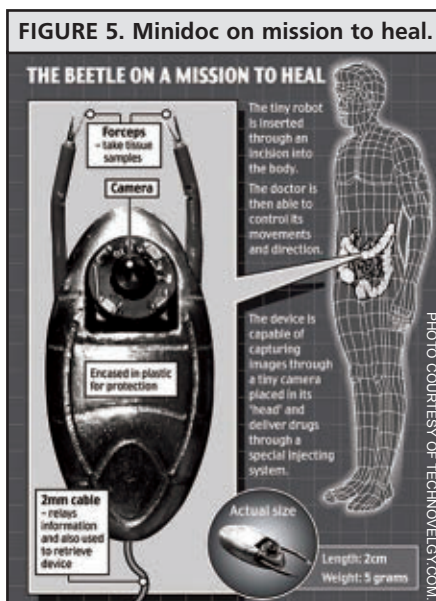


FIGURE 5. Minidoc on mission to heal.

ROVING ROBOT CAMERA

The radio-controlled robot can roam the stomach or abdomen, acting as an extra eye for surgeons (schematic only, not to scale)

Grooved tubular wheels grip slippery surfaces, allowing the camera to be steered remotely

Enclosed motor and universal joints

Retractable biopsy needle

CAMERA

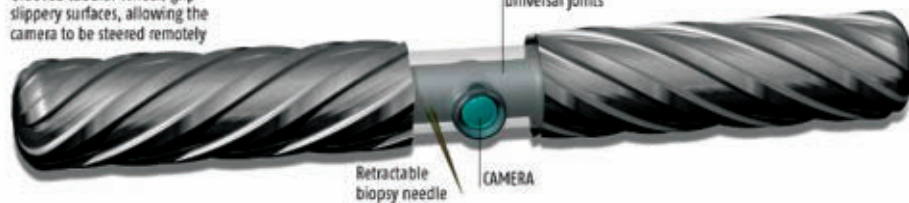


FIGURE 6. Remotely-controlled medical robot.

gastrointestinal system and possibly detect polyps or tumors inside. They provide a fantastic look at the inside of the body, but do not allow the doctor to actually perform any sort of maneuver, operation, or administration of medicine.

Figure 3 is a photo of a unique camera pill developed by the Fraunhofer-Institut in Germany and manufactured by Given Imaging. In the past, camera pills have been somewhat useless in taking decent pictures of the upper GI tract due to the high speeds that the pills travel from the mouth to the lower stomach. However, this pill has been used in examining the esophagus and the sphincter muscle area at the top of the stomach and is steered and held in place for up to 10 minutes by external magnetic fields from a device the "size of a bar of chocolate."

Two years ago, Japanese researchers at the Ritsumeikan University and the Shiga University of

Medical Science developed a tiny prototype robot that can be inserted through an incision into a patient's body. Figures 4 & 5 show the beetle-sized robot held in a hand and a chart showing how it is used as a medical device within the human body. The tiny robot is about 2 cm (0.79 in) long and 1 cm (0.4 in) across and weighs about 5 grams. Maneuvering the robot is accomplished by varying a magnetic field outside the body to tug the robot in the desired direction, just like the camera pill mentioned previously.

Five different versions were developed to test different medical procedures such as taking tissue samples, capturing images, and delivering drugs at precise locations via an injector. The robot is guided through the body using an MRI image taken prior to the procedure as a reference map. An attached mini (2 mm) cable relays video and sensor information back to a computer. If the robot gets stuck or cannot be maneuvered by the magnetic fields, it can be pulled out by the cable.

In early '06, Dmitry Oleynikov, M.D., of the University of Nebraska Medical Center unveiled a unique mini robot that he felt might allow doctors on the ground to perform surgery on astronauts in space. Again, this robot is not like the da Vinci System but an actual robot to enter a patient's body to view the internal organs and possibly take biopsies of suspected areas.

These robots are the size of a lipstick case and have TV cameras and lighting to allow a doctor to observe while taking a sample, cauterizing, or clamping a wound or delivering

medications. The three inch long robots use specialized spiral wheels for propulsion and so as not to tear or damage delicate tissue. The medical team feels that the robot's size could be halved when full scale production allows for specialized miniature circuitry. Figure 6 shows the remotely-controlled robot.

"It's a stark contrast to existing laparoscopic

techniques, which allow surgeons to perform operations through small incisions. The benefits of laparoscopy are limited to less complex procedures, however, because of losses in imaging and dexterity compared to conventional surgery," says Shane Farritor, associate professor of Mechanical Engineering at the University of Nebraska, Lincoln. "We want to be the Microsoft leader in this technology and be the state that changes the way surgery is done."

They will probably start out as single-use devices — much the same as the camera pills now in use — but could be sterilized in the future when temperature-stable components are developed. As with all new technology, costs will plummet when refinement and mass production are in place.

Small Robots for the Rest of Us

Several years ago, I'd heard about the iSoBot made by Tomy (Takaratomy Group in Japan) and had the chance to experiment (play) with one a while back. Touted by the *Guinness Book of World Records* as "the smallest humanoid robot in production," my wife recently saw it on sale at Hammacher Schlemmer for only \$99.95 and decided to get it for me for Christmas. The little guy is only 6-1/2" tall, but has 17 servos and thus, 17 degrees of freedom. The servos that Tomy uses are not the higher quality actuators we've used for years in experimental robots, but they are perfect for a sub-one pound robot such as this (see Figure 7).

FIGURE 7. iSoBot.



iSoBot has a gyro-sensor that stabilizes his walks and balance. Powered by 3-1.2V NiMH batteries, I had no trouble getting almost an hour of action out of iSoBot before he needed 'dinner.' These are the Sanyo Eneloop low discharge AA cells that retain 90% of their charge after six months and can be replaced for under \$10 for a pack of four cells. Fast charging is not recommended with this type of battery. The robot had originally been priced around \$299.

There are 180 pre-programmed motions that allow sequences such as a reenactment of a cowboy gun duel, martial arts movements, somersaults, and doing pushups, but you can also program it yourself to perform 240 movements in sequence. The robot can talk and can respond to voice commands such as "stop," "back up," "go forward," or to questions such as "how are you."

The operator enters an alpha-numeric code into the remote control to access different commands that are represented on the LCD panel in the control unit. You can operate in one of four modes: voice control, manual control, programmed sequences mode, and special actions mode. It's a cute little fellow.

Rope Climbing Robot

"Little" seems to be more of the norm these days as students in schools and colleges learn mechanical, electric, and computer sciences through construction of simple robotic projects. I was recently invited to a contest at Washington State University, Vancouver where the contestants were small robots designed by freshmen students to climb a rope up to and stop at seven feet. The robots had to operate on one or two nine-volt batteries; had to use a single 12 volt motor; and weigh no more than two pounds.

This was an excellent exercise in simple mechanics and was part of the school's Mech 101 Introduction to Mechanical Engineering course. Using various scrounged parts not to exceed \$50, I was amazed at the ingenuity of the students' many approaches to the

task. Simple odometry with an encoder, a tethered string, and even a timer were used to have the robots stop at the top of the rope. Figure 8 shows the members of Team Awesome watching their machine climb up the rope on a practice run. (The robot is the red-white-green little thing near eye level, not the black and yellow starting block.)

The shortest time was just one of the factors considered in judging there was also a written final report required and points were given for other mechanical accomplishments.

M&M Sorting Robot

One of the most interesting and absolutely cool robots that I've ever seen is Ted Griebing's M&M sorting robot. Not only did the robot have to steer with its two wheels driven in a differential configuration, but it needed a color vision system to distinguish between different colored M&Ms. Figure 9 shows the tiny 1" cube going about its business of sorting colored M&M candies.

That's A Wrap

I've just touched on a few of the small robots that have made recent headlines. There are many experimenters who specialize in only the very smallest types of robots such as the BEAM robots that might use two tiny pager/cell phone vibrator motors for movement and a solar cell charging a capacitor for power. The various military agencies of the US

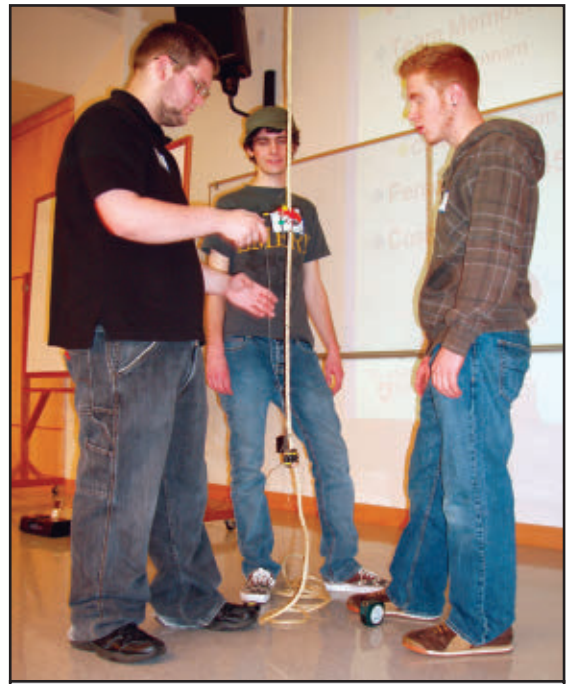


FIGURE 8. Rope-climbing robot.

have long been examining flying robot concepts that might be the proverbial "fly on the wall," to go totally unnoticed. Using motors even smaller than pager motors (such as watch motors), tiny robot mechanisms carrying miniscule video cameras and transmitters will infiltrate enemy lines to give military leaders valuable intel. Citizen Watch Company built the tiny robot in Figure 10 as a start for research on even smaller robots. There really appears to be no end to the smallness in robots. I'd appreciate readers sending me more information on small experimental robots that they've built or have seen. **SV**

Tom Carroll can be reached via email at TWCarroll@aol.com.

FIGURE 9. Griebing's M & M robot.

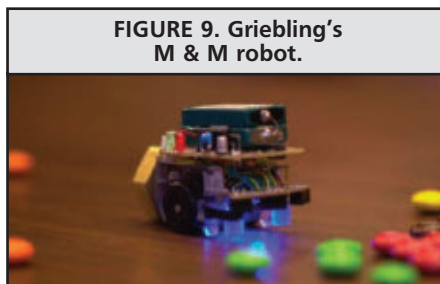


FIGURE 10. Citizen Watch robot.



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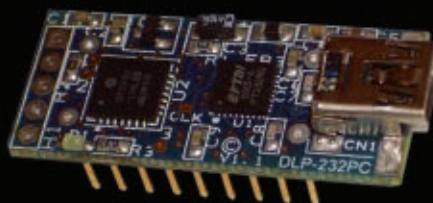
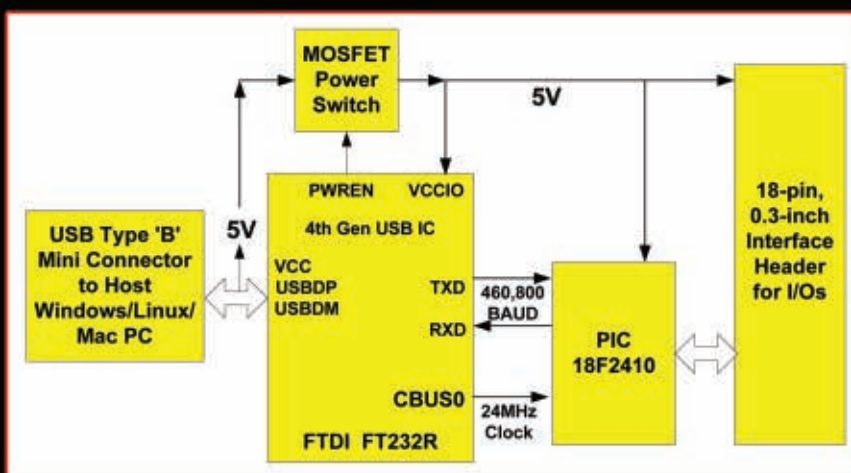
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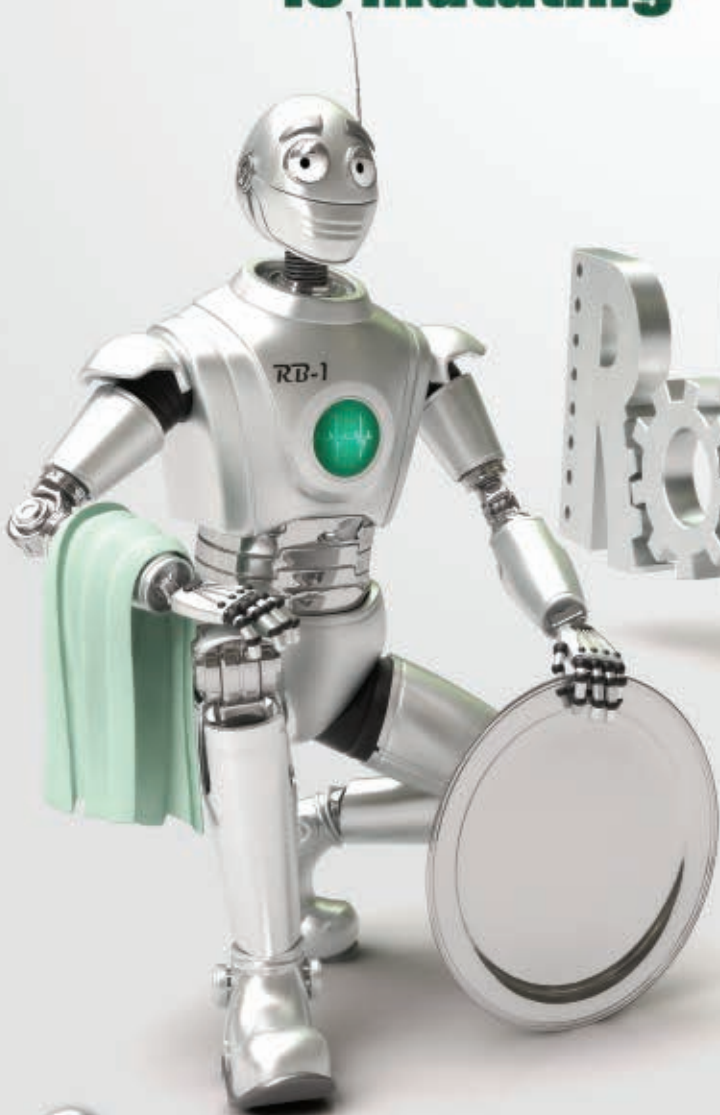
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